



FIGURE 7.1
Me playing
with my ding-
a-ling in the
Holimarine Talent
Show. Note the
groupies queuing
up at the front

7.1. What will this chapter tell me? ①

Although none of us can know the future, predicting it is so important that organisms are hard wired to learn about predictable events in their environment. We saw in the previous chapter that I received a guitar for Christmas when I was 8. My first foray into public performance was a weekly talent show at a holiday camp called ‘Holimarine’ in Wales (it doesn’t exist anymore because I am old and this was 1981). I sang a Chuck Berry song called ‘My ding-a-ling’¹ and to my absolute amazement I won the competition.² Suddenly other 8 year olds across the land (well, a ballroom in Wales) worshipped me (I made lots of friends after the competition). I had tasted success, it tasted like praline chocolate, and so I wanted to enter the competition in the second week of our holiday. To ensure success, I needed to know why I had won in the first week. One way to do this would have been to collect data and to use these data to predict people’s evaluations of children’s performances in the contest

¹ It appears that even then I had a passion for lowering the tone of things that should be taken seriously.

² I have a very grainy video of this performance recorded by my dad’s friend on a video camera the size of a medium-sized dog that had to be accompanied at all times by a ‘battery pack’ the size and weight of a tank. Maybe I’ll put it up on the companion website ...

from certain variables: the age of the performer, what type of performance they gave (singing, telling a joke, magic tricks), and maybe how cute they looked. A regression analysis on these data would enable us to predict future evaluations (success in next week’s competition) based on values of the predictor variables. If, for example, singing was an important factor in getting a good audience evaluation, then I could sing again the following week; however, if jokers tended to do better then I could switch to a comedy routine. When I was 8 I wasn’t the sad geek that I am today, so I didn’t know about regression analysis (nor did I wish to know); however, my dad thought that success was due to the winning combination of a cherub-looking 8 year old singing songs that can be interpreted in a filthy way. He wrote me a song to sing in the competition about the keyboard player in the Holimarine Band ‘messing about with his organ’, and first place was mine again. There’s no accounting for taste.

7.2. An introduction to regression ①

In the previous chapter we looked at how to measure relationships between two variables. These correlations can be very useful but we can take this process a step further and predict one variable from another. A simple example might be to try to predict levels of stress from the amount of time until you have to give a talk. You’d expect this to be a negative relationship (the smaller the amount of time until the talk, the larger the anxiety). We could then extend this basic relationship to answer a question such as ‘if there’s 10 minutes to go until someone has to give a talk, how anxious will they be?’ This is the essence of regression analysis: we fit a model to our data and use it to predict values of the dependent variable (DV) from one or more independent variables (IVs).³ Regression analysis is a way of predicting an **outcome variable** from one **predictor variable** (**simple regression**) or several predictor variables (**multiple regression**). This tool is incredibly useful because it allows us to go a step beyond the data that we collected.

In section 2.4.3 I introduced you to the idea that we can predict any data using the following general equation:

$$\text{outcome}_i = (\text{model}) + \text{error}_i \tag{7.1}$$

This just means that the outcome we’re trying to predict for a particular person can be predicted by whatever model we fit to the data plus some kind of error. In regression, the model we fit is linear, which means that we summarize a data set with a straight line (think back to Jane Superbrain Box 2.1). As such, the word ‘model’ in the equation above simply gets replaced by ‘things’ that define the line that we fit to the data (see the next section).

With any data set there are several lines that could be used to summarize the general trend and so we need a way to decide which of many possible lines to choose. For the sake of making accurate predictions we want to fit a model that *best* describes the data. The simplest way to do this would be to use your eye to gauge a line that looks as though it summarizes the data well. You don’t need to be a genius to realize that the ‘eyeball’ method is very subjective and so offers no assurance that the model is the best one that could have been chosen. Instead, we use a mathematical technique called the *method of least squares* to establish the line that best describes the data collected.

How do I fit a straight line to my data?



³ I want to remind you here of something I discussed in Chapter 1: SPSS refers to regression variables as dependent and independent variables (as in controlled experiments). However, correlational research by its nature seldom controls the independent variables to measure the effect on a dependent variable and so I will talk about ‘independent variables’ as *predictors*, and the ‘dependent variable’ as the *outcome*.

7.2.1. Some important information about straight lines ①

I mentioned above that in our general equation the word ‘model’ gets replaced by ‘things that define the line that we fit to the data’. In fact, any straight line can be defined by two things: (1) the slope (or gradient) of the line (usually denoted by b_1); and (2) the point at which the line crosses the vertical axis of the graph (known as the *intercept* of the line, b_0). In fact, our general model becomes equation (7.2) below in which Y_i is the outcome that we want to predict and X_i is the i th participant’s score on the predictor variable.⁴ Here b_1 is the gradient of the straight line fitted to the data and b_0 is the intercept of that line. These parameters b_1 and b_0 are known as the *regression coefficients* and will crop up time and time again in this book, where you may see them referred to generally as b (without any subscript) or b_i (meaning the b associated with variable i). There is a residual term, ε_i , which represents the difference between the score predicted by the line for participant i and the score that participant i actually obtained. The equation is often conceptualized without this residual term (so, ignore it if it’s upsetting you); however, it is worth knowing that this term represents the fact our model will not fit the data collected perfectly:

$$Y_i = (b_0 + b_1X_i) + \varepsilon_i$$

(7.2)

A particular line has a specific intercept and gradient. Figure 7.2 shows a set of lines that have the same intercept but different gradients, and a set of lines that have the same gradient but different intercepts. Figure 7.2 also illustrates another useful point: the gradient of the line tells us something about the nature of the relationship being described. In Chapter 6 we saw how relationships can be either positive or negative (and I don’t mean the difference between getting on well with your girlfriend and arguing all the time!). A line that has a gradient with a positive value describes a positive relationship, whereas a line with a negative gradient describes a negative relationship. So, if you look at the graph in Figure 7.2

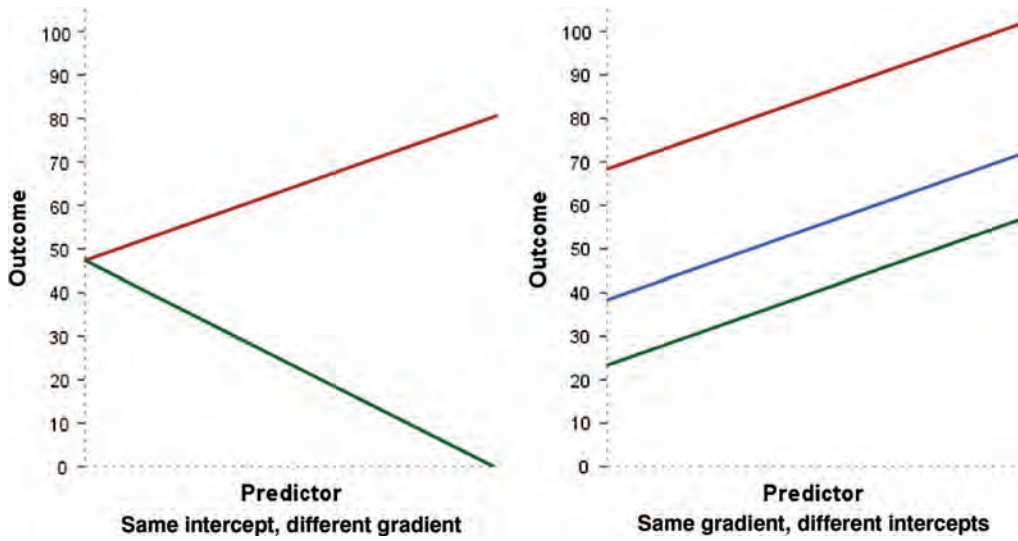


FIGURE 7.2
Lines with the same gradients but different intercepts, and lines that share the same intercept but have different gradients

⁴ You’ll sometimes see this equation written as:

$$Y_i = (\beta_0 + \beta_1X_i) + \varepsilon_i$$

The only difference is that this equation has got β s in it instead of b s and in fact both versions are the same thing, they just use different letters to represent the coefficients. When SPSS estimates the coefficients in this equation it labels them b and to be consistent with the SPSS output that you’ll end up looking at, I’ve used b s instead of β s.

in which the gradients differ but the intercepts are the same, then the red line describes a positive relationship whereas the green line describes a negative relationship. Basically then, the gradient (b) tells us what the model looks like (its shape) and the intercept (b_0) tells us where the model is (its location in geometric space).

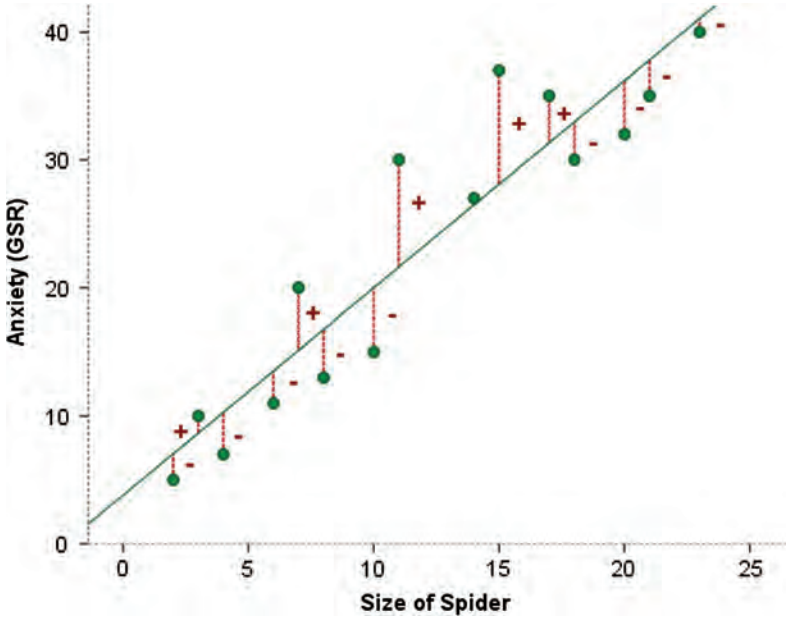
If it is possible to describe a line knowing only the gradient and the intercept of that line, then we can use these values to describe our model (because in linear regression the model we use is a straight line). So, the model that we fit to our data in linear regression can be conceptualized as a straight line that can be described mathematically by equation (7.2). With regression we strive to find the line that best describes the data collected, then estimate the gradient and intercept of that line. Having defined these values, we can insert different values of our predictor variable into the model to estimate the value of the outcome variable.

7.2.2. The method of least squares ①

I have already mentioned that the method of least squares is a way of finding the line that best fits the data (i.e. finding a line that goes through, or as close to, as many of the data points as possible). This ‘line of best fit’ is found by ascertaining which line, of all of the possible lines that could be drawn, results in the least amount of difference between the observed data points and the line. Figure 7.3 shows that when any line is fitted to a set of data, there will be small differences between the values predicted by the line and the data that were actually observed.

Back in Chapter 2 we saw that we could assess the fit of a model (the example we used was the mean) by looking at the deviations between the model and the actual data collected. These deviations were the vertical distances between what the model predicted and each data point that was actually observed. We can do exactly the same to assess the fit of a regression line (which, like the mean, is a statistical model). So, again we are interested in the vertical differences between the line and the actual data because the line is our model: we use it to predict values of Y from values of the X variable. In regression these differences are usually called **residuals** rather than deviations, but they are the same thing. As with the mean, data points fall both above (the model underestimates their value) and below (the

FIGURE 7.3
This graph shows a scatterplot of some data with a line representing the general trend. The vertical lines (dotted) represent the differences (or residuals) between the line and the actual data



model overestimates their value) the line yielding both positive and negative differences. In the discussion of variance in section 2.4.2 I explained that if we sum positive and negative differences then they tend to cancel each other out and that to circumvent this problem we square the differences before adding them up. We do the same thing here. The resulting squared differences provide a gauge of how well a particular line fits the data: if the squared differences are large, the line is not representative of the data; if the squared differences are small, the line is representative.

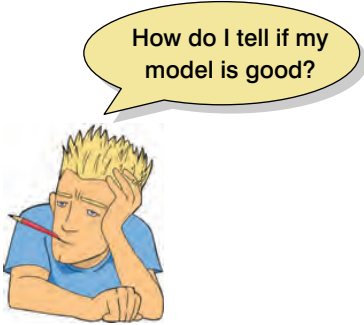
You could, if you were particularly bored, calculate the sum of squared differences (or SS for short) for every possible line that is fitted to your data and then compare these ‘goodness-of-fit’ measures. The one with the lowest SS is the line of best fit. Fortunately we don’t have to do this because the method of least squares does it for us: it selects the line that has the lowest sum of squared differences (i.e. the line that best represents the observed data). How exactly it does this is by using a mathematical technique for finding maxima and minima and this technique is used to find the line that minimizes the sum of squared differences. I don’t really know much more about it than that, to be honest, so I tend to think of the process as a little bearded wizard called Nephwick the Line Finder who just magically finds lines of best fit. Yes, he lives inside your computer. The end result is that Nephwick estimates the value of the slope and intercept of the ‘line of best fit’ for you. We tend to call this line of best fit a regression line.

7.2.3. Assessing the goodness of fit: sums of squares, R and R^2 ①

Once Nephwick the Line Finder has found the line of best fit it is important that we assess how well this line fits the actual data (we assess the **goodness of fit** of the model). We do this because even though this line is the best one available, it can still be a lousy fit to the data! In section 2.4.2 we saw that one measure of the adequacy of a model is the sum of squared differences (or more generally we assess models using equation (7.3) below). If we want to assess the line of best fit, we need to compare it against something, and the thing we choose is the most basic model we can find. So we use equation (7.3) to calculate the fit of the most basic model, and then the fit of the best model (the line of best fit), and basically if the best model is any good then it should fit the data significantly better than our basic model:

deviation = $\sum (\text{observed} - \text{model})^2$ (7.3)

This is all quite abstract so let’s look at an example. Imagine that I was interested in predicting record sales (Y) from the amount of money spent advertising that record (X). One day my boss came in to my office and said ‘Andy, I know you wanted to be a rock star and you’ve ended up working as my stats-monkey, but how many records will we sell if we spend £100,000 on advertising?’ If I didn’t have an accurate model of the relationship between record sales and advertising, what would my best guess be? Well, probably the best answer I could give would be the mean number of record sales (say, 200,000) because on average that’s how many records we expect to sell. This response might well satisfy a brainless record company executive (who didn’t offer my band a record contract). However, what if he had asked ‘How many records will we sell if we spend £1 on advertising?’ Again, in the absence of any accurate information, my best guess would be to give the average number of sales (200,000). There is a problem: whatever amount of money is spent on advertising I



always predict the same levels of sales. As such, the mean is a model of ‘no relationship’ at all between the variables. It should be pretty clear then that the mean is fairly useless as a model of a relationship between two variables – but it is the simplest model available.

So, as a basic strategy for predicting the outcome, we might choose to use the mean, because on average it will be a fairly good guess of an outcome. Using the mean as a model, we can calculate the difference between the observed values, and the values predicted by the mean (equation (7.3)). We saw in section 2.4.2 that we square all of these differences to give us the sum of squared differences. This sum of squared differences is known as the **total sum of squares** (denoted SS_T) because it is the total amount of differences present when the most basic model is applied to the data. This value represents how good the mean is as a model of the observed data. Now, if we fit the more sophisticated model to the data, such as a line of best fit, we can again work out the differences between this new model and the observed data (again using equation (7.3)). In the previous section we saw that the method of least squares finds the best possible line to describe a set of data by minimizing the difference between the model fitted to the data and the data themselves. However, even with this optimal model there is still some inaccuracy, which is represented by the differences between each observed data point and the value predicted by the regression line. As before, these differences are squared before they are added up so that the directions of the differences do not cancel out. The result is known as the *sum of squared residuals* or **residual sum of squares** (SS_R). This value represents the degree of inaccuracy when the best model is fitted to the data. We can use these two values to calculate how much better the regression line (the line of best fit) is than just using the mean as a model (i.e. how much better is the best possible model than the worst model?). The improvement in prediction resulting from using the regression model rather than the mean is calculated by calculating the difference between SS_T and SS_R . This difference shows us the reduction in the inaccuracy of the model resulting from fitting the regression model to the data. This improvement is the **model sum of squares** (SS_M). Figure 7.4 shows each sum of squares graphically.

If the value of SS_M is large then the regression model is very different from using the mean to predict the outcome variable. This implies that the regression model has made a big improvement to how well the outcome variable can be predicted. However, if SS_M is small then using the regression model is little better than using the mean (i.e. the regression model is no better than taking our ‘best guess’). A useful measure arising from these sums of squares is the proportion of improvement due to the model. This is easily calculated by dividing the sum of squares for the model by the total sum of squares. The resulting value is called R^2 and to express this value as a percentage you should multiply it by 100. R^2 represents the amount of variance in the outcome explained by the model (SS_M) relative to how much variation there was to explain in the first place (SS_T). Therefore, as a percentage, it represents the percentage of the variation in the outcome that can be explained by the model:

$$R^2 = \frac{SS_M}{SS_T} \quad (7.4)$$

This R^2 is the same as the one we met in Chapter 6 (section 6.5.2.3) and you might have noticed that it is interpreted in the same way. Therefore, in simple regression we can take the square root of this value to obtain Pearson’s correlation coefficient. As such, the correlation coefficient provides us with a good estimate of the overall fit of the regression model, and R^2 provides us with a good gauge of the substantive size of the relationship.

A second use of the sums of squares in assessing the model is through the F -test. I mentioned way back in Chapter 2 that test statistics (like F) are usually the amount of systematic variance divided by the amount of unsystematic variance, or, put another way, the model compared against the error in the model. This is true here: F is based upon the ratio of the improvement due to the model (SS_M) and the difference between the model and the observed data (SS_R). Actually, because the sums of squares depend on the number

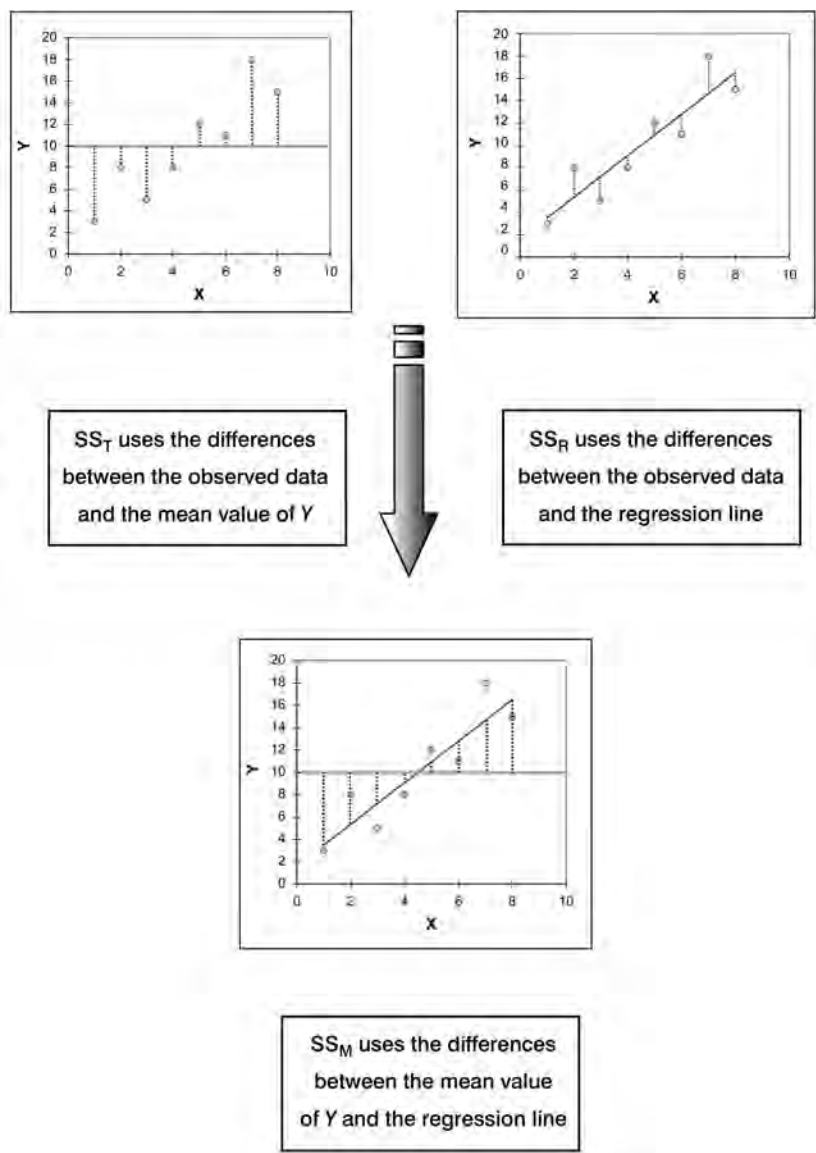


FIGURE 7.4
Diagram showing
from where the
regression sums
of squares derive

of differences that we have added up, we use the average sums of squares (referred to as the **mean squares** or MS). To work out the mean sums of squares we divide by the degrees of freedom (this is comparable to calculating the variance from the sums of squares – see section 2.4.2). For SS_M the degrees of freedom are simply the number of variables in the model, and for SS_R they are the number of observations minus the number of parameters being estimated (i.e. the number of beta coefficients including the constant). The result is the mean squares for the model (MS_M) and the residual mean squares (MS_R). At this stage it isn't essential that you understand how the mean squares are derived (it is explained in Chapter 10). However, it is important that you understand that the **F-ratio** (equation (7.5)) is a measure of how much the model has improved the prediction of the outcome compared to the level of inaccuracy of the model:

$$F = \frac{MS_M}{MS_R}$$

(7.5)

If a model is good, then we expect the improvement in prediction due to the model to be large (so, MS_M will be large) and the difference between the model and the observed data to be small (so, MS_R will be small). In short, a good model should have a large F -ratio (greater than 1 at least) because the top of equation (7.5) will be bigger than the bottom. The exact magnitude of this F -ratio can be assessed using critical values for the corresponding degrees of freedom (as in the Appendix).

7.2.4. Assessing individual predictors ①

We've seen that the predictor in a regression model has a coefficient (b_1), which in simple regression represents the gradient of the regression line. The value of b represents the change in the outcome resulting from a unit change in the predictor. If the model was useless at predicting the outcome, then if the value of the predictor changes, what might we expect the change in the outcome to be? Well, if the model is very bad then we would expect the change in the outcome to be zero. Think back to Figure 7.4 (see the panel representing SS_T) in which we saw that using the mean was a very bad way of predicting the outcome. In fact, the line representing the mean is flat, which means that as the predictor variable changes, the value of the outcome does *not* change (because for each level of the predictor variable, we predict that the outcome will equal the mean value). The important point here is that a bad model (such as the mean) will have regression coefficients of 0 for the predictors. A regression coefficient of 0 means: (1) a unit change in the predictor variable results in no change in the predicted value of the outcome (the predicted value of the outcome does not change at all); and (2) the gradient of the regression line is 0, meaning that the regression line is flat. Hopefully, you'll see that it logically follows that if a variable significantly predicts an outcome, then it should have a b -value significantly different from zero. This hypothesis is tested using a t -test (see Chapter 9). The **t -statistic** tests the null hypothesis that the value of b is 0: therefore, if it is significant we gain confidence in the hypothesis that the b -value is significantly different from 0 and that the predictor variable contributes significantly to our ability to estimate values of the outcome.

Like F , the t -statistic is also based on the ratio of explained variance against unexplained variance or error. Well, actually, what we're interested in here is not so much variance but whether the b we have is big compared to the amount of error in that estimate. To estimate how much error we could expect to find in b we use the standard error. The standard error tells us something about how different b -values would be across different samples. We could take lots and lots of samples of data regarding record sales and advertising budgets and calculate the b -values for each sample. We could plot a frequency distribution of these samples to discover whether the b -values from all samples would be relatively similar, or whether they would be very different (think back to section 2.5.1). We can use the standard deviation of this distribution (known as the *standard error*) as a measure of the similarity of b -values across samples. If the standard error is very small, then it means that most samples are likely to have a b -value similar to the one in our sample (because there is little variation across samples). The t -test tells us whether the b -value is different from 0 relative to the variation in b -values across samples. When the standard error is small even a small deviation from zero can reflect a meaningful difference because b is representative of the majority of possible samples.

Equation (7.6) shows how the t -test is calculated and you'll find a general version of this equation in Chapter 9 (equation (9.1)). The b_{expected} is simply the value of b that we would expect to obtain if the null hypothesis were true. I mentioned earlier that the null hypothesis is that b is 0 and so this value can be replaced by 0. The equation simplifies to become the observed value of b divided by the standard error with which it is associated:

$$\begin{aligned} t &= \frac{b_{\text{observed}} - b_{\text{expected}}}{SE_b} \\ &= \frac{b_{\text{observed}}}{SE_b} \end{aligned} \tag{7.6}$$

The values of t have a special distribution that differs according to the degrees of freedom for the test. In regression, the degrees of freedom are $N - p - 1$, where N is the total sample size and p is the number of predictors. In simple regression when we have only one predictor, this reduces down to $N - 2$. Having established which t -distribution needs to be used, the observed value of t can then be compared to the values that we would expect to find if there was no effect (i.e. $b = 0$): if t is very large then it is unlikely to have occurred when there is no effect (these values can be found in the Appendix). SPSS provides the exact probability that the observed value or a larger value of t would occur if the value of b was, in fact, 0. As a general rule, if this observed significance is less than .05, then scientists assume that b is significantly different from 0; put another way, the predictor makes a significant contribution to predicting the outcome.

7.3. Doing simple regression on SPSS ①

So far, we have seen a little of the theory behind regression, albeit restricted to the situation in which there is only one predictor. To help clarify what we have learnt so far, we will go through an example of a simple regression on SPSS. Earlier on I asked you to imagine that I worked for a record company and that my boss was interested in predicting record sales from advertising. There are some data for this example in the file **Record1.sav**. This data file has 200 rows, each one representing a different record. There are also two columns, one representing the sales of each record in the week after release and the other representing the amount (in pounds) spent promoting the record before release. This is the format for entering regression data: the outcome variable and any predictors should be entered in different columns, and each row should represent independent values of those variables.



The pattern of the data is shown in Figure 7.5 and it should be clear that a positive relationship exists: so, the more money spent advertising the record, the more it is likely to sell. Of course there are some records that sell well regardless of advertising (top left of scatterplot), but there are none that sell badly when advertising levels are high (bottom right of scatterplot). The scatterplot also shows the line of best fit for these data: bearing in mind that the mean would be represented by a flat line at around the 200,000 sales mark, the regression line is noticeably different.

To find out the parameters that describe the regression line, and to see whether this line is a useful model, we need to run a regression analysis. To do the analysis you need to access the main dialog box by selecting **Analyze > Regression > Linear**. Figure 7.6 shows the resulting dialog box. There is a space labelled *Dependent* in which you should place the outcome variable (in this example **sales**). So, select **sales** from the list on the left-hand side, and transfer it by dragging it or clicking on the arrow button. There is another space labelled *Independent(s)* in which any predictor variable should be placed. In simple regression we use only one predictor (in this example, **adverts**) and so you should select **adverts** from the list and click on the arrow button to transfer it to the list of predictors. There are a variety of options available, but these will be explored within the context of multiple regression. For the time being just click on **OK** to run the basic analysis.

FIGURE 7.5
Scatterplot showing the relationship between record sales and the amount spent promoting the record

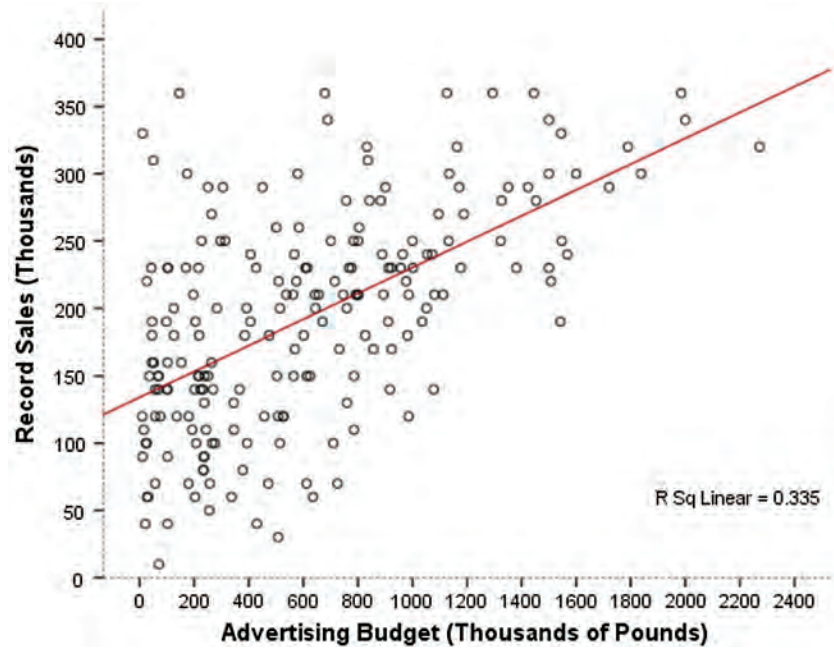
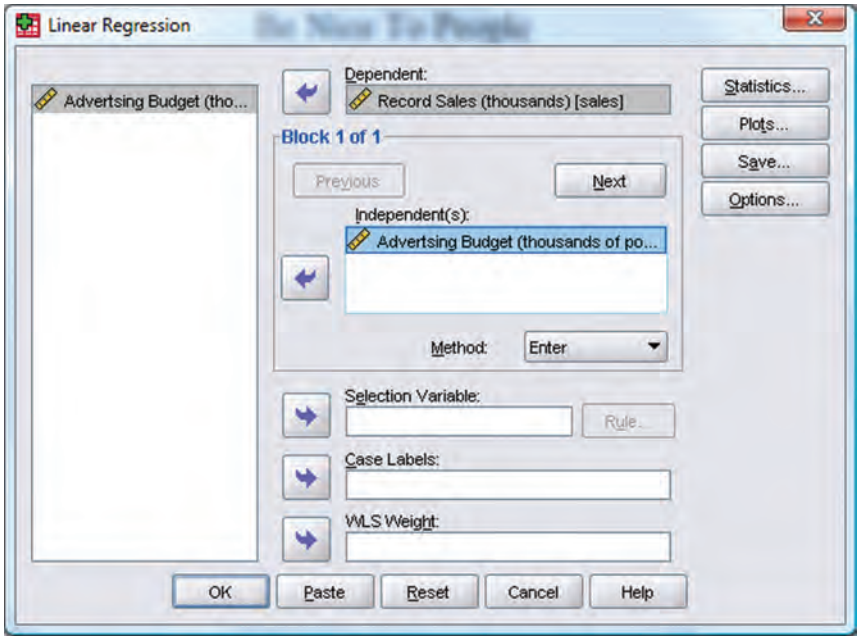


FIGURE 7.6
Main dialog box for regression



7.4. Interpreting a simple regression ①

7.4.1. Overall fit of the model ①

The first table provided by SPSS is a summary of the model (SPSS Output 7.1). This summary table provides the value of R and R^2 for the model that has been derived. For these data, R has

a value of .578 and because there is only one predictor, this value represents the simple correlation between advertising and record sales (you can confirm this by running a correlation using what you were taught in Chapter 6). The value of R^2 is .335, which tells us that advertising expenditure can account for 33.5% of the variation in record sales. In other words, if we are trying to explain why some records sell more than others, we can look at the variation in sales of different records. There might be many factors that can explain this variation, but our model, which includes only advertising expenditure, can explain approximately 33% of it. This means that 66% of the variation in record sales cannot be explained by advertising alone. Therefore, there must be other variables that have an influence also.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.578 ^a	.335	.331	65.991

a. Predictors: (Constant), Advertising Budget (thousands of pounds)

SPSS OUTPUT 7.1

The next part of the output (SPSS Output 7.2) reports an analysis of variance (ANOVA – see Chapter 10). The summary table shows the various sums of squares described in Figure 7.4 and the degrees of freedom associated with each. From these two values, the average sums of squares (the mean squares) can be calculated by dividing the sums of squares by the associated degrees of freedom. The most important part of the table is the F -ratio, which is calculated using equation (7.5), and the associated significance value of that F -ratio. For these data, F is 99.59, which is significant at $p < .001$ (because the value in the column labelled *Sig.* is less than .001). This result tells us that there is less than a 0.1% chance that an F -ratio this large would happen if the null hypothesis were true. Therefore, we can conclude that our regression model results in significantly better prediction of record sales than if we used the mean value of record sales. In short, the regression model overall predicts record sales significantly well.

ANOVA^b

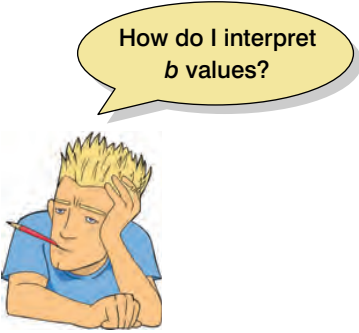
Model	Sum of Squares	df	Mean Square	F	Sig.
1 Regression	433687.833	1	433687.833	99.587	.000 ^a
Residual	882264.167	198	4354.870		
Total	1295952.000	199			

a. Predictors: (Constant), Advertising Budget (thousands of pounds)
b. Dependent Variable: Record Sales (thousands)

SPSS OUTPUT 7.2

7.4.2. Model parameters ①

The ANOVA tells us whether the model, overall, results in a significantly good degree of prediction of the outcome variable. However, the ANOVA doesn't tell us about the individual contribution of variables in the model (although in this simple case there is only one variable in the model and so we can infer that this variable is a good predictor). The table in SPSS Output 7.3 provides details of the model parameters (the beta values) and the significance of these values. We saw in equation (7.2) that b_0 was the Y intercept and this value is the value B (in the SPSS output) for the constant. So, from the table, we can say that b_0 is 134.14, and this can be interpreted as meaning that when no money is spent on



advertising (when $X = 0$), the model predicts that 134,140 records will be sold (remember that our unit of measurement was thousands of records). We can also read off the value of b_1 from the table and this value represents the gradient of the regression line. It is 0.096. Although this value is the slope of the regression line, it is more useful to think of this value as representing *the change in the outcome associated with a unit change in the predictor*. Therefore, if our predictor variable is increased by one unit (if the advertising budget is increased by 1), then our model predicts that 0.096 extra records will be sold. Our units of measurement were thousands of pounds and thousands of records sold, so we can say that for an increase in advertising of £1000 the model predicts 96 ($0.096 \times 1000 = 96$) extra record sales. As you might imagine, this investment is pretty bad for the record company: it invests £1000 and gets only 96 extra sales. Fortunately, as we already know, advertising accounts for only one-third of record sales.

SPSS OUTPUT 7.3

Coefficients ^a					
Model		Unstandardized Coefficients		Standardized Coefficients	t
		B	Std. Error	Beta	
1	(Constant)	134.140	7.537		17.799
	Advertising Budget (thousands of pounds)	.096	.010	.578	9.979

a. Dependent Variable: Record Sales (thousands)

We saw earlier that, in general, values of the regression coefficient b represent the change in the outcome resulting from a unit change in the predictor and that if a predictor is having a significant impact on our ability to predict the outcome then this b should be different from 0 (and big relative to its standard error). We also saw that the t -test tells us whether the b -value is different from 0. SPSS provides the exact probability that the observed value of t would occur if the value of b in the population were 0. If this observed significance is less than .05, then scientists agree that the result reflects a genuine effect (see Chapter 2). For these two values, the probabilities are .000 (zero to 3 decimal places) and so we can say that the probability of these t -values or larger occurring if the values of b in the population were 0 is less than .001. Therefore, the b s are different from 0 and we can conclude that the advertising budget makes a significant contribution ($p < .001$) to predicting record sales.



SELF-TEST How is the t in SPSS Output 7.3 calculated? Use the values in the table to see if you can get the same value as SPSS.

7.4.3. Using the model ①

So far, we have discovered that we have a useful model, one that significantly improves our ability to predict record sales. However, the next stage is often to use that model to make some predictions. The first stage is to define the model by replacing the b -values in equation (7.2) with the values from SPSS Output 7.3. In addition, we can replace the X and Y with the variable names so that the model becomes:

record sales_{*i*} = b_0 + b_1 advertising budget_{*i*}
= 134.14 + (0.096 × advertising budget_{*i*})

(7.7)

It is now possible to make a prediction about record sales, by replacing the advertising budget with a value of interest. For example, imagine a record executive wanted to spend £100,000 on advertising a new record. Remembering that our units are already in thousands of pounds, we can simply replace the advertising budget with 100. He would discover that record sales should be around 144,000 for the first week of sales:

$$\begin{aligned} \text{record sales}_i &= 134.14 + (0.096 \times \text{advertising budget}_i) \\ &= 134.14 + (0.096 \times 100) \\ &= 143.74 \end{aligned}$$

(7.8)



SELF-TEST How many records would be sold if we spent £666,000 on advertising the latest CD by black metal band Abgott?



CRAMMING SAM'S TIPS

Simple regression

- Simple regression is a way of predicting values of one variable from another.
- We do this by fitting a statistical model to the data in the form of a straight line.
- This line is the line that best summarizes the pattern of the data.
- We have to assess how well the line fits the data using:
 - R^2 which tells us how much variance is explained by the model compared to how much variance there is to explain in the first place. It is the proportion of variance in the outcome variable that is shared by the predictor variable.
 - F , which tells us how much variability the model can explain relative to how much it can't explain (i.e. it's the ratio of how good the model is compared to how bad it is).
- The b -value tells us the gradient of the regression line and the strength of the relationship between a predictor and the outcome variable. If it is significant ($Sig. < .05$ in the SPSS table) then the predictor variable significantly predicts the outcome variable.

7.5. Multiple regression: the basics ②

To summarize what we have learnt so far, in simple linear regression the outcome variable Y is predicted using the equation of a straight line (equation (7.2)). Given that we have collected several values of Y and X , the unknown parameters in the equation can be calculated. They are calculated by fitting a model to the data (in this case a straight line) for which the sum of the squared differences between the line and the actual data points is minimized. This method is called the method of least

What is the difference between simple and multiple regression?



squares. Multiple regression is a logical extension of these principles to situations in which there are several predictors. Again, we still use our basic equation of:

$$\text{outcome}_i = (\text{model}) + \text{error}_i$$

but this time the model is slightly more complex. It is basically the same as for simple regression except that for every extra predictor you include, you have to add a coefficient; so, each predictor variable has its own coefficient, and the outcome variable is predicted from a combination of all the variables multiplied by their respective coefficients plus a residual term (see equation (7.9) – the brackets aren’t necessary, they’re just to make the connection to the general equation above):

$$Y_i = (b_0 + b_1X_{i1} + b_2X_{i2} + \dots + b_nX_{in}) + \varepsilon_i \tag{7.9}$$

Y is the outcome variable, b_1 is the coefficient of the first predictor (X_1), b_2 is the coefficient of the second predictor (X_2), b_n is the coefficient of the n th predictor (X_n), and ε_i is the difference between the predicted and the observed value of Y for the i th participant. In this case, the model fitted is more complicated, but the basic principle is the same as simple regression. That is, we seek to find the linear combination of predictors that correlate maximally with the outcome variable. Therefore, when we refer to the regression model in multiple regression, we are talking about a model in the form of equation (7.9).

7.5.1. An example of a multiple regression model ②

Imagine that our record company executive was interested in extending his model of record sales to incorporate another variable. We know already that advertising accounts for 33% of variation in record sales, but a much larger 67% remains unexplained. The record executive could measure a new predictor in an attempt to explain some of the unexplained variation in record sales. He decides to measure the number of times the record is played on Radio 1 (the UK’s biggest national radio station) during the week prior to release. The existing model that we derived using SPSS (see equation (7.7)) can now be extended to include this new variable (**airplay**):

$$\text{record sales}_i = b_0 + b_1\text{advertising budget}_i + b_2\text{airplay}_i + \varepsilon_i \tag{7.10}$$

The new model is based on equation (7.9) and includes a b -value for both predictors (and, of course, the constant). If we calculate the b -values, we could make predictions about record sales based not only on the amount spent on advertising but also in terms of radio play. There are only two predictors in this model and so we could display this model graphically in three dimensions (Figure 7.7).

Equation (7.9) describes the tinted trapezium in the diagram (this is known as the regression *plane*) and the dots represent the observed data points. Like simple regression, the plane fitted to the data aims to best-predict the observed data. However, there are invariably some differences between the model and the real-life data (this fact is evident because some of the dots do not lie exactly on the tinted area of the graph). The b -value for advertising describes the slope of the left and right sides of the regression plane, whereas the b -value for airplay describes the slope of the top and bottom of the regression plane. Just like simple regression, knowledge of these two slopes tells us about the shape of the model (what it looks like) and the intercept locates the regression plane in space.

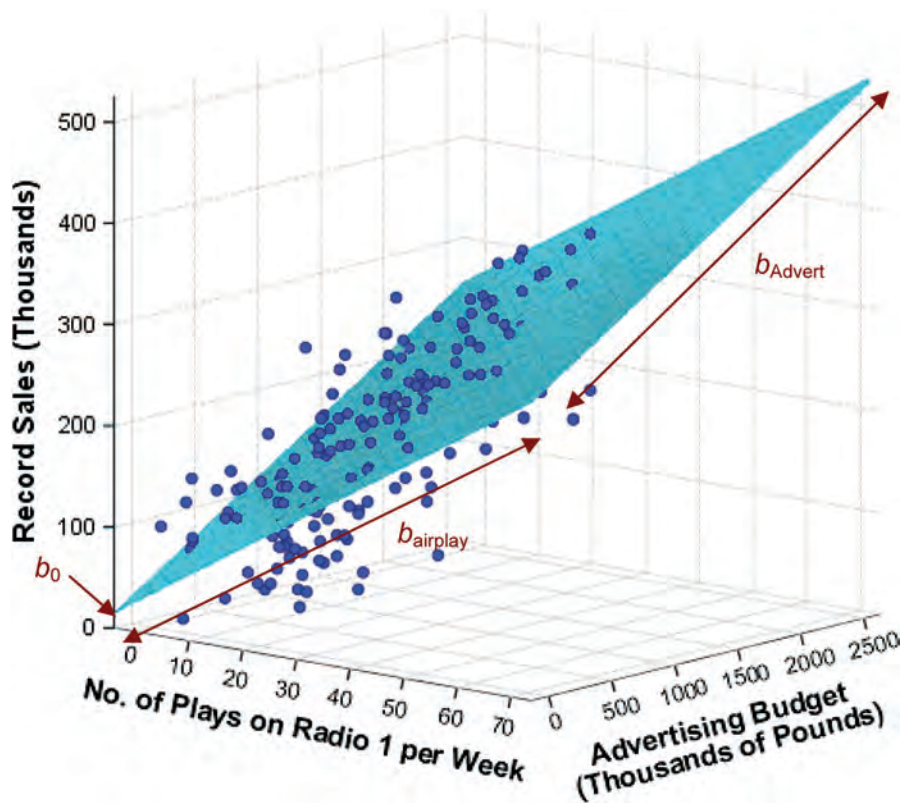


FIGURE 7.7
Scatterplot of
the relationship
between record
sales, advertising
budget and
radio play

It is fairly easy to visualize a regression model with two predictors, because it is possible to plot the regression plane using a 3-D scatterplot. However, multiple regression can be used with three, four or even ten or more predictors. Although you can't immediately visualize what such complex models look like, or visualize what the b -values represent, you should be able to apply the principles of these basic models to more complex scenarios.

7.5.2. Sums of squares, R and R^2 ②

When we have several predictors, the partitioning of sums of squares is the same as in the single variable case except that the model we refer to takes the form of equation (7.9) rather than simply being a 2-D straight line. Therefore, SS_T can be calculated that represents the difference between the observed values and the mean value of the outcome variable. SS_R still represents the difference between the values of Y predicted by the model and the observed values. Finally, SS_M can still be calculated and represents the difference between the values of Y predicted by the model and the mean value. Although the computation of these values is much more complex than in simple regression, conceptually these values are the same.

When there are several predictors it does not make sense to look at the simple correlation coefficient and instead SPSS produces a multiple correlation coefficient (labelled **Multiple R**). Multiple R is the correlation between the observed values of Y and the values of Y predicted by the multiple regression model. Therefore, large values of the multiple R represent a large correlation between the predicted and observed values of the outcome. A multiple R of 1 represents a situation in which the model perfectly predicts the observed

data. As such, multiple R is a gauge of how well the model predicts the observed data. It follows that the resulting R^2 can be interpreted in the same way as simple regression: it is the amount of variation in the outcome variable that is accounted for by the model.

7.5.3. Methods of regression ②

If we are interested in constructing a complex model with several predictors, how do we decide which predictors to use? A great deal of care should be taken in selecting predictors for a model because the values of the regression coefficients depend upon the variables in the model. Therefore, the predictors included and the way in which they are entered into the model can have a great impact. In an ideal world, predictors should be selected based on past research.⁵ If new predictors are being added to existing models then select these new variables based on the substantive *theoretical* importance of these variables. One thing *not* to do is select hundreds of random predictors, bung them all into a regression analysis and hope for the best. In addition to the problem of selecting predictors, there are several ways in which variables can be entered into a model. When predictors are all completely uncorrelated the order of variable entry has very little effect on the parameters calculated; however, we rarely have uncorrelated predictors and so the method of predictor selection is crucial.

7.5.3.1. Hierarchical (blockwise entry) ②

In **hierarchical regression** predictors are selected based on past work and the experimenter decides in which order to enter the predictors into the model. As a general rule, known predictors (from other research) should be entered into the model first in order of their importance in predicting the outcome. After known predictors have been entered, the experimenter can add any new predictors into the model. New predictors can be entered either all in one go, in a stepwise manner, or hierarchically (such that the new predictor suspected to be the most important is entered first).

7.5.3.2. Forced entry ②

Forced entry (or *Enter* as it is known in SPSS) is a method in which all predictors are forced into the model simultaneously. Like hierarchical, this method relies on good theoretical reasons for including the chosen predictors, but unlike hierarchical the experimenter makes no decision about the order in which variables are entered. Some researchers believe that this method is the only appropriate method for theory testing (Studenmund & Cassidy, 1987) because stepwise techniques are influenced by random variation in the data and so seldom give replicable results if the model is retested.

7.5.3.3. Stepwise methods ②

In **stepwise regressions** decisions about the order in which predictors are entered into the model are based on a purely mathematical criterion. In the *forward* method, an initial model is defined that contains only the constant (b_0). The computer then searches for the predictor (out of the ones available) that best predicts the outcome variable – it does this

⁵ I might cynically qualify this suggestion by proposing that predictors be chosen based on past research that has utilized good methodology. If basing such decisions on regression analyses, select predictors based only on past research that has used regression appropriately and yielded reliable, generalizable models!

by selecting the predictor that has the highest simple correlation with the outcome. If this predictor significantly improves the ability of the model to predict the outcome, then this predictor is retained in the model and the computer searches for a second predictor. The criterion used for selecting this second predictor is that it is the variable that has the largest semi-partial correlation with the outcome. Let me explain this in plain English. Imagine that the first predictor can explain 40% of the variation in the outcome variable; then there is still 60% left unexplained. The computer searches for the predictor that can explain the biggest part of the remaining 60% (so, it is not interested in the 40% that is already explained). As such, this semi-partial correlation gives a measure of how much ‘new variance’ in the outcome can be explained by each remaining predictor (see section 6.6). The predictor that accounts for the most new variance is added to the model and, if it makes a significant contribution to the predictive power of the model, it is retained and another predictor is considered.

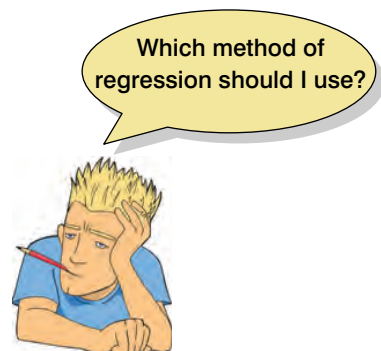
The *stepwise* method in SPSS is the same as the forward method, except that each time a predictor is added to the equation, a removal test is made of the least useful predictor. As such the regression equation is constantly being reassessed to see whether any redundant predictors can be removed. The *backward* method is the opposite of the forward method in that the computer begins by placing all predictors in the model and then calculating the contribution of each one by looking at the significance value of the *t*-test for each predictor. This significance value is compared against a removal criterion (which can be either an absolute value of the test statistic or a probability value for that test statistic). If a predictor meets the removal criterion (i.e. if it is not making a statistically significant contribution to how well the model predicts the outcome variable) it is removed from the model and the model is re-estimated for the remaining predictors. The contribution of the remaining predictors is then reassessed.

If you do decide to use a stepwise method then the backward method is preferable to the forward method. This is because of **suppressor effects**, which occur when a predictor has a significant effect but only when another variable is held constant. Forward selection is more likely than backward elimination to exclude predictors involved in suppressor effects. As such, the forward method runs a higher risk of making a Type II error (i.e. missing a predictor that does in fact predict the outcome).

7.5.3.4. Choosing a method ②

SPSS allows you to opt for any one of these methods and it is important to select an appropriate one. The forward, backward and stepwise methods all come under the general heading of *stepwise methods* because they all rely on the computer selecting variables based upon mathematical criteria. Many writers argue that this takes many important methodological decisions out of the hands of the researcher. What’s more, the models derived by computer often take advantage of random sampling variation and so decisions about which variables should be included will be based upon slight differences in their semi-partial correlation. However, these slight statistical differences may contrast dramatically with the theoretical importance of a predictor to the model. There is also the danger of over-fitting (having too many variables in the model that essentially make little contribution to predicting the outcome) and under-fitting (leaving out important predictors) the model. For this reason stepwise methods are best avoided except for exploratory model building. If you must do a stepwise regression then it is advisable to **cross-validate** your model by splitting the data (see section 7.6.2.2).

When there is a sound theoretical literature available, then base your model upon what past research tells you. Include any meaningful variables in the model in their order of importance. After this initial analysis, repeat the regression but exclude any variables that



were statistically redundant the first time around. There are important considerations in deciding which predictors should be included. First, it is important not to include too many predictors. As a general rule, the fewer predictors the better, and certainly include only predictors for which you have a good theoretical grounding (it is meaningless to measure hundreds of variables and then put them all into a regression model). So, be selective and remember that you should have a decent sample size – see section 7.6.2.3.

7.6. How accurate is my regression model? ②

How do I tell if my model is accurate?



When we have produced a model based on a sample of data there are two important questions to ask: (1) does the model fit the observed data well, or is it influenced by a small number of cases; and (2) can my model generalize to other samples? These questions are vital to ask because they affect how we use the model that has been constructed. These questions are also, in some sense, hierarchical because we wouldn't want to generalize a bad model. However, it is a mistake to think that because a model fits the observed data well we can draw conclusions beyond our sample. **Generalization** is a critical additional step and if we find that our model is not generalizable, then we must restrict any conclusions based on the model to the sample used. First, we will look at how we establish whether a model is an accurate representation of the actual data, and in section 7.6.2 we move on to look at how we assess whether a model can be used to make inferences beyond the sample of data that has been collected.

7.6.1. Assessing the regression model I: diagnostics ②

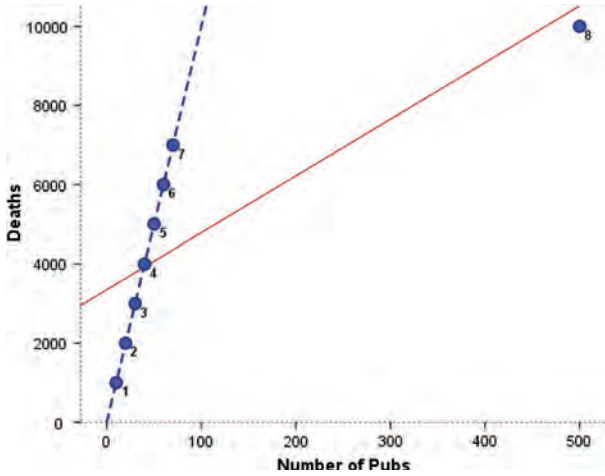
To answer the question of whether the model fits the observed data well, or if it is influenced by a small number of cases, we can look for outliers and influential cases (the difference is explained in Jane Superbrain Box 7.1). We will look at these in turn.



JANE SUPERBRAIN 7.1

The difference between residuals and influence statistics ③

In this section I've described two ways to look for cases that might bias the model: residual and influence statistics. To illustrate how these measures differ, imagine that the Mayor



of London at the turn of the last century was interested in how drinking affected mortality. London is divided up into different regions called boroughs, and so he might measure the number of pubs and the number of deaths over a period of time in eight of his boroughs. The data are in a file called **pubs.sav**.



The scatterplot of these data reveals that without the last case there is a perfect linear relationship (the dashed straight line). However, the presence of the last case (case 8) changes the line of best fit dramatically

(although this line is still a significant fit of the data – do the regression analysis and see for yourself).

What’s interesting about these data is when we look at the residuals and influence statistics. The standardized residual for case 8 is the second *smallest*: this outlier produces a very small residual (most of the non-outliers have larger residuals) because it sits very close to the line that has been fitted to the data. How can this be? Look at the influence statistics below and you’ll see that they’re massive for case 8: it exerts a huge influence over the model.

Case Summaries^a

	Standardized Residual	Mahalanobis Distance	Cook's Distance	Centered Leverage Value	DFFIT	DFBETA Intercept	DFBETA pubs
1	-.133839	.28515	.21328	.04074	-.495.72692	-.509.65184	1.39249
2	-.87895	.22370	.08530	.03196	-.305.09716	-.321.12768	.80153
3	-.41950	.16969	.01814	.02424	-.137.20167	-.147.10661	.33016
4	.03995	.12314	.00015	.01759	12.38769	13.45081	-.02658
5	.49940	.08403	.02294	.01200	147.81622	161.44976	-.27267
6	.95885	.05237	.08092	.00748	273.00807	297.67748	-.41116
7	1.41830	.02817	.17107	.00402	391.72124	422.81664	-.44422
8	-.27966	6.03375	227.14286	.86196	-39478.58473	3351.95531	-85.66108
Total N	8	8	8	8	8	8	8

a. Limited to first 100 cases.

As always when you see a statistical oddity you should ask what was happening in the real world. The last data point represents the City of London, a tiny area of only 1 square mile in the centre of London where very few people lived but where thousands of commuters (even then) came to work and had lunch in the pubs. Hence the pubs didn’t rely on the resident population for their business and the

residents didn’t consume all of their beer! Therefore, there was a massive number of pubs.

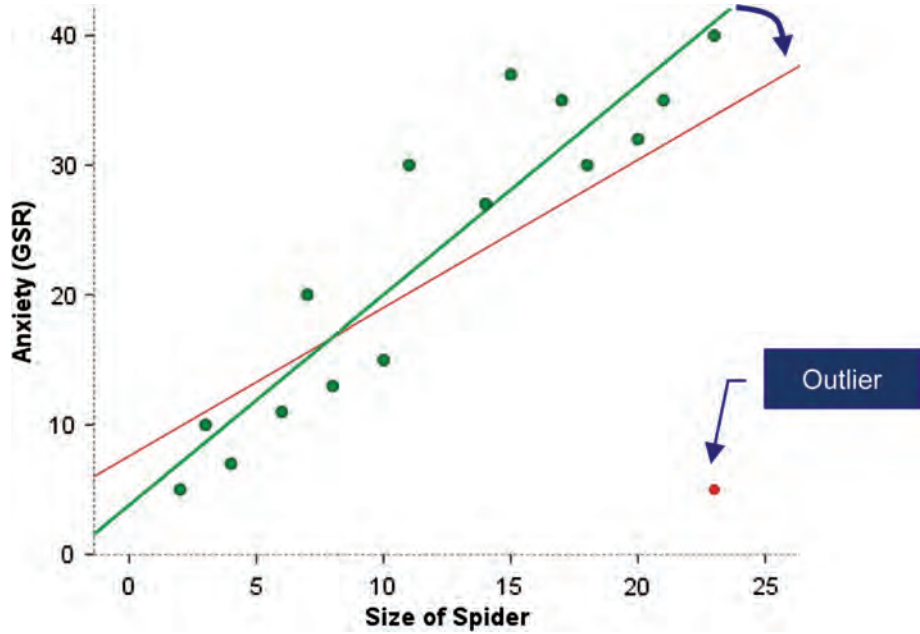
This illustrates that a case exerting a massive influence can produce a small residual – so look at both! (I’m very grateful to David Hitchin for this example, and he in turn got it from Dr Richard Roberts.)

7.6.1.1. Outliers and residuals ②

An outlier is a case that differs substantially from the main trend of the data (see Jane Superbrain Box 4.1). Figure 7.8 shows an example of such a case in regression. Outliers can cause your model to be biased because they affect the values of the estimated regression coefficients. For example, Figure 7.8 uses the same data as Figure 7.3 except that the score of one participant has been changed to be an outlier (in this case a person who was very calm in the presence of a very big spider). The change in this one point has had a dramatic effect on the regression model chosen to fit the data. With the outlier present, the regression model changes: its gradient is reduced (the line becomes flatter) and the intercept increases (the new line will cross the Y-axis at a higher point). It should be clear from this diagram that it is important to try to detect outliers to see whether the model is biased in this way.

How do you think that you might detect an outlier? Well, we know that an outlier, by its nature, is very different from all of the other scores. This being true, do you think that the model will predict that person’s score very accurately? The answer is *no*: looking at Figure 7.8 it is evident that even though the outlier has biased the model, the model still predicts that one value very badly (the regression line is long way from the outlier). Therefore, if we were to work out the differences between the data values that were

FIGURE 7.8
Graph demonstrating the effect of an outlier. The green line represents the original regression line for these data (see Figure 7.3), whereas the red line represents the regression line when an outlier is present



collected, and the values predicted by the model, we could detect an outlier by looking for large differences. This process is the same as looking for cases that the model predicts inaccurately. The differences between the values of the outcome predicted by the model and the values of the outcome observed in the sample are known as *residuals*. These residuals represent the error present in the model. If a model fits the sample data well then all residuals will be small (if the model was a perfect fit of the sample data – all data points fall on the regression line – then all residuals would be zero). If a model is a poor fit of the sample data then the residuals will be large. Also, if any cases stand out as having a large residual, then they could be outliers.

The *normal* or **unstandardized residuals** described above are measured in the same units as the outcome variable and so are difficult to interpret across different models. What we can do is to look for residuals that stand out as being particularly large. However, we cannot define a universal cut-off point for what constitutes a large residual. To overcome this problem, we use **standardized residuals**, which are the residuals divided by an estimate of their standard deviation. We came across standardization in section 6.3.2 as a means of converting variables into a standard unit of measurement (the standard deviation); we also came across *z*-scores (see section 1.7.4) in which variables are converted into standard deviation units (i.e. they're converted into scores that are distributed around a mean of 0 with a standard deviation of 1). By converting residuals into *z*-scores (standardized residuals) we can compare residuals from different models and use what we know about the properties of *z*-scores to devise universal guidelines for what constitutes an acceptable (or unacceptable) value. For example, we know from Chapter 1 that in a normally distributed sample, 95% of *z*-scores should lie between -1.96 and $+1.96$, 99% should lie between -2.58 and $+2.58$, and 99.9% (i.e. nearly all of them) should lie between -3.29 and $+3.29$. Some general rules for standardized residuals are derived from these facts: (1) standardized residuals with an absolute value greater than 3.29 (we can use 3 as an approximation) are cause for concern because in an average sample case a value this high is unlikely to happen by chance; (2) if more than 1% of our sample cases have standardized residuals with an absolute value greater than 2.58 (we usually just say 2.5) there is evidence that the level of error within our model is unacceptable (the model is a fairly poor fit of the sample data); and (3) if more than 5% of cases have standardized residuals with an absolute value greater than 1.96 (we can use 2 for convenience) then there is also evidence that the model is a poor representation of the actual data.

A third form of residual is the **Studentized residual**, which is the unstandardized residual divided by an estimate of its standard deviation that varies point by point. These residuals have the same properties as the standardized residuals but usually provide a more precise estimate of the error variance of a specific case.

7.6.1.2. Influential cases ③

As well as testing for outliers by looking at the error in the model, it is also possible to look at whether certain cases exert undue influence over the parameters of the model. So, if we were to delete a certain case, would we obtain different regression coefficients? This type of analysis can help to determine whether the regression model is stable across the sample, or whether it is biased by a few influential cases. Again, this process will unveil outliers.

There are several residual statistics that can be used to assess the influence of a particular case. One statistic is the **adjusted predicted value** for a case when that case is excluded from the analysis. In effect, the computer calculates a new model without a particular case and then uses this new model to predict the value of the outcome variable for the case that was excluded. If a case does not exert a large influence over the model then we would expect the adjusted predicted value to be very similar to the predicted value when the case is included. Put simply, if the model is stable then the predicted value of a case should be the same regardless of whether or not that case was used to calculate the model. The difference between the adjusted predicted value and the original predicted value is known as **DFFit** (see below). We can also look at the residual based on the adjusted predicted value: that is, the difference between the adjusted predicted value and the original observed value. This is the **deleted residual**. The deleted residual can be divided by the standard deviation to give a standardized value known as the **Studentized deleted residual**. This residual can be compared across different regression analyses because it is measured in standard units.



The deleted residuals are very useful to assess the influence of a case on the ability of the model to predict that case. However, they do not provide any information about how a case influences the model as a whole (i.e. the impact that a case has on the model’s ability to predict *all* cases). One statistic that does consider the effect of a single case on the model as a whole is **Cook’s distance**. Cook’s distance is a measure of the overall influence of a case on the model and Cook and Weisberg (1982) have suggested that values greater than 1 may be cause for concern.

A second measure of influence is **leverage** (sometimes called **hat values**), which gauges the influence of the observed value of the outcome variable over the predicted values. The average leverage value is defined as $(k + 1)/n$ in which k is the number of predictors in the model and n is the number of participants.⁶ Leverage values can lie between 0 (indicating that the case has no influence whatsoever) and 1 (indicating that the case has complete influence over prediction). If no cases exert undue influence over the model then we would expect all of the leverage values to be close to the average value $((k + 1)/n)$. Hoaglin and Welsch (1978) recommend investigating cases with values greater than twice the average $(2(k + 1)/n)$ and Stevens (2002) recommends using three times the average $(3(k + 1)/n)$ as a cut-off point for identifying cases having undue influence. We will see how to use these cut-off points later. However, cases with large leverage values will not necessarily have a large influence on the regression coefficients because they are measured on the outcome variables rather than the predictors.

Related to the leverage values are the **Mahalanobis distances** (Figure 7.9), which measure the distance of cases from the mean(s) of the predictor variable(s). You need to look for the cases with the highest values. It is not easy to establish a cut-off point at which to worry,

⁶ You may come across the average leverage denoted as p/n in which p is the number of parameters being estimated. In multiple regression, we estimate parameters for each predictor and also for a constant and so p is equivalent to the number of predictors plus one, $(k + 1)$.

FIGURE 7.9
Prasanta
Chandra
Mahalanobis
staring into his
distances



although Barnett and Lewis (1978) have produced a table of critical values dependent on the number of predictors and the sample size. From their work it is clear that even with large samples ($N = 500$) and 5 predictors, values above 25 are cause for concern. In smaller samples ($N = 100$) and with fewer predictors (namely 3) values greater than 15 are problematic, and in very small samples ($N = 30$) with only 2 predictors values greater than 11 should be examined. However, for more specific advice, refer to Barnett and Lewis's table.

It is possible to run the regression analysis with a case included and then rerun the analysis with that same case excluded. If we did this, undoubtedly there would be some difference between the b coefficients in the two regression equations. This difference would tell us how much influence a particular case has on the parameters of the regression model. To take a hypothetical example, imagine two variables that had a perfect negative relationship except for a single case (case 30). If a regression analysis was done on the 29 cases that were perfectly linearly related then we would get a model in which the predictor variable X perfectly predicts the outcome variable Y , and there are no errors. If we then ran the analysis but this time include the case that didn't conform (case 30), then the resulting model has different parameters. Some data are stored in the file `dfbeta.sav` which illustrate such a situation. Try running a simple regression first with all the cases included and then with case 30 deleted. The results are summarized in Table 7.1, which shows: (1) the parameters for the regression model when the extreme case is included or excluded; (2) the resulting regression equations; and (3) the value of Y predicted from participant 30's score on the X variable (which is obtained by replacing the X in the regression equation with participant 30's score for X , which was 1).

When case 30 is excluded, these data have a perfect negative relationship; hence the coefficient for the predictor (b_1) is -1 (remember that in simple regression this term is the same as Pearson's correlation coefficient), and the coefficient for the constant (the intercept, b_0) is 31. However, when case 30 is included, both parameters are reduced⁷ and the difference between the parameters is also displayed. The difference between a parameter estimated using all cases and estimated when one case is excluded is known as the **DFBeta** in SPSS. DFBeta is calculated for every case and for each of the parameters in the model. So, in our hypothetical example, the DFBeta for the constant is -2 , and the DFBeta for the predictor variable is 0.1. By looking at the values of DFBeta, it is possible to identify cases that have a large influence on the parameters of the regression model. Again, the units of measurement used will affect these values and so SPSS produces a **standardized DFBeta**. These standardized



⁷ The value of b_1 is reduced because the data no longer have a perfect linear relationship and so there is now variance that the model cannot explain.

TABLE 7.1 The difference in the parameters of the regression model when one case is excluded

Parameter (<i>b</i>)	Case 30 Included	Case 30 Excluded	Difference
Constant (intercept)	29.00	31.00	−2.00
Predictor (gradient)	−0.90	−1.00	0.10
Model (regression line):	$Y = (-0.9) X + 29$	$Y = (-1) X + 31$	
Predicted <i>Y</i>	28.10	30.00	−1.90

values are easier to use because universal cut-off points can be applied. In this case absolute values above 1 indicate cases that substantially influence the model parameters (although Stevens, 2002, suggests looking at cases with absolute values greater than 2).

A related statistic is the **DFFit**, which is the difference between the predicted value for a case when the model is calculated including that case and when the model is calculated excluding that case: in this example the value is −1.90 (see Table 7.1). If a case is not influential then its DFFit should be zero – hence, we expect non-influential cases to have small DFFit values. However, we have the problem that this statistic depends on the units of measurement of the outcome and so a DFFit of 0.5 will be very small if the outcome ranges from 1 to 100, but very large if the outcome varies from 0 to 1. Therefore, SPSS also produces standardized versions of the DFFit values (**Standardized DFFit**). A final measure is that of the **covariance ratio (CVR)**, which is a measure of whether a case influences the variance of the regression parameters. A description of the computation of this statistic would leave most readers dazed and confused, so suffice to say that when this ratio is close to 1 the case is having very little influence on the variances of the model parameters. Belsey, Kuh and Welsch (1980) recommend the following:

- If $CVR_i > 1 + [3(k + 1)/n]$ then deleting the *i*th case will damage the precision of some of the model’s parameters.
- If $CVR_i < 1 - [3(k + 1)/n]$ then deleting the *i*th case will improve the precision of some of the model’s parameters.

In both equations, *k* is the number of predictors, CVR_i is the covariance ratio for the *i*th participant, and *n* is the sample size.



7.6.1.3. A final comment on diagnostic statistics ②

There are a lot of diagnostic statistics that should be examined after a regression analysis, and it is difficult to summarize this wealth of material into a concise conclusion. However, one thing I would like to stress is a point made by Belsey et al. (1980) who noted the dangers inherent in these procedures. The point is that diagnostics are tools that enable you to see how good or bad your model is in terms of fitting the sampled data. They are a way of assessing your model. They are *not*, however, a way of justifying the removal of data points to effect some desirable change in the regression parameters (e.g. deleting a case that changes a non-significant *b*-value into a significant one). Stevens (2002), as ever, offers excellent advice:

If a point is a significant outlier on *Y*, but its Cook’s distance is < 1, there is no real need to delete that point since it does not have a large effect on the regression analysis. However, one should still be interested in studying such points further to understand why they did not fit the model. (p. 135)

7.6.2. Assessing the regression model II: generalization ②

When a regression analysis is done, an equation can be produced that is correct for the sample of observed values. However, in the social sciences we are usually interested in generalizing our findings outside of the sample. So, although it can be useful to draw conclusions about a particular sample of people, it is usually more interesting if we can then assume that our conclusions are true for a wider population. For a regression model to generalize we must be sure that underlying assumptions have been met, and to test whether the model does generalize we can look at cross-validating it.

7.6.2.1. Checking assumptions ②

To draw conclusions about a population based on a regression analysis done on a sample, several assumptions must be true (see Berry, 1993):

- **Variable types:** All predictor variables must be quantitative or categorical (with two categories), and the outcome variable must be quantitative, continuous and unbounded. By quantitative I mean that they should be measured at the interval level and by unbounded I mean that there should be no constraints on the variability of the outcome. If the outcome is a measure ranging from 1 to 10 yet the data collected vary between 3 and 7, then these data are constrained.
- **Non-zero variance:** The predictors should have some variation in value (i.e. they do not have variances of 0).
- **No perfect multicollinearity:** There should be no perfect linear relationship between two or more of the predictors. So, the predictor variables should not correlate too highly (see section 7.6.2.4).
- **Predictors are uncorrelated with ‘external variables’:** *External variables* are variables that haven’t been included in the regression model which influence the outcome variable.⁸ These variables can be thought of as similar to the ‘third variable’ that was discussed with reference to correlation. This assumption means that there should be no external variables that correlate with any of the variables included in the regression model. Obviously, if external variables do correlate with the predictors, then the conclusions we draw from the model become unreliable (because other variables exist that can predict the outcome just as well).
- **Homoscedasticity:** At each level of the predictor variable(s), the variance of the residual terms should be constant. This just means that the residuals at each level of the predictor(s) should have the same variance (**homoscedasticity**); when the variances are very unequal there is said to be **heteroscedasticity** (see section 5.6 as well).
- **Independent errors:** For any two observations the residual terms should be uncorrelated (or independent). This eventuality is sometimes described as a lack of **autocorrelation**. This assumption can be tested with the **Durbin-Watson test**, which tests for serial correlations between errors. Specifically, it tests whether adjacent residuals are correlated. The test statistic can vary between 0 and 4 with a value of 2 meaning that the residuals are uncorrelated. A value greater than 2 indicates a negative correlation between

⁸ Some authors choose to refer to these external variables as part of an error term that includes any random factor in the way in which the outcome varies. However, to avoid confusion with the residual terms in the regression equations I have chosen the label ‘external variables’. Although this term implicitly washes over any random factors, I acknowledge their presence here!

adjacent residuals, whereas a value below 2 indicates a positive correlation. The size of the Durbin–Watson statistic depends upon the number of predictors in the model and the number of observations. For accuracy, you should look up the exact acceptable values in Durbin and Watson’s (1951) original paper. As a very conservative rule of thumb, values less than 1 or greater than 3 are definitely cause for concern; however, values closer to 2 may still be problematic depending on your sample and model.

- **Normally distributed errors:** It is assumed that the residuals in the model are random, normally distributed variables with a mean of 0. This assumption simply means that the differences between the model and the observed data are most frequently zero or very close to zero, and that differences much greater than zero happen only occasionally. Some people confuse this assumption with the idea that predictors have to be normally distributed. In fact, predictors do not need to be normally distributed (see section 7.11).
- **Independence:** It is assumed that all of the values of the outcome variable are independent (in other words, each value of the outcome variable comes from a separate entity).
- **Linearity:** The mean values of the outcome variable for each increment of the predictor(s) lie along a straight line. In plain English this means that it is assumed that the relationship we are modelling is a linear one. If we model a non-linear relationship using a linear model then this obviously limits the generalizability of the findings.

This list of assumptions probably seems pretty daunting but as we saw in Chapter 5, assumptions are important. When the assumptions of regression are met, the model that we get for a sample can be accurately applied to the population of interest (the coefficients and parameters of the regression equation are said to be *unbiased*). Some people assume that this means that when the assumptions are met the regression model from a sample is always identical to the model that would have been obtained had we been able to test the entire population. Unfortunately, this belief isn’t true. What an unbiased model does tell us is that *on average* the regression model from the sample is the same as the population model. However, you should be clear that even when the assumptions are met, it is possible that a model obtained from a sample may not be the same as the population model – but the likelihood of them being the same is increased.

7.6.2.2. Cross-validation of the model ③

Even if we can’t be confident that the model derived from our sample accurately represents the entire population, there are ways in which we can assess how well our model can predict the outcome in a different sample. Assessing the accuracy of a model across different samples is known as cross-validation. If a model can be generalized, then it must be capable of accurately predicting the same outcome variable from the same set of predictors in a different group of people. If the model is applied to a different sample and there is a severe drop in its predictive power, then the model clearly does *not* generalize. As a first rule of thumb, we should aim to collect enough data to obtain a reliable regression model (see the next section). Once we have a regression model there are two main methods of cross-validation:

- **Adjusted R^2 :** In SPSS, not only are the values of R and R^2 calculated, but also an **adjusted R^2** . This adjusted value indicates the loss of predictive power or **shrinkage**. Whereas R^2 tells us how much of the variance in Y is accounted for by the regression model from our sample, the adjusted value tells us how much variance in Y would be accounted for if the model had been derived from the population from which the sample was taken. SPSS derives the adjusted R^2 using Wherry’s equation. However, this equation has been criticized because it tells us nothing about how well the regression model would predict an entirely different set of data (how well can the model predict scores of a different sample of data from the same population?). One version

of R^2 that does tell us how well the model cross-validates uses Stein’s formula which is shown in equation (7.11) (see Stevens, 2002):

$$\text{adjusted } R^2 = 1 - \left[\left(\frac{n-1}{n-k-1} \right) \left(\frac{n-2}{n-k-2} \right) \left(\frac{n+1}{n} \right) \right] (1 - R^2) \tag{7.11}$$

In Stein’s equation, R^2 is the unadjusted value, n is the number of participants and k is the number of predictors in the model. For the more mathematically minded of you, it is worth using this equation to cross-validate a regression model.

- **Data splitting:** This approach involves randomly splitting your data set, computing a regression equation on both halves of the data and then comparing the resulting models. When using stepwise methods, cross-validation is a good idea; you should run the stepwise regression on a random selection of about 80% of your cases. Then force this model on the remaining 20% of the data. By comparing the values of R^2 and b in the two samples you can tell how well the original model generalizes (see Tabachnick & Fidell, 2007, for more detail).

7.6.2.3. Sample size in regression ③



In the previous section I said that it’s important to collect enough data to obtain a reliable regression model. Well, how much is enough? You’ll find a lot of rules of thumb floating about, the two most common being that you should have 10 cases of data for each predictor in the model, or 15 cases of data per predictor. So, with five predictors, you’d need 50 or 75 cases respectively (depending on the rule you use). These rules are very pervasive (even I used the 15 cases per predictor rule in the first edition of this book) but they over simplify the issue considerably. In fact, the sample size required will depend on the size of effect that we’re trying to detect (i.e. how strong the relationship is that we’re trying to measure) and how much power we want to detect these effects.

The simplest rule of thumb is that the bigger the sample size, the better! The reason is that the estimate of R that we get from regression is dependent on the number of predictors, k , and the sample size, N . In fact the expected R for random data is $k/(N - 1)$ and so with small sample sizes random data can appear to show a strong effect: for example, with six predictors and 21 cases of data, $R = 6/(21 - 1) = .3$ (a medium effect size by Cohen’s criteria described in section 6.3.2). Obviously for random data we’d want the expected R to be 0 (no effect) and for this to be true we need large samples (to take the previous example, if we had 100 cases not 21, then the expected R would be a more acceptable .06).

It’s all very well knowing that larger is better, but researchers usually need some more concrete guidelines (much as we’d all love to collect 1000 cases of data it isn’t always practical!). Green (1991) makes two rules of thumb for the *minimum* acceptable sample size, the first based on whether you want to test the overall fit of your regression model (i.e. test the R^2), and the second based on whether you want to test the individual predictors within the model (i.e. test b -values of the model). If you want to test the model overall, then he recommends a minimum sample size of $50 + 8k$, where k is the number of predictors. So, with five predictors, you’d need a sample size of $50 + 40 = 90$. If you want to test the individual predictors then he suggests a minimum sample size of $104 + k$, so again taking the example of 5 predictors you’d need a sample size of $104 + 5 = 109$. Of course, in most cases we’re interested both in the overall fit and in the contribution of individual predictors, and in this situation Green recommends you calculate both of the minimum sample sizes I’ve just described, and use the one that has the largest value (so, in the five-predictor example, we’d use 109 because it is bigger than 90).

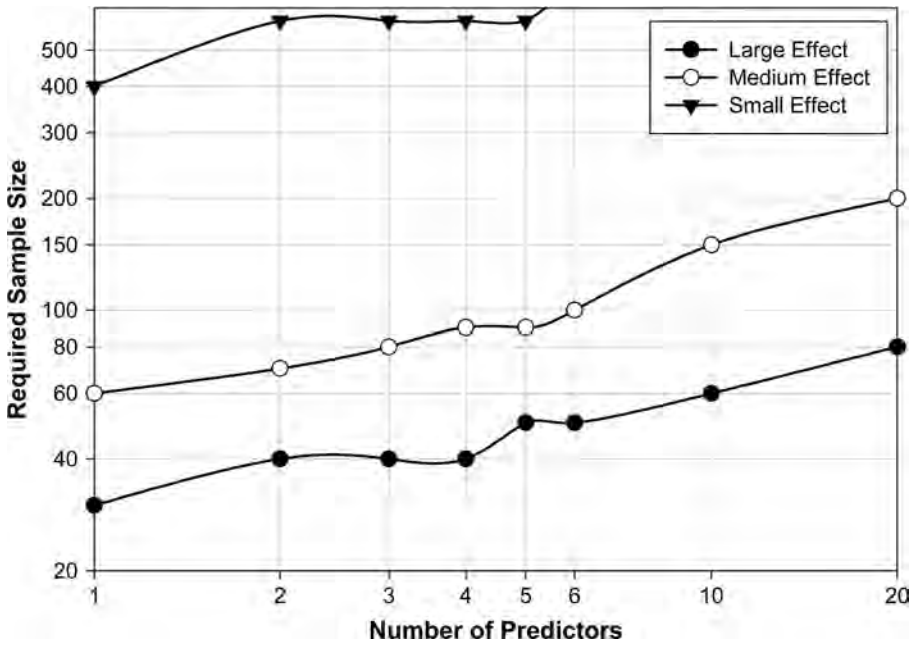


FIGURE 7.10
Graph to show the sample size required in regression depending on the number of predictors and the size of expected effect

Now, these guidelines are all right as a rough and ready guide, but they still oversimplify the problem. As I’ve mentioned, the sample size required actually depends on the size of the effect (i.e. how well our predictors predict the outcome) and how much statistical power we want to detect these effects. Miles and Shevlin (2001) produce some extremely useful graphs that illustrate the sample sizes needed to achieve different levels of power, for different effect sizes, as the number of predictors vary. For precise estimates of the sample size you should be using, I recommend using these graphs. I’ve summarized some of the general findings in Figure 7.10. This diagram shows the sample size required to achieve a high level of power (I’ve taken Cohen’s, 1988, benchmark of .8) depending on the number of predictors and the size of expected effect. To summarize the graph very broadly: (1) if you expect to find a large effect then a sample size of 80 will always suffice (with up to 20 predictors) and if there are fewer predictors then you can afford to have a smaller sample; (2) if you’re expecting a medium effect, then a sample size of 200 will always suffice (up to 20 predictors), you should always have a sample size above 60, and with six or fewer predictors you’ll be fine with a sample of 100; and (3) if you’re expecting a small effect size then just don’t bother unless you have the time and resources to collect at least 600 cases of data (and many more if you have six or more predictors).

7.6.2.4. Multicollinearity ②

Multicollinearity exists when there is a strong correlation between two or more predictors in a regression model. Multicollinearity poses a problem only for multiple regression because (without wishing to state the obvious) simple regression requires only one predictor. **Perfect collinearity** exists when at least one predictor is a perfect linear combination of the others (the simplest example being two predictors that are perfectly correlated – they have a correlation coefficient of 1). If there is perfect collinearity between predictors it becomes impossible to obtain unique estimates of the regression coefficients because there are an infinite number of combinations of coefficients that would work equally well. Put simply, if we have two predictors that are perfectly correlated, then the values of *b* for each variable are interchangeable. The good news is that perfect collinearity is rare in

real-life data. The bad news is that less than perfect collinearity is virtually unavoidable. Low levels of collinearity pose little threat to the models generated by SPSS, but as collinearity increases there are three problems that arise:

- **Untrustworthy b s:** As collinearity increases so do the standard errors of the b coefficients. If you think back to what the standard error represents, then big standard errors for b coefficients means that these b s are more variable across samples. Therefore, it means that the b coefficient in our sample is less likely to represent the population. Crudely put, multicollinearity means that the b -values are less trustworthy. Don't lend them money and don't let them go for dinner with your boy- or girlfriend. Of course if the b s are variable from sample to sample then the resulting predictor equations will be unstable across samples too.
- **It limits the size of R :** Remember that R is a measure of the multiple correlation between the predictors and the outcome and that R^2 indicates the variance in the outcome for which the predictors account. Imagine a situation in which a single variable predicts the outcome variable fairly successfully (e.g. $R = .80$) and a second predictor variable is then added to the model. This second variable might account for a lot of the variance in the outcome (which is why it is included in the model), but the variance it accounts for is the same variance accounted for by the first variable. In other words, once the variance accounted for by the first predictor has been removed, the second predictor accounts for very little of the remaining variance (the second variable accounts for very little *unique variance*). Hence, the overall variance in the outcome accounted for by the two predictors is little more than when only one predictor is used (so R might increase from .80 to .82). This idea is connected to the notion of partial correlation that was explained in Chapter 6. If, however, the two predictors are completely uncorrelated, then the second predictor is likely to account for different variance in the outcome to that accounted for by the first predictor. So, although in itself the second predictor might account for only a little of the variance in the outcome, the variance it does account for is different to that of the other predictor (and so when both predictors are included, R is substantially larger, say .95). Therefore, having uncorrelated predictors is beneficial.
- **Importance of predictors:** Multicollinearity between predictors makes it difficult to assess the individual importance of a predictor. If the predictors are highly correlated, and each accounts for similar variance in the outcome, then how can we know which of the two variables is important? Quite simply we can't tell which variable is important – the model could include either one, interchangeably.

One way of identifying multicollinearity is to scan a correlation matrix of all of the predictor variables and see if any correlate very highly (by very highly I mean correlations of above .80 or .90). This is a good 'ball park' method but misses more subtle forms of multicollinearity. Luckily, SPSS produces various collinearity diagnostics, one of which is the **variance inflation factor (VIF)**. The VIF indicates whether a predictor has a strong linear relationship with the other predictor(s). Although there are no hard and fast rules about what value of the VIF should cause concern, Myers (1990) suggests that a value of 10 is a good value at which to worry. What's more, if the average VIF is greater than 1, then multicollinearity may be biasing the regression model (Bowerman & O'Connell, 1990). Related to the VIF is the **tolerance** statistic, which is its reciprocal ($1/\text{VIF}$). As such, values below 0.1 indicate serious problems although Menard (1995) suggests that values below 0.2 are worthy of concern.

Other measures that are useful in discovering whether predictors are dependent are the *eigenvalues of the scaled, uncentred cross-products matrix*, the *condition indexes* and the *variance proportions*. These statistics are extremely complex and will be covered as part of the interpretation of SPSS output (see section 7.8.5). If none of this has made any sense then have a look at Hucheson and Sofroniou (1999: 78–85) who give a really clear explanation of multicollinearity.

7.7. How to do multiple regression using SPSS ②

7.7.1. Some things to think about before the analysis ②

A good strategy to adopt with regression is to measure predictor variables for which there are sound theoretical reasons for expecting them to predict the outcome. Run a regression analysis in which all predictors are entered into the model and examine the output to see which predictors contribute substantially to the model’s ability to predict the outcome. Once you have established which variables are important, rerun the analysis including only the important predictors and use the resulting parameter estimates to define your regression model. If the initial analysis reveals that there are two or more significant predictors then you could consider running a forward step-wise analysis (rather than forced entry) to find out the individual contribution of each predictor.

I have spent a lot of time explaining the theory behind regression and some of the diagnostic tools necessary to gauge the accuracy of a regression model. It is important to remember that SPSS may appear to be very clever, but in fact it is not. Admittedly, it can do lots of complex calculations in a matter of seconds, but what it can’t do is control the quality of the model that is generated – to do this requires a human brain (and preferably a trained one). SPSS will happily generate output based on any garbage you decide to feed into the data editor and SPSS will not judge the results or give any indication of whether the model can be generalized or if it is valid. However, SPSS provides the statistics necessary to judge these things, and at this point our brains must take over the job – which is slightly worrying (especially if your brain is as small as mine)!

7.7.2. Main options ②

Imagine that the record company executive was now interested in extending the model of record sales to incorporate other variables. He decides to measure two new variables: (1) the number of times the record is played on Radio 1 during the week prior to release (**airplay**); and (2) the attractiveness of the band (**attract**). Before a record is released, the executive notes the amount spent on advertising, the number of times the record is played on radio the week before release, and the attractiveness of the band. He does this for 200 different records (each made by a different band). Attractiveness was measured by asking a random sample of the target audience to rate the attractiveness of each band on a scale from 0 (hideous potato-heads) to 10 (gorgeous sex objects). The mode attractiveness given by the sample was used in the regression (because he was interested in what the majority of people thought, rather than the average of people’s opinions).

These data are in the file **Record2.sav** and you should note that each variable has its own column (the same layout as for correlation) and each row represents a different record. So, the first record had £10,260 spent advertising it, sold 330,000 copies, received 43 plays on Radio 1 the week before release, and was made by a band that the majority of people rated as gorgeous sex objects (Figure 7.11).

The executive has past research indicating that advertising budget is a significant predictor of record sales, and so he should include this variable in the model first. His new variables (**airplay** and **attract**) should, therefore, be entered into the model *after* advertising budget. This method is hierarchical (the researcher decides in which order to enter variables into the model based on past research). To do a hierarchical regression in SPSS we have to enter the variables in blocks (each block representing one step in the hierarchy). To get to the main regression dialog box (Figure 7.12) select **Analyze Regression**

R **Linear...**; this dialog box is the same as when we encountered it for simple regression.

The main dialog box is fairly self-explanatory in that there is a space to specify the dependent variable (outcome) and a space to place one or more independent variables



FIGURE 7.11
Data layout
for multiple
regression

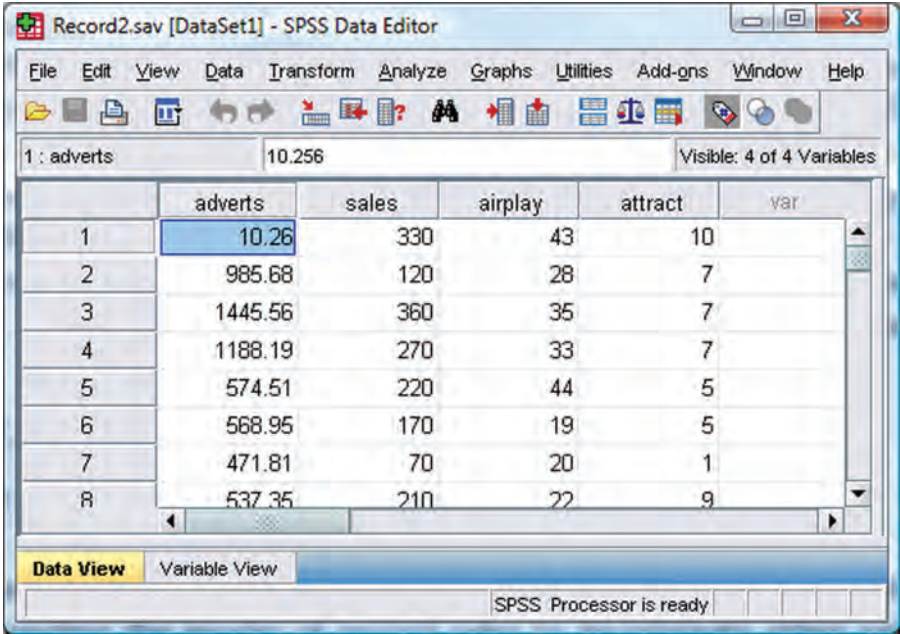
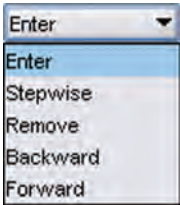
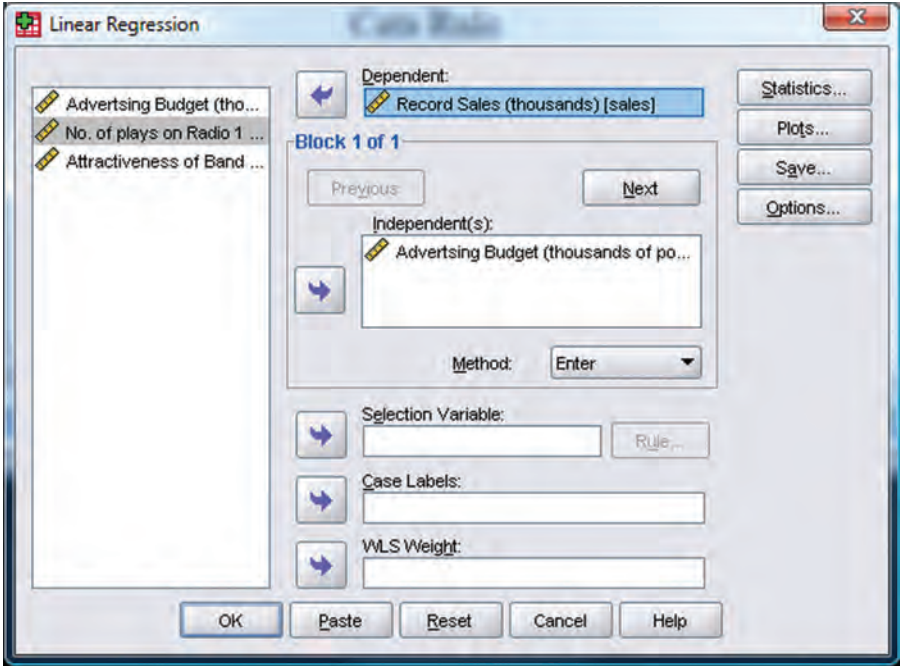


FIGURE 7.12
Main dialog
box for block 1
of the multiple
regression



(predictor variables). As usual, the variables in the data editor are listed on the left-hand side of the box. Highlight the outcome variable (record sales) in this list by clicking on it and then drag it to the box labelled *Dependent* or click on . We also need to specify the predictor variable for the first block. We decided that advertising budget should be entered into the model first (because past research indicates that it is an important predictor), so highlight this variable in the list and drag it to the box labelled *Independent(s)* or click on . Underneath the *Independent(s)* box, there

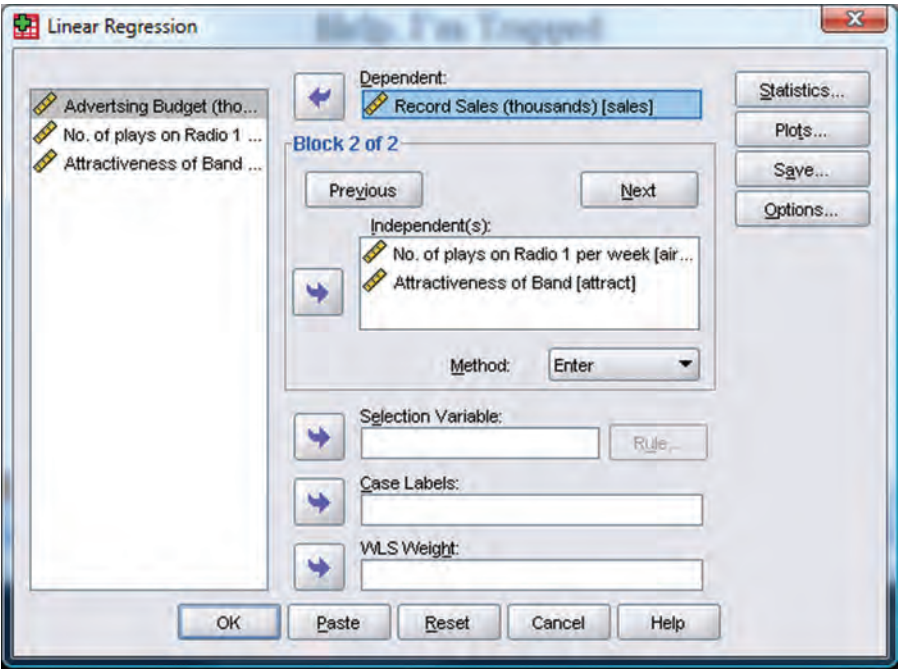


FIGURE 7.13
Main dialog
box for block 2
of the multiple
regression

is a drop-down menu for specifying the *Method* of regression (see section 7.5.3). You can select a different method of variable entry for each block by clicking on *Enter* next to where it says *Method*. The default option is forced entry, and this is the option we want, but if you were carrying out more exploratory work, you might decide to use one of the stepwise methods (forward, backward, stepwise or remove).

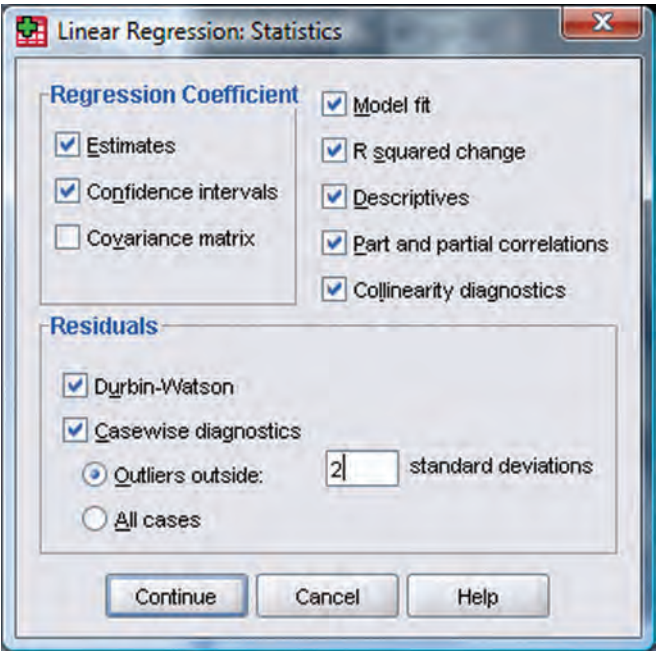
Having specified the first block in the hierarchy, we need to move on to the second. To tell the computer that you want to specify a new block of predictors you must click on *Next*. This process clears the *Independent(s)* box so that you can enter the new predictors (you should also note that above this box it now reads *Block 2 of 2* indicating that you are in the second block of the two that you have so far specified). We decided that the second block would contain both of the new predictors and so you should click on **airplay** and **attract** (while holding down *Ctrl*) in the variables list and drag them to the *Independent(s)* box or click on . The dialog box should now look like Figure 7.13. To move between blocks use the *Previous* and *Next* buttons (so, for example, to move back to block 1, click on *Previous*).

It is possible to select different methods of variable entry for different blocks in a hierarchy. So although we specified forced entry for the first block, we could now specify a stepwise method for the second. Given that we have no previous research regarding the effects of attractiveness and airplay on record sales, we might be justified in requesting a stepwise method for this block. However, because of the problems with stepwise methods, I am going to stick with forced entry for both blocks in this example.

7.7.3. Statistics ②

In the main regression dialog box click on *Statistics...* to open a dialog box for selecting various important options relating to the model (Figure 7.14). Most of these options relate to the parameters of the model; however, there are procedures available for checking the assumptions of no multicollinearity (collinearity diagnostics) and serial independence of errors

FIGURE 7.14
Statistics
dialog box for
regression
analysis



(Durbin–Watson). When you have selected the statistics you require (I recommend all but the covariance matrix as a general rule) click on **Continue** to return to the main dialog box:

- *Estimates*: This option is selected by default because it gives us the estimated coefficients of the regression model (i.e. the estimated b -values). Test statistics and their significance are produced for each regression coefficient: a t -test is used to see whether each b differs significantly from zero (see section 7.2.4).
- *Confidence intervals*: This option produces confidence intervals for each of the unstandardized regression coefficients. Confidence intervals can be a very useful tool in assessing the likely value of the regression coefficients in the population – I will describe their exact interpretation later.
- *Covariance matrix*: This option will display a matrix of the covariances, correlation coefficients and variances between the regression coefficients of each variable in the model. A variance–covariance matrix is produced with variances displayed along the diagonal and covariances displayed as off-diagonal elements. The correlations are produced in a separate matrix.
- *Model fit*: This option is vital and so is selected by default. It provides not only a statistical test of the model’s ability to predict the outcome variable (the F -test, see section 7.2.3), but also the value of R (or multiple R), the corresponding R^2 and the adjusted R^2 .
- *R squared change*: This option displays the change in R^2 resulting from the inclusion of a new predictor (or block of predictors). This measure is a useful way to assess the contribution of new predictors (or blocks) to explaining variance in the outcome.
- *Descriptives*: If selected, this option displays a table of the mean, standard deviation and number of observations of all of the variables included in the analysis. A correlation matrix is also displayed showing the correlation between all of the variables and the one-tailed probability for each correlation coefficient. This option is extremely useful because the correlation matrix can be used to assess whether predictors are interrelated (which can be used to establish whether there is multicollinearity).
- *Part and partial correlations*: This option produces the zero-order correlation (the Pearson correlation) between each predictor and the outcome variable. It also

produces the partial correlation between each predictor and the outcome, controlling for all other predictors in the model. Finally, it produces the part correlation (or semi-partial correlation) between each predictor and the outcome. This correlation represents the relationship between each predictor and the part of the outcome that is not explained by the other predictors in the model. As such, it measures the unique relationship between a predictor and the outcome (see section 6.6).

- *Collinearity diagnostics*: This option is for obtaining collinearity statistics such as the VIF, tolerance, eigenvalues of the scaled, uncentred cross-products matrix, condition indexes and variance proportions (see section 7.6.2.3).
- *Durbin-Watson*: This option produces the **Durbin-Watson test** statistic, which tests the assumption of independent errors. Unfortunately, SPSS does not provide the significance value of this test, so you must decide for yourself whether the value is different enough from 2 to be cause for concern (see section 7.6.2.1).
- *Casewise diagnostics*: This option, if selected, lists the observed value of the outcome, the predicted value of the outcome, the difference between these values (the residual) and this difference standardized. Furthermore, it will list these values either for all cases, or just for cases for which the standardized residual is greater than 3 (when the $\pm$ sign is ignored). This criterion value of 3 can be changed, and I recommend changing it to 2 for reasons that will become apparent. A summary table of residual statistics indicating the minimum, maximum, mean and standard deviation of both the values predicted by the model and the residuals (see section 7.7.5) is also produced.

7.7.4. Regression plots ②

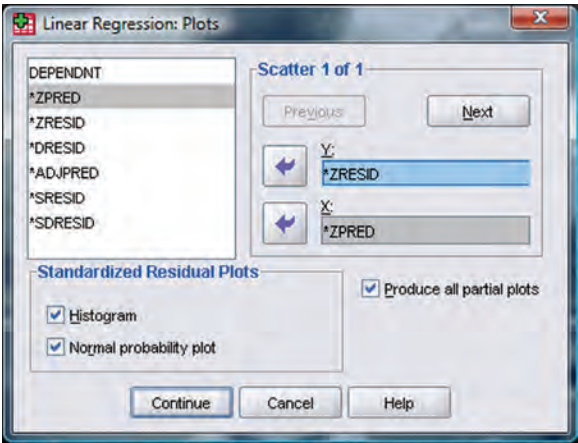
Once you are back in the main dialog box, click on  to activate the regression plots dialog box shown in Figure 7.15. This dialog box provides the means to specify several graphs, which can help to establish the validity of some regression assumptions. Most of these plots involve various *residual* values, which will be described in more detail in section 7.7.5.

On the left-hand side of the dialog box is a list of several variables.

- **DEPENDNT** (*the outcome variable*).
- ***ZPRED** (*the standardized predicted values* of the dependent variable based on the model). These values are standardized forms of the values predicted by the model.
- ***ZRESID** (*the standardized residuals, or errors*). These values are the standardized differences between the observed data and the values that the model predicts).
- ***DRESID** (*the deleted residuals*). See section 7.6.1.1 for details.
- ***ADJPRED** (*the adjusted predicted values*). See section 7.6.1.1 for details.
- ***SRESID** (*the Studentized residual*). See section 7.6.1.1 for details.
- ***SDRESID** (*the Studentized deleted residual*). This value is the deleted residual divided by its standard deviation.

The variables listed in this dialog box all come under the general heading of residuals, and were discussed in detail in section 7.6.1.1. For a basic analysis it is worth plotting ***ZRESID** (Y-axis) against ***ZPRED** (X-axis), because this plot is useful to determine whether the assumptions of random errors and homoscedasticity have been met. A plot of ***SRESID** (y-axis) against ***ZPRED** (x-axis) will show up any heteroscedasticity also. Although often these two plots are virtually identical, the latter is more sensitive on a case-by-case basis. To create these plots simply select a variable from the list, and transfer it to the space labelled

FIGURE 7.15
*Linear
Regression:
Plots dialog box*



either x or y (which refer to the axes) by clicking on . When you have selected two variables for the first plot (as is the case in Figure 7.15) you can specify a new plot by clicking on . This process clears the spaces in which variables are specified. If you click on and would like to return to the plot that you last specified, then simply click on . You can specify up to nine plots.

You can also select the tick-box labelled *Produce all partial plots* which will produce scatterplots of the residuals of the outcome variable and each of the predictors when both variables are regressed separately on the remaining predictors. Regardless of whether the previous sentence made any sense to you, these plots have several important characteristics that make them worth inspecting. First, the gradient of the regression line between the two residual variables is equivalent to the coefficient of the predictor in the regression equation. As such, any obvious outliers on a partial plot represent cases that might have undue influence on a predictor’s regression coefficient. Second, non-linear relationships between a predictor and the outcome variable are much more detectable using these plots. Finally, they are a useful way of detecting collinearity. For these reasons, I recommend requesting them.

There are several options for plots of the standardized residuals. First, you can select a histogram of the standardized residuals (this is extremely useful for checking the assumption of normality of errors). Second, you can ask for a normal probability plot, which also provides information about whether the residuals in the model are normally distributed. When you have selected the options you require, click on to take you back to the main regression dialog box.

7.7.5. Saving regression diagnostics ②

In section 7.6 we met two types of regression diagnostics: those that help us assess how well our model fits our sample and those that help us detect cases that have a large influence on the model generated. In SPSS we can choose to save these diagnostic variables in the data editor (so, SPSS will calculate them and then create new columns in the data editor in which the values are placed).

To save regression diagnostics you need to click on in the main regression dialog box. This process activates the *save new variables* dialog box (see Figure 7.16). Once this dialog box is active, it is a simple matter to tick the boxes next to the required statistics. Most of the available options were explained in section 7.6 and Figure 7.16 shows, what I consider to be, a fairly basic set of diagnostic statistics. Standardized (and Studentized) versions of these diagnostics are generally easier to interpret and so I suggest selecting them in

preference to the unstandardized versions. Once the regression has been run, SPSS creates a column in your data editor for each statistic requested and it has a standard set of variable names to describe each one. After the name, there will be a number that refers to the analysis that has been run. So, for the first regression run on a data set the variable names will be followed by a 1, if you carry out a second regression it will create a new set of variables with names followed by a 2, and so on. The names of the variables that will be created are below. When you have selected the diagnostics you require (by clicking in the appropriate boxes), click on **Continue** to return to the main regression dialog box:

- **pre_1**: unstandardized predicted value.
- **zpr_1**: standardized predicted value.
- **adj_1**: adjusted predicted value.
- **sep_1**: standard error of predicted value.
- **res_1**: unstandardized residual.
- **zre_1**: standardized residual.
- **sre_1**: Studentized residual.
- **dre_1**: deleted residual.
- **sdr_1**: Studentized deleted residual.
- **mah_1**: Mahalanobis distance.
- **coo_1**: Cook's distance.
- **lev_1**: centred leverage value.
- **sdb0_1**: standardized DFBETA (intercept).
- **sdb1_1**: standardized DFBETA (predictor 1).
- **sdb2_1**: standardized DFBETA (predictor 2).
- **sdf_1**: standardized DFFIT.
- **cov_1**: covariance ratio.

7.7.6. Further options ②

As a final step in the analysis, you can click on **Options** to take you to the options dialog box (Figure 7.17). The first set of options allows you to change the criteria used for entering variables in a stepwise regression. If you insist on doing stepwise regression, then it's probably best that you leave the default criterion of .05 probability for entry alone. However, you can make this criterion more stringent (.01). There is also the option to build a model that doesn't include a constant (i.e. has no Y intercept). This option should also be left alone (almost always). Finally, you can select a method for dealing with missing data points (see SPSS Tip 6.1). By default, SPSS excludes cases listwise, which in regression means that if a person has a missing value for any variable, then they are excluded from the whole analysis. So, for example, if our record company executive didn't have an attractiveness score for one of his bands, their data would not be used in the regression model. Another option is to excluded cases on a pairwise basis, which means that if a participant has a score missing for a particular variable, then their data are excluded only from calculations involving the variable for which they have no score. So, data for the band for which there was no attractiveness rating would still be used to calculate the relationships between advertising budget, airplay and record sales. However, if you do this many of your variables may not make sense, and you can end up with absurdities such as R^2 either negative or greater than 1.0. So, it's not a good option.

FIGURE 7.16

Dialog box
for regression
diagnostics

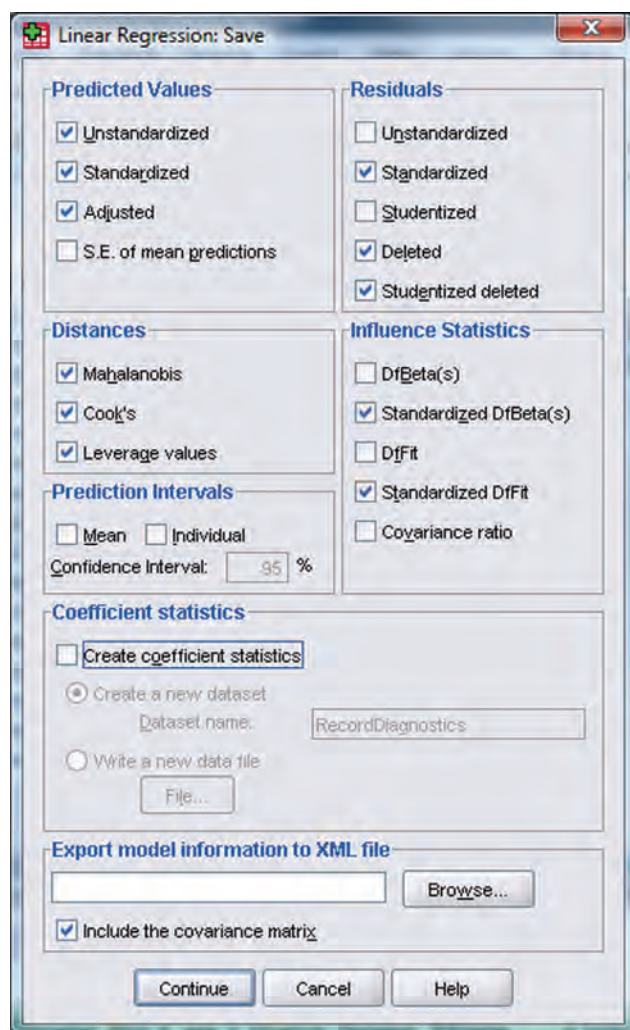
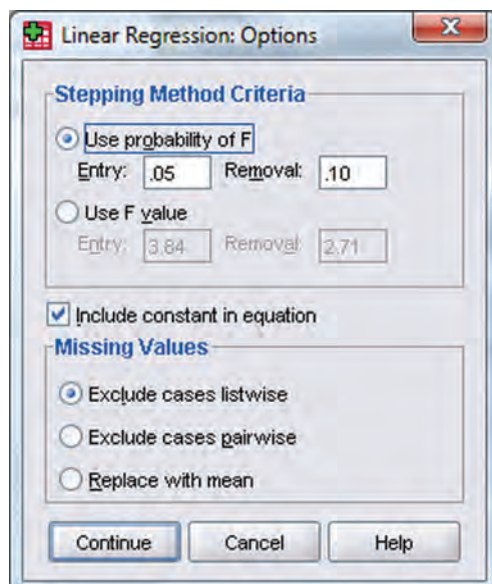


FIGURE 7.17

Options for linear
regression



Another possibility is to replace the missing score with the average score for this variable and then include that case in the analysis (so, our example band would be given an attractiveness rating equal to the average attractiveness of all bands). The problem with this final choice is that it is likely to suppress the true value of the standard deviation (and more importantly the standard error). The standard deviation will be suppressed because for any replaced case there will be no difference between the mean and the score, whereas if data had been collected for that case there would, almost certainly, have been some difference between the score and the mean. Obviously, if the sample is large and the number of missing values small then this is not a serious consideration. However, if there are many missing values this choice is potentially dangerous because smaller standard errors are more likely to lead to significant results that are a product of the data replacement rather than a genuine effect. The final option is to use the Missing Value Analysis routine in SPSS. This is for experts. It makes use of the fact that if two or more variables are present and correlated for most cases in the file, and an occasional value is missing, you can replace the missing values with estimates far better than the mean (Tabachnick & Fidell, 2007: Chapter 4, describe some of these procedures).

7.8. Interpreting multiple regression ②

Having selected all of the relevant options and returned to the main dialog box, we need to click on **OK** to run the analysis. SPSS will spew out copious amounts of output in the viewer window, and we now turn to look at how to make sense of this information.

7.8.1. Descriptives ②

The output described in this section is produced using the options in the *Linear Regression: Statistics* dialog box (see Figure 7.14). To begin with, if you selected the *Descriptives* option, SPSS will produce the table seen in SPSS Output 7.4. This table tells us the mean and standard deviation of each variable in our data set, so we now know that the average number of record sales was 193,000. This table isn't necessary for interpreting the regression model, but it is a useful summary of the data. In addition to the descriptive statistics, selecting this option produces a correlation matrix too. This table shows three things. First, the table shows the value of Pearson's correlation coefficient between every pair of variables (e.g. we can see that the advertising budget had a large positive correlation with record sales, $r = .578$). Second, the one-tailed significance of each correlation is displayed (e.g. the correlation above is significant, $p < .001$). Finally, the number of cases contributing to each correlation ($N = 200$) is shown.

You might notice that along the diagonal of the matrix the values for the correlation coefficients are all 1.00 (i.e. a perfect positive correlation). The reason for this is that these values represent the correlation of each variable with itself, so obviously the resulting values are 1. The correlation matrix is extremely useful for getting a rough idea of the relationships between predictors and the outcome, and for a preliminary look for multicollinearity. If there is no multicollinearity in the data then there should be no substantial correlations ($r > .9$) between predictors.

If we look only at the predictors (ignore record sales) then the highest correlation is between the attractiveness of the band and the amount of airplay which is significant at a .01 level ($r = .182$, $p = .005$). Despite the significance of this correlation, the coefficient is small and so it looks as though our predictors are measuring different things (there is little collinearity). We can see also that of all of the predictors the number of plays on Radio 1 correlates best with the outcome ($r = .599$, $p < .001$) and so it is likely that this variable will best predict record sales.



GRAMMING SAM’S TIPS

Use the descriptive statistics to check the correlation matrix for multicollinearity; that is, predictors that correlate too highly with each other, $R > .9$.

SPSS OUTPUT 7.4

Descriptive statistics
for regression
analysis

Descriptive Statistics

	Mean	Std. Deviation	N
Record Sales (thousands)	193.2000	80.6990	200
Advertising Budget (thousands of pounds)	614.4123	485.6552	200
No. of plays on Radio 1 per week	27.5000	12.2696	200
Attractiveness of Band	6.7700	1.3953	200

Correlations

		Record Sales (thousands)	Advertising Budget (thousands of pounds)	No. of plays on Radio 1 per week	Attractiveness of Band
Pearson Correlation	Record Sales (thousands)	1.000	.578	.599	.326
	Advertising Budget (thousands of pounds)	.578	1.000	.102	.081
	No. of plays on Radio 1 per week	.599	.102	1.000	.182
	Attractiveness of Band	.326	.081	.182	1.000
Sig. (1-tailed)	Record Sales (thousands)	.	.000	.000	.000
	Advertising Budget (thousands of pounds)	.000	.	.076	.128
	No. of plays on Radio 1 per week	.000	.076	.	.005
	Attractiveness of Band	.000	.128	.005	.
N	Record Sales (thousands)	200	200	200	200
	Advertising Budget (thousands of pounds)	200	200	200	200
	No. of plays on Radio 1 per week	200	200	200	200
	Attractiveness of Band	200	200	200	200

7.8.2. Summary of model ②

The next section of output describes the overall model (so it tells us whether the model is successful in predicting record sales). Remember that we chose a hierarchical method and so each set of summary statistics is repeated for each stage in the hierarchy. In SPSS Output 7.5 you should note that there are two models. Model 1 refers to the first stage in the hierarchy when only advertising budget is used as a predictor. Model 2 refers to when all three predictors are used. SPSS Output 7.5 is the *model summary* and this table was produced using the *Model fit* option. This option is selected by default in SPSS because it provides us with some very important information about the model: the values of R , R^2 and the

adjusted R^2 . If the *R squared change* and *Durbin-Watson* options were selected, then these values are included also (if they weren't selected you'll find that you have a smaller table).

The model summary table is shown in SPSS Output 7.5 and you should notice that under this table SPSS tells us what the dependent variable (outcome) was and what the predictors were in each of the two models. In the column labelled *R* are the values of the multiple correlation coefficient between the predictors and the outcome. When only advertising budget is used as a predictor, this is the simple correlation between advertising and record sales (0.578). In fact all of the statistics for model 1 are the same as the simple regression model earlier (see section 7.4). The next column gives us a value of R^2 , which we already know is a measure of how much of the variability in the outcome is accounted for by the predictors. For the first model its value is .335, which means that advertising budget accounts for 33.5% of the variation in record sales. However, when the other two predictors are included as well (model 2), this value increases to .665 or 66.5% of the variance in record sales. Therefore, if advertising accounts for 33.5%, we can tell that attractiveness and radio play account for an additional 33%.⁹ So, the inclusion of the two new predictors has explained quite a large amount of the variation in record sales.

SPSS OUTPUT 7.5
Regression model
summary

Model Summary^c

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics					Durbin-Watson
					R Square Change	F Change	df1	df2	Sig. F Change	
1	.578 ^a	.335	.331	65.9914	.335	99.587	1	198	.000	
2	.815 ^b	.665	.660	47.0873	.330	96.447	2	196	.000	1.950

a. Predictors: (Constant), Advertising Budget (thousands of pounds)
b. Predictors: (Constant), Advertising Budget (thousands of pounds), Attractiveness of Band, No. of plays on Radio 1 per week
c. Dependent Variable: Record Sales (thousands)

The adjusted R^2 gives us some idea of how well our model generalizes and ideally we would like its value to be the same, or very close to, the value of R^2 . In this example the difference for the final model is small (in fact the difference between the values is $.665 - .660 = .005$ (about 0.5%). This shrinkage means that if the model were derived from the population rather than a sample it would account for approximately 0.5% less variance in the outcome. Advanced students might like to apply Stein's formula to the R^2 to get some idea of its likely value in different samples. Stein's formula was given in equation (7.11) and can be applied by replacing n with the sample size (200) and k with the number of predictors (3):

$$\begin{aligned} \text{adjusted } R^2 &= 1 - \left[\left(\frac{200 - 1}{200 - 3 - 1} \right) \left(\frac{200 - 2}{200 - 3 - 2} \right) \left(\frac{200 + 1}{200} \right) \right] (1 - 0.665) \\ &= 1 - [(1.015)(1.015)(1.005)](0.335) \\ &= 1 - 0.347 \\ &= 0.653 \end{aligned}$$

This value is very similar to the observed value of R^2 (.665) indicating that the cross-validity of this model is very good.

The change statistics are provided only if requested and these tell us whether the change in R^2 is significant. The significance of R^2 can actually be tested using an F -ratio, and this F is calculated from the following equation (in which N is the number of cases or participants, and k is the number of predictors in the model):

$$F = \frac{(N - k - 1)R^2}{k(1 - R^2)}$$

⁹ That is, 33% = 66.5% – 33.5% (this value is the *R Square Change* in the table).

In SPSS Output 7.5, the change in this F is reported for each block of the hierarchy. So, model 1 causes R^2 to change from 0 to .335, and this change in the amount of variance explained gives rise to an F -ratio of 99.587, which is significant with a probability less than .001. Bearing in mind for this first model that we have only one predictor (so $k = 1$) and 200 cases ($N = 200$), this F comes from the equation above:¹⁰

$$F_{Model1} = \frac{(200 - 1 - 1)0.334648}{1(1 - 0.334648)} = 99.587$$

The addition of the new predictors (model 2) causes R^2 to increase by .330 (see above). We can calculate the F -ratio for this change using the same equation, but because we're looking at the change in models we use the change in R^2 , R^2_{Change} , and the R^2 in the new model (model 2 in this case so I've called it R^2_2) and we also use the change in the number of predictors, k_{Change} (model 1 had one predictor and model 2 had three predictors, so the change in the number of predictors is $3 - 1 = 2$), and the number of predictors in the new model, k_2 (in this case because we're looking at model 2). Again, if we use a few more decimal places than in the SPSS table, we get approximately the same answer as SPSS:

$$\begin{aligned} F_{Change} &= \frac{(N - k_2 - 1)R^2_{Change}}{k_{Change}(1 - R^2_2)} \\ &= \frac{(200 - 3 - 1) \times 0.330}{2(1 - 0.664668)} \\ &= 96.44 \end{aligned}$$

As such, the change in the amount of variance that can be explained gives rise to an F -ratio of 96.44, which is again significant ($p < .001$). The change statistics therefore tell us about the difference made by adding new predictors to the model.

Finally, if you requested the Durbin-Watson statistic it will be found in the last column of the table in SPSS Output 7.5. This statistic informs us about whether the assumption of independent errors is tenable (see section 7.6.2.1). As a conservative rule I suggested that values less than 1 or greater than 3 should definitely raise alarm bells (although I urge you to look up precise values for the situation of interest). The closer to 2 that the value is, the better, and for these data the value is 1.950, which is so close to 2 that the assumption has almost certainly been met.

SPSS Output 7.6 shows the next part of the output, which contains an ANOVA that tests whether the model is significantly better at predicting the outcome than using the mean as a 'best guess'. Specifically, the F -ratio represents the ratio of the improvement in prediction that results from fitting the model, relative to the inaccuracy that still exists in the model (see section 7.2.3). This table is again split into two sections: one for each model. We are told the value of the sum of squares for the model (this value is SS_M in section 7.2.3 and represents the improvement in prediction resulting from fitting a regression line to the data rather than using the mean as an estimate of the outcome). We are also told the residual sum of squares (this value is SS_R in section 7.2.3 and represents the total difference between the model and the observed data). We are also told the degrees of freedom (df) for each term. In the case of the improvement due to the model, this value is equal to the number of predictors (1 for the first model and 3 for the second), and for SS_R it is the number of observations (200) minus the number of coefficients in the regression model. The first

¹⁰ To get the same values as SPSS we have to use the exact value of R^2 , which is 0.3346480676231 (if you don't believe me double-click on the table in the SPSS output that reports this value, then double-click on the cell of the table containing the value of R^2 and you'll see that .335 becomes the value that I've just typed!).

model has two coefficients (one for the predictor and one for the constant) whereas the second has four (one for each of the three predictors and one for the constant). Therefore, model 1 has 198 degrees of freedom whereas model 2 has 196. The average sum of squares (MS) is then calculated for each term by dividing the SS by the *df*. The *F*-ratio is calculated by dividing the average improvement in prediction by the model (MS_M) by the average difference between the model and the observed data (MS_R). If the improvement due to fitting the regression model is much greater than the inaccuracy within the model then the value of *F* will be greater than 1 and SPSS calculates the exact probability of obtaining the value of *F* by chance. For the initial model the *F*-ratio is 99.587, which is very unlikely to have happened by chance ($p < .001$). For the second model the value of *F* is even higher (129.498), which is also highly significant ($p < .001$). We can interpret these results as meaning that the initial model significantly improved our ability to predict the outcome variable, but that the new model (with the extra predictors) was even better (because the *F*-ratio is more significant).

SPSS OUTPUT 7.6

ANOVA ^c						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	433687.833	1	433687.833	99.587	.000 ^a
	Residual	862264.167	198	4354.870		
	Total	1295952.0	199			
2	Regression	861377.418	3	287125.806	129.498	.000 ^b
	Residual	434574.582	196	2217.217		
	Total	1295952.0	199			

- a. Predictors: (Constant), Advertising Budget (thousands of pounds)
- b. Predictors: (Constant), Advertising Budget (thousands of pounds), Attractiveness of Band, No. of Plays on Radio 1 per Week
- c. Dependent Variable: Record Sales (thousands)



CRAMMING SAM'S TIPS

The fit of the regression model can be assessed using the **Model Summary** and **ANOVA** tables from SPSS. Look for the R^2 to tell you the proportion of variance explained by the model. If you have done a hierarchical regression then you can assess the improvement of the model at each stage of the analysis by looking at the change in R^2 and whether this change is significant (look for values less than .05 in the column labelled *Sig F Change*.). The ANOVA also tells us whether the model is a significant fit of the data overall (look for values less than .05 in the column labelled *Sig*.). Finally, there is an assumption that errors in regression are independent; this assumption is likely to be met if the Durbin–Watson statistic is close to 2 (and between 1 and 3).

7.8.3. Model parameters ②

So far we have looked at several summary statistics telling us whether or not the model has improved our ability to predict the outcome variable. The next part of the output is concerned with the parameters of the model. SPSS Output 7.7 shows the model

parameters for both steps in the hierarchy. Now, the first step in our hierarchy was to include advertising budget (as we did for the simple regression earlier in this chapter) and so the parameters for the first model are identical to the parameters obtained in SPSS Output 7.3. Therefore, we will be concerned only with the parameters for the final model (in which all predictors were included). The format of the table of coefficients will depend on the options selected. The confidence interval for the *b*-values, collinearity diagnostics and the part and partial correlations will be present only if selected in the dialog box in Figure 7.14.

SPSS OUTPUT 7.7
Coefficients of the regression model¹¹

Coefficients ^a								
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	134.140	7.537		17.799	.000	119.278	149.002
	Advertising Budget (thousands of pounds)	.096	.010	.578	9.979	.000	.077	.115
2	(Constant)	-26.613	17.350		-1.534	.127	-60.830	7.604
	Advertising Budget (thousands of pounds)	.085	.007	.511	12.261	.000	.071	.099
	No. of plays on Radio 1 per week	3.367	.278	.512	12.123	.000	2.820	3.915
	Attractiveness of Band	11.086	2.438	.192	4.548	.000	6.279	15.894

a. Dependent Variable: Record Sales (thousands)

Coefficients ^a						
Model		Correlations			Collinearity Statistics	
		Zero-order	Partial	Part	Tolerance	VIF
1	Advertising Budget (thousands of pounds)	.578	.578	.578	1.000	1.000
2	Advertising Budget (thousands of pounds)	.578	.659	.507	.986	1.015
	No. of plays on Radio 1 per week	.599	.655	.501	.959	1.043
	Attractiveness of Band	.326	.309	.188	.963	1.038

a. Dependent Variable: Record Sales (thousands)

Remember that in multiple regression the model takes the form of equation (7.9) and in that equation there are several unknown quantities (the *b*-values). The first part of the table gives us estimates for these *b*-values and these values indicate the individual contribution of each predictor to the model. If we replace the *b*-values in equation (7.9) we find that we can define the model as follows:

sales_{*i*} = *b*₀ + *b*₁advertising_{*i*} + *b*₂airplay_{*i*} + *b*₃attractiveness_{*i*}
= − 26.61 + (0.08advertising_{*i*}) + (3.37airplay_{*i*})
+ (11.09attractiveness_{*i*})

(7.12)

The *b*-values tell us about the relationship between record sales and each predictor. If the value is positive we can tell that there is a positive relationship between the predictor and the outcome, whereas a negative coefficient represents a negative relationship. For these data all three predictors have positive *b*-values indicating positive relationships. So, as advertising budget increases, record sales increase; as plays on the radio increase, so do record sales; and finally more attractive bands will sell more records. The *b*-values tell us more than this, though. They tell us to what degree each predictor affects the outcome *if the effects of all other predictors are held constant*:

¹¹ To spare your eyesight I have split this part of the output into two tables; however, it should appear as one long table in the SPSS Viewer.

- **Advertising budget** ($b = 0.085$): This value indicates that as advertising budget increases by one unit, record sales increase by 0.085 units. Both variables were measured in thousands; therefore, for every £1000 more spent on advertising, an extra 0.085 thousand records (85 records) are sold. This interpretation is true only if the effects of attractiveness of the band and airplay are held constant.
- **Airplay** ($b = 3.367$): This value indicates that as the number of plays on radio in the week before release increases by one, record sales increase by 3.367 units. Therefore, every additional play of a song on radio (in the week before release) is associated with an extra 3.367 thousand records (3367 records) being sold. This interpretation is true only if the effects of attractiveness of the band and advertising are held constant.
- **Attractiveness** ($b = 11.086$): This value indicates that a band rated one unit higher on the attractiveness scale can expect additional record sales of 11.086 units. Therefore, every unit increase in the attractiveness of the band is associated with an extra 11.086 thousand records (11,086 records) being sold. This interpretation is true only if the effects of radio airplay and advertising are held constant.

Each of these beta values has an associated standard error indicating to what extent these values would vary across different samples, and these standard errors are used to determine whether or not the b -value differs significantly from zero. As we saw in section 7.4.2, a t -statistic can be derived that tests whether a b -value is significantly different from 0. In simple regression, a significant value of t indicates that the slope of the regression line is significantly different from horizontal, but in multiple regression, it is not so easy to visualize what the value tells us. Well, it is easiest to conceptualize the t -tests as measures of whether the predictor is making a significant contribution to the model. Therefore, if the t -test associated with a b -value is significant (if the value in the column labelled *Sig.* is less than .05) then the predictor is making a significant contribution to the model. The smaller the value of *Sig.* (and the larger the value of t), the greater the contribution of that predictor. For this model, the advertising budget ($t(196) = 12.26, p < .001$), the amount of radio play prior to release ($t(196) = 12.12, p < .001$) and attractiveness of the band ($t(196) = 4.55, p < .001$) are all significant predictors of record sales.¹² From the magnitude of the t -statistics we can see that the advertising budget and radio play had a similar impact, whereas the attractiveness of the band had less impact.

The b -values and their significance are important statistics to look at; however, the standardized versions of the b -values are in many ways easier to interpret (because they are not dependent on the units of measurement of the variables). The standardized beta values are provided by SPSS (labelled as Beta, β) and they tell us the number of standard deviations that the outcome will change as a result of one standard deviation change in the predictor. The standardized beta values are all measured in standard deviation units and so are directly comparable: therefore, they provide a better insight into the ‘importance’ of a predictor in the model. The standardized beta values for airplay and advertising budget are virtually identical (.512 and .511 respectively) indicating that both variables have a comparable degree of importance in the model (this concurs with what the magnitude of the t -statistics told us). To interpret these values literally, we need to know the standard deviations of all of the variables and these values can be found in SPSS Output 7.4.

¹² For all of these predictors I wrote $t(196)$. The number in brackets is the degrees of freedom. We saw in section 7.2.4 that in regression the degrees of freedom are $N - p - 1$, where N is the total sample size (in this case 200) and p is the number of predictors (in this case 3). For these data we get $200 - 3 - 1 = 196$.

- **Advertising budget** (*standardized β* = .511): This value indicates that as advertising budget increases by one standard deviation (£485,655), record sales increase by 0.511 standard deviations. The standard deviation for record sales is 80,699 and so this constitutes a change of 41,240 sales ($0.511 \times 80,699$). Therefore, for every £485,655 more spent on advertising, an extra 41,240 records are sold. This interpretation is true only if the effects of attractiveness of the band and airplay are held constant.
- **Airplay** (*standardized β* = .512): This value indicates that as the number of plays on radio in the week before release increases by 1 standard deviation (12.27), record sales increase by 0.512 standard deviations. The standard deviation for record sales is 80,699 and so this constitutes a change of 41,320 sales ($0.512 \times 80,699$). Therefore, if Radio 1 plays the song an extra 12.27 times in the week before release, 41,320 extra record sales can be expected. This interpretation is true only if the effects of attractiveness of the band and advertising are held constant.
- **Attractiveness** (*standardized β* = .192): This value indicates that a band rated one standard deviation (1.40 units) higher on the attractiveness scale can expect additional record sales of 0.192 standard deviations units. This constitutes a change of 15,490 sales ($0.192 \times 80,699$). Therefore, a band with an attractiveness rating 1.40 higher than another band can expect 15,490 additional sales. This interpretation is true only if the effects of radio airplay and advertising are held constant.



SELF-TEST Think back to what the confidence interval of the mean represented (section 2.5.2). Can you guess what the confidence interval for b represents?

Imagine that we collected 100 samples of data measuring the same variables as our current model. For each sample we could create a regression model to represent the data. If the model is reliable then we hope to find very similar parameters in all samples. Therefore, each sample should produce approximately the same b -values. The confidence intervals of the unstandardized beta values are boundaries constructed such that in 95% of these samples these boundaries will contain the true value of b (see section 2.5.2). Therefore, if we'd collected 100 samples, and calculated the confidence intervals for b , we are saying that 95% of these confidence intervals would contain the true value of b . Therefore, we can be fairly confident that the confidence interval we have constructed for this sample will contain the true value of b in the population. This being so, a good model will have a small confidence interval, indicating that the value of b in this sample is close to the true value of b in the population. The sign (positive or negative) of the b -values tells us about the direction of the relationship between the predictor and the outcome. Therefore, we would expect a very bad model to have confidence intervals that cross zero, indicating that in some samples the predictor has a negative relationship to the outcome whereas in others it has a positive relationship. In this model, the two best predictors (advertising and airplay) have very tight confidence intervals indicating that the estimates for the current model are likely to be representative of the true population values. The interval for attractiveness is wider (but still does not cross zero) indicating that the parameter for this variable is less representative, but nevertheless significant.

If you asked for part and partial correlations, then they will appear in the output in separate columns of the table. The zero-order correlations are the simple Pearson's correlation coefficients (and so correspond to the values in SPSS Output 7.4). The partial correlations represent the relationships between each predictor and the outcome variable, controlling for the effects of the other two predictors. The part correlations represent the relationship between each predictor and the outcome, controlling for the effect that the other two variables have on the outcome. In effect, these part correlations represent the unique relationship that each predictor has with the outcome. If you opt to do a stepwise regression, you would

find that variable entry is based initially on the variable with the largest zero-order correlation and then on the part correlations of the remaining variables. Therefore, airplay would be entered first (because it has the largest zero-order correlation), then advertising budget (because its part correlation is bigger than attractiveness) and then finally attractiveness. Try running a forward stepwise regression on these data to see if I'm right! Finally, we are given details of the collinearity statistics, but these will be discussed in section 7.8.5.



CRAMMING SAM'S TIPS

The individual contribution of variables to the regression model can be found in the **Coefficients** table from SPSS. If you have done a hierarchical regression then look at the values for the final model. For each predictor variable, you can see if it has made a significant contribution to predicting the outcome by looking at the column labelled *Sig.* (values less than .05 are significant). You should also look at the standardized beta values because these tell you the importance of each predictor (bigger absolute value = more important). The Tolerance and VIF values will also come in handy later on, so make a note of them!

7.8.4. Excluded variables ②

At each stage of a regression analysis SPSS provides a summary of any variables that have not yet been entered into the model. In a hierarchical model, this summary has details of the variables that have been specified to be entered in subsequent steps, and in stepwise regression this table contains summaries of the variables that SPSS is considering entering into the model. For this example, there is a summary of the excluded variables (SPSS Output 7.8) for the first stage of the hierarchy (there is no summary for the second stage because all predictors are in the model). The summary gives an estimate of each predictor's beta value if it was entered into the equation at this point and calculates a *t*-test for this value. In a stepwise regression, SPSS should enter the predictor with the highest *t*-statistic and will continue entering predictors until there are none left with *t*-statistics that have significance values less than .05. The partial correlation also provides some indication as to what contribution (if any) an excluded predictor would make if it were entered into the model.

SPSS OUTPUT 7.8

Excluded Variables^b

		Beta In	t	Sig.	Partial Correlation	Collinearity Statistics		
						Tolerance	VIF	Minimum Tolerance
1	No. of plays on Radio 1 per week	.546 ^a	12.513	.000	.665	.990	1.010	.990
	Attractiveness of Band	.281 ^a	5.136	.000	.344	.993	1.007	.993

a. Predictors in the Model: (Constant), Advertising Budget (thousands of pounds)
b. Dependent Variable: Record Sales (thousands)

7.8.5. Assessing the assumption of no multicollinearity ②

SPSS Output 7.7 provided some measures of whether there is collinearity in the data. Specifically, it provides the VIF and tolerance statistics (with tolerance being 1 divided

by the VIF). There are a few guidelines from section 7.6.2.3 that can be applied here:

- If the largest VIF is greater than 10 then there is cause for concern (Bowerman & O’Connell, 1990; Myers, 1990).
- If the average VIF is substantially greater than 1 then the regression may be biased (Bowerman & O’Connell, 1990).
- Tolerance below 0.1 indicates a serious problem.
- Tolerance below 0.2 indicates a potential problem (Menard, 1995).

For our current model the VIF values are all well below 10 and the tolerance statistics all well above 0.2; therefore, we can safely conclude that there is no collinearity within our data. To calculate the average VIF we simply add the VIF values for each predictor and divide by the number of predictors (*k*):

$$\overline{\text{VIF}} = \frac{\sum_{i=1}^k \text{VIF}_i}{k} = \frac{1.015 + 1.043 + 1.038}{3} = 1.032$$

The average VIF is very close to 1 and this confirms that collinearity is not a problem for this model. SPSS also produces a table of eigenvalues of the scaled, uncentred cross-products matrix, condition indexes and variance proportions (see Jane Superbrain Box 7.2). There is a lengthy discussion, and example, of collinearity in section 8.8.1 and how to detect it using variance proportions, so I will limit myself now to saying that we are looking for large variance proportions on the same *small* eigenvalues. Therefore, in SPSS Output 7.9 we look at the bottom few rows of the table (these are the small eigenvalues) and look for any variables that both have high variance proportions for that eigenvalue. The variance proportions vary between 0 and 1, and for each predictor should be distributed across different dimensions (or eigenvalues). For this model, you can see that each predictor has most of its variance loading onto a different dimension (advertising has 96% of variance on dimension 2, airplay has 93% of variance on dimension 3 and attractiveness has 92% of variance on dimension 4). These data represent a classic example of no multicollinearity. For an example of when collinearity exists in the data and some suggestions about what can be done, see Chapters 8 (section 8.8.1) and 17 (section 17.3.3.3).

SPSS OUTPUT 7.9

Collinearity Diagnostics^a

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions			
				(Constant)	Advertising Budget (thousands of pounds)	No. of plays on Radio 1 per week	Attractiveness of Band
1	1	1.785	1.000	.11	.11		
	2	.215	2.883	.89	.89		
2	1	3.562	1.000	.00	.02	.01	.00
	2	.308	3.401	.01	.96	.05	.01
	3	.109	5.704	.05	.02	.93	.07
	4	2.039E-02	13.219	.94	.00	.00	.92

^a. Dependent Variable: Record Sales (thousands)



CRAMMING SAM’S TIPS

To check the assumption of no multicollinearity, use the VIF values from the table **Labelled** coefficients in the SPSS output. If these values are less than 10 then that indicates there probably isn’t cause for concern. If you take the average of VIF values, and this average is not substantially greater than 1, then that also indicates that there’s no cause for concern.

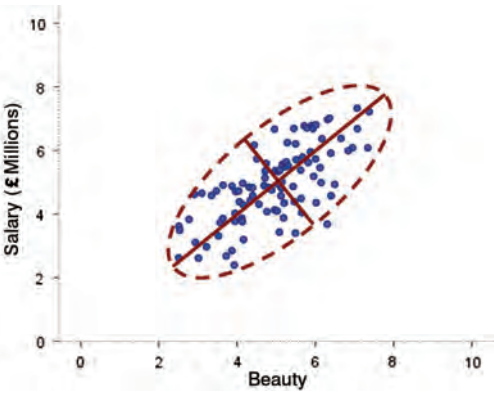
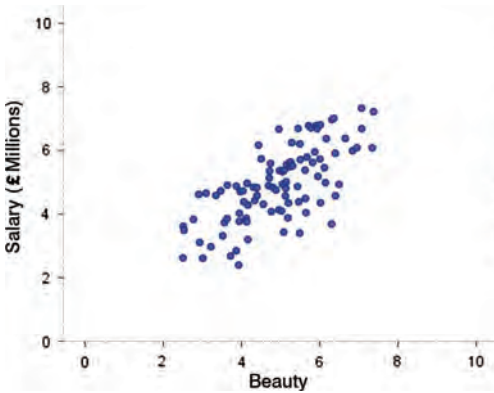


JANE SUPERBRAIN 7.2

What are eigenvectors and eigenvalues? ④

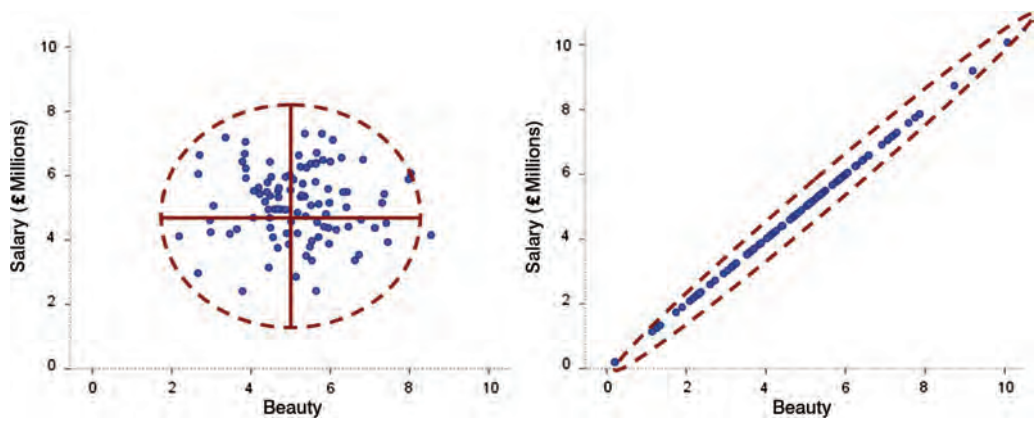
The definitions and mathematics of eigenvalues and eigenvectors are very complicated and most of us need not worry about them (although they do crop up again in Chapters 16 and 17). However, although the mathematics of them is hard, they are quite easy to visualize! Imagine we have two variables: the salary a supermodel earns in a year, and how attractive she is. Also imagine these two variables are normally distributed and so can be considered together as a bivariate normal distribution. If these variables are correlated, then their scatterplot forms an ellipse. This is shown in the scatterplots below: if we draw a dashed line around the outer values of the scatterplot we get something oval shaped. Now, we can draw two lines to measure the length and height of this ellipse. These

lines are the *eigenvectors* of the original correlation matrix for these two variables (a vector is just a set of numbers that tells us the location of a line in geometric space). Note that the two lines we've drawn (one for height and one for width of the oval) are perpendicular; that is, they are at 90 degrees, which means that they are independent of one another). So, with two variables, eigenvectors are just lines measuring the length and height of the ellipse that surrounds the scatterplot of data for those variables. If we add a third variable (e.g. experience of the supermodel) then all that happens is our scatterplot gets a third dimension, the ellipse turns into something shaped like a rugby ball (or American football), and because we now have a third dimension (height, width and depth) we get an extra eigenvector to measure this extra dimension. If we add a fourth variable, a similar logic applies (although it's harder to visualize): we get an extra dimension, and an eigenvector to measure that dimension. Now, each eigenvector has an *eigenvalue* that tells us its length (i.e. the distance from one end of the eigenvector to the other). So, by looking at all of the eigenvalues for a data set, we know the dimensions of the ellipse or rugby ball: put more generally, we know the dimensions of the data. Therefore, the eigenvalues show how evenly (or otherwise) the variances of the matrix are distributed.



In the case of two variables, the *condition* of the data is related to the ratio of the larger eigenvalue to the smaller. Let's look at the two extremes: when there is no relationship at all between variables, and when there is a perfect relationship. When there is no relationship, the scatterplot will, more or less, be contained within a circle (or a sphere if we had three variables). If we again draw lines that measure the height and width of this circle we'll find that these lines are the same length. The eigenvalues measure the length, therefore the eigenvalues will also be the same. So, when we divide the

largest eigenvalue by the smallest we'll get a value of 1 (because the eigenvalues are the same). When the variables are perfectly correlated (i.e. there is perfect collinearity) then the scatterplot forms a straight line and the ellipse surrounding it will also collapse to a straight line. Therefore, the height of the ellipse will be very small indeed (it will approach zero). Therefore, when we divide the largest eigenvalue by the smallest we'll get a value that tends to infinity (because the smallest eigenvalue is close to zero). Therefore, an infinite condition index is a sign of deep trouble.



7.8.6. Casewise diagnostics ②

SPSS produces a summary table of the residual statistics and these should be examined for extreme cases. SPSS Output 7.10 shows any cases that have a standardized residual less than -2 or greater than 2 (remember that we changed the default criterion from 3 to 2 in Figure 7.14). I mentioned in section 7.6.1.1 that in an ordinary sample we would expect 95% of cases to have standardized residuals within about ± 2 . We have a sample of 200, therefore it is reasonable to expect about 10 cases (5%) to have standardized residuals outside of these limits. From SPSS Output 7.10 we can see that we have 12 cases (6%) that are outside of the limits: therefore, our sample is within 1% of what we would expect. In addition, 99% of cases should lie within ± 2.5 and so we would expect only 1% of cases to lie outside of these limits. From the cases listed here, it is clear that two cases (1%) lie outside of the limits (cases 164 and 169). Therefore, our sample appears to conform to what we would expect for a fairly accurate model. These diagnostics give us no real cause for concern except that case 169 has a standardized residual greater than 3, which is probably large enough for us to investigate this case further.

SPSS OUTPUT 7.10

Casewise Diagnostics^a

Case Number	Std. Residual	Record Sales (thousands)	Predicted Value	Residual
1	2.125	330.00	229.9203	100.0797
2	-2.314	120.00	228.9490	-108.9490
10	2.114	300.00	200.4662	99.5338
47	-2.442	40.00	154.9698	-114.9698
52	2.069	190.00	92.5973	97.4027
55	-2.424	190.00	304.1231	-114.1231
61	2.098	300.00	201.1897	98.8103
68	-2.345	70.00	180.4156	-110.4156
100	2.066	250.00	152.7133	97.2867
164	-2.577	120.00	241.3240	-121.3240
169	3.061	360.00	215.8675	144.1325
200	-2.064	110.00	207.2061	-97.2061

a. Dependent Variable: Record Sales (thousands)

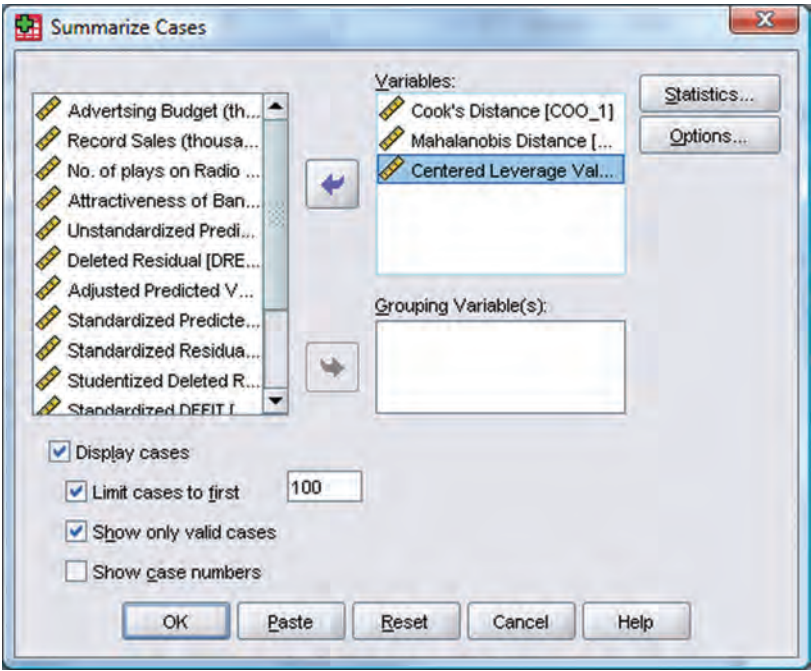


FIGURE 7.18
The summarize cases dialog box

You may remember that in section 7.7.5 we asked SPSS to save various diagnostic statistics. You should find that the data editor now contains columns for these variables. It is perfectly acceptable to check these values in the data editor, but you can also get SPSS to list the values in your viewer window too. To list variables you need to use the *Case Summaries* command, which can be found by selecting **Analyze > Reports > Case Summaries...** Figure 7.18 shows the dialog box for this function. Simply select the variables that you want to list and transfer them to the box labelled *Variables* by clicking on . By default, SPSS will limit the output to the first 100 cases, but if you want to list all of your cases then simply deselect this option. It is also very important to select the *Show case numbers* option because otherwise you might not be able to identify a problem case.

One useful strategy is to use the casewise diagnostics to identify cases that you want to investigate further. So, to save space, I created a coding variable (1 = include, 0 = exclude) so that I could specify the 12 cases listed in SPSS Output 7.11 in one group, and all other cases in the other. By using this coding variable and specifying it as a grouping variable in the *Summarize Cases* dialog box, I could look at those 12 cases together and discard all others.

SPSS Output 7.11 shows the influence statistics for the 12 cases that I selected. None of them have a Cook's distance greater than 1 (even case 169 is well below this criterion) and so none of the cases is having an undue influence on the model. The average leverage can be calculated as 0.02 ($k + 1/n = 4/200$) and so we are looking for values either twice as large as this (0.04) or three times as large (0.06) depending on which statistician you trust most (see section 7.6.1.2)! All cases are within the boundary of three times the average and only case 1 is close to two times the average. Finally, from our guidelines for the Mahalanobis distance we saw that with a sample of 100 and three predictors, values greater than 15 were problematic. We have three predictors and a larger sample size, so this value will be a conservative cut-off, yet none of our cases comes close to exceeding this criterion. The evidence suggests that there are no influential cases within our data (although all cases would need to be examined to confirm this fact).

We can look also at the DFBeta statistics to see whether any case would have a large influence on the regression parameters. An absolute value greater than 1 is a problem and

SPSS OUTPUT 7.11

Case Summaries							
	Case Number	Standardized DFBETA Intercept	Standardized DFBETA ADVERTS	Standardized DFBETA AIRPLAY	Standardized DFBETA ATTRACT	Standardized DFFIT	COVRATIO
1	1	-.31554	-.24235	.15774	.35329	.48929	.97127
2	2	.01259	-.12637	.00942	-.01868	-.21110	.92018
3	10	-.01256	-.15612	.16772	.00672	.26896	.94392
4	47	.06645	.19602	.04829	-.17857	-.31469	.91458
5	52	.35291	-.02881	-.13667	-.26965	.36742	.95995
6	55	.17427	-.32649	-.02307	-.12435	-.40736	.92486
7	61	.00082	-.01539	.02793	.02054	.15562	.93654
8	68	-.00281	.21146	-.14766	-.01760	-.30216	.92370
9	100	.06113	.14523	-.29984	.06766	.35732	.95888
10	164	.17983	.28988	-.40088	-.11706	-.54029	.92037
11	169	-.16819	-.25765	.25739	.16968	.46132	.85325
12	200	.16633	-.04639	.14213	-.25907	-.31985	.95435
Total	N	12	12	12	12	12	12

Case Summaries				
	Case Number	Cook's Distance	Mahalanobis Distance	Centered Leverage Value
1	1	.05870	8.39591	.04219
2	2	.01089	.59830	.00301
3	10	.01776	2.07154	.01041
4	47	.02412	2.12475	.01068
5	52	.03316	4.81841	.02421
6	55	.04042	4.19960	.02110
7	61	.00595	.06880	.00035
8	68	.02229	2.13106	.01071
9	100	.03136	4.53310	.02278
10	164	.07077	6.83538	.03435
11	169	.05087	3.14841	.01582
12	200	.02513	3.49043	.01754
Total	N	12	12	12

in all cases the values lie within ± 1 , which shows that these cases have no undue influence over the regression parameters. There is also a column for the covariance ratio. We saw in section 7.6.1.2 that we need to use the following criteria:

- $CVR_i > 1 + [3(k + 1)/n] = 1 + [3(3 + 1)/200] = 1.06$.
- $CVR_i < 1 - [3(k + 1)/n] = 1 - [3(3 + 1)/200] = 0.94$.

Therefore, we are looking for any cases that deviate substantially from these boundaries. Most of our 12 potential outliers have CVR values within or just outside these boundaries. The only case that causes concern is case 169 (again!) whose CVR is some way below the bottom limit. However, given the Cook's distance for this case, there is probably little cause for alarm.

You would have requested other diagnostic statistics and from what you know from the earlier discussion of them you would be well advised to glance over them in case of any unusual cases in the data. However, from this minimal set of diagnostics we appear to have a fairly reliable model that has not been unduly influenced by any subset of cases.



CRAMMING SAM'S TIPS

You need to look for cases that might be influencing the regression model:

- Look at standardized residuals and check that no more than 5% of cases have absolute values above 2, and that no more than about 1% have absolute values above 2.5. Any case with the value above about 3, could be an outlier.
- Look in the data editor for the values of Cook's distance: any value above 1 indicates a case that might be influencing the model.
- Calculate the average leverage (the number of predictors plus 1, divided by the sample size) and then look for values greater than twice or three times this average value.
- For Mahalanobis distance, a crude check is to look for values above 25 in large samples (500) and values above 15 in smaller samples (100). However, Barnett and Lewis (1978) should be consulted for more detailed analysis.
- Look for absolute values of DFBeta greater than 1.
- Calculate the upper and lower limit of acceptable values for the covariance ratio, CVR. The upper limit is 1 plus three times the average leverage, whereas the lower limit is 1 minus three times the average leverage. Cases that have a CVR that fall outside of these limits may be problematic.

7.8.7. Checking assumptions ②

As a final stage in the analysis, you should check the assumptions of the model. We have already looked for collinearity within the data and used Durbin–Watson to check whether the residuals in the model are independent. In section 7.7.4 we asked for a plot of *ZRESID against *ZPRED and for a histogram and normal probability plot of the residuals. The graph of *ZRESID and *ZPRED should look like a random array of dots evenly dispersed around zero. If this graph funnels out, then the chances are that there is heteroscedasticity in the data. If there is any sort of curve in this graph then the chances are that the data have broken the assumption of linearity. Figure 7.19 shows several examples of the plot of standardized residuals against standardized predicted values. Panel (a) shows the graph for the data in our record sales example. Note how the points are randomly and evenly dispersed throughout the plot. This pattern is indicative of a situation in which the assumptions of linearity and homoscedasticity have been met. Panel (b) shows a similar plot for a data set that violates the assumption of homoscedasticity. Note that the points form the shape of a funnel so they become more spread out across the graph. This funnel shape is typical of heteroscedasticity and indicates increasing variance across the residuals. Panel (c) shows a plot of some data in which there is a non-linear relationship between the outcome and the predictor. This pattern is shown up by the residuals. A line illustrating the curvilinear relationship has been drawn over the top of the graph to illustrate the trend in the data. Finally, panel (d) represents a situation in which the data not only represent a non-linear relationship, but also show heteroscedasticity. Note first the curved trend in the data, and then also note that at one end of the plot the points are very close together whereas at the other end they are widely dispersed. When these assumptions have been violated you will not see these exact patterns, but hopefully these plots will help you to understand the types of anomalies you should look out for.

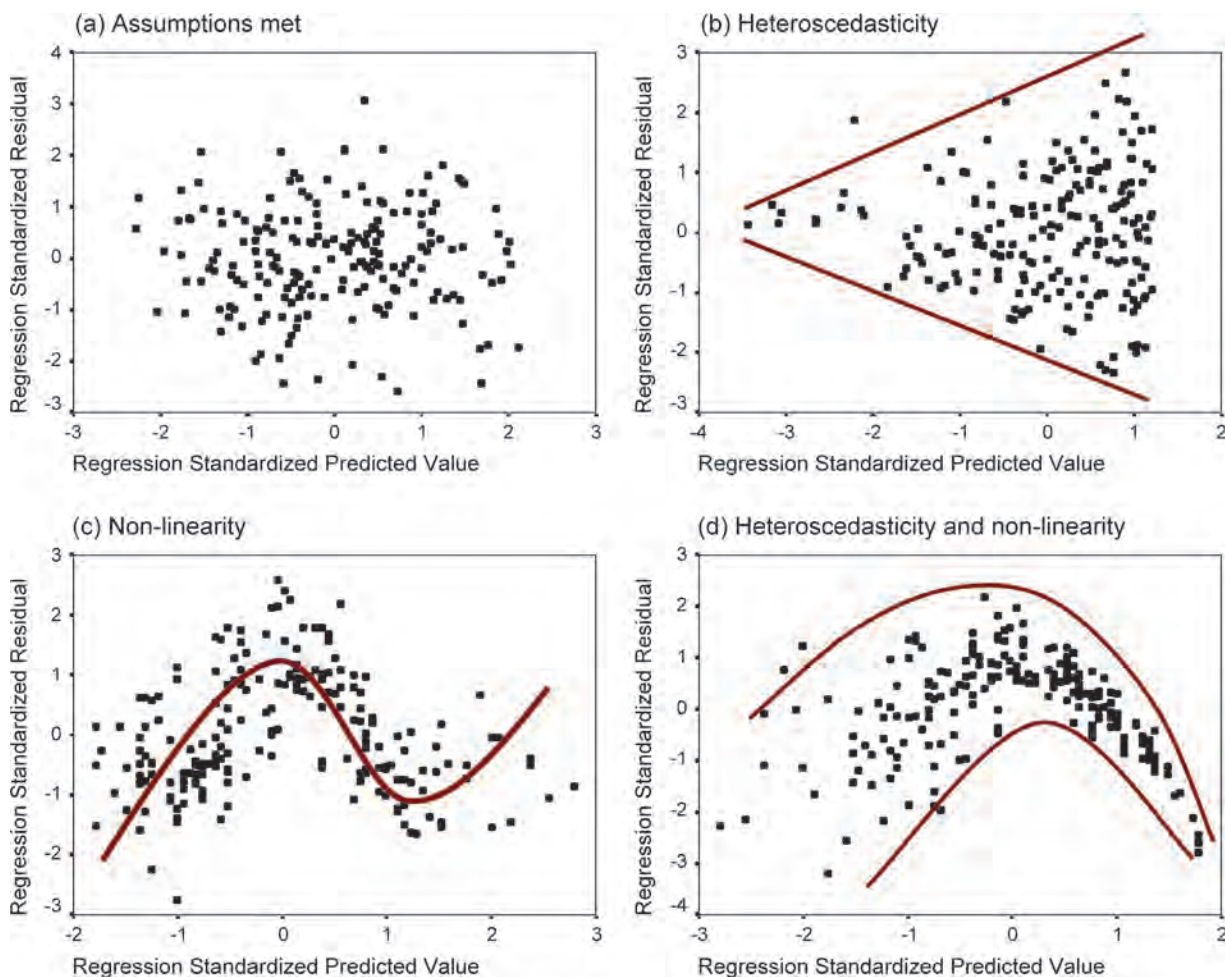


FIGURE 7.19 Plots of *ZRESID against *ZPRED

To test the normality of residuals, we must look at the histogram and normal probability plot selected in Figure 7.15. Figure 7.20 shows the histogram and normal probability plot of the data for the current example (left-hand side). The histogram should look like a normal distribution (a bell-shaped curve). SPSS draws a curve on the histogram to show the shape of the distribution. For the record company data, the distribution is roughly normal (although there is a slight deficiency of residuals exactly on zero). Compare this histogram to the extremely non-normal histogram next to it and it should be clear that the non-normal distribution is extremely skewed (unsymmetrical). So, you should look for a curve that has the same shape as the one for the record sales data: any deviation from this curve is a sign of non-normality and the greater the deviation, the more non-normally distributed the residuals. The normal probability plot also shows up deviations from normality (see Chapter 5). The straight line in this plot represents a normal distribution, and the points represent the observed residuals. Therefore, in a perfectly normally distributed data set, all points will lie on the line. This is pretty much what we see for the record sales data. However, next to the normal probability plot of the record sales data is an example of an extreme deviation from normality. In this plot, the dots

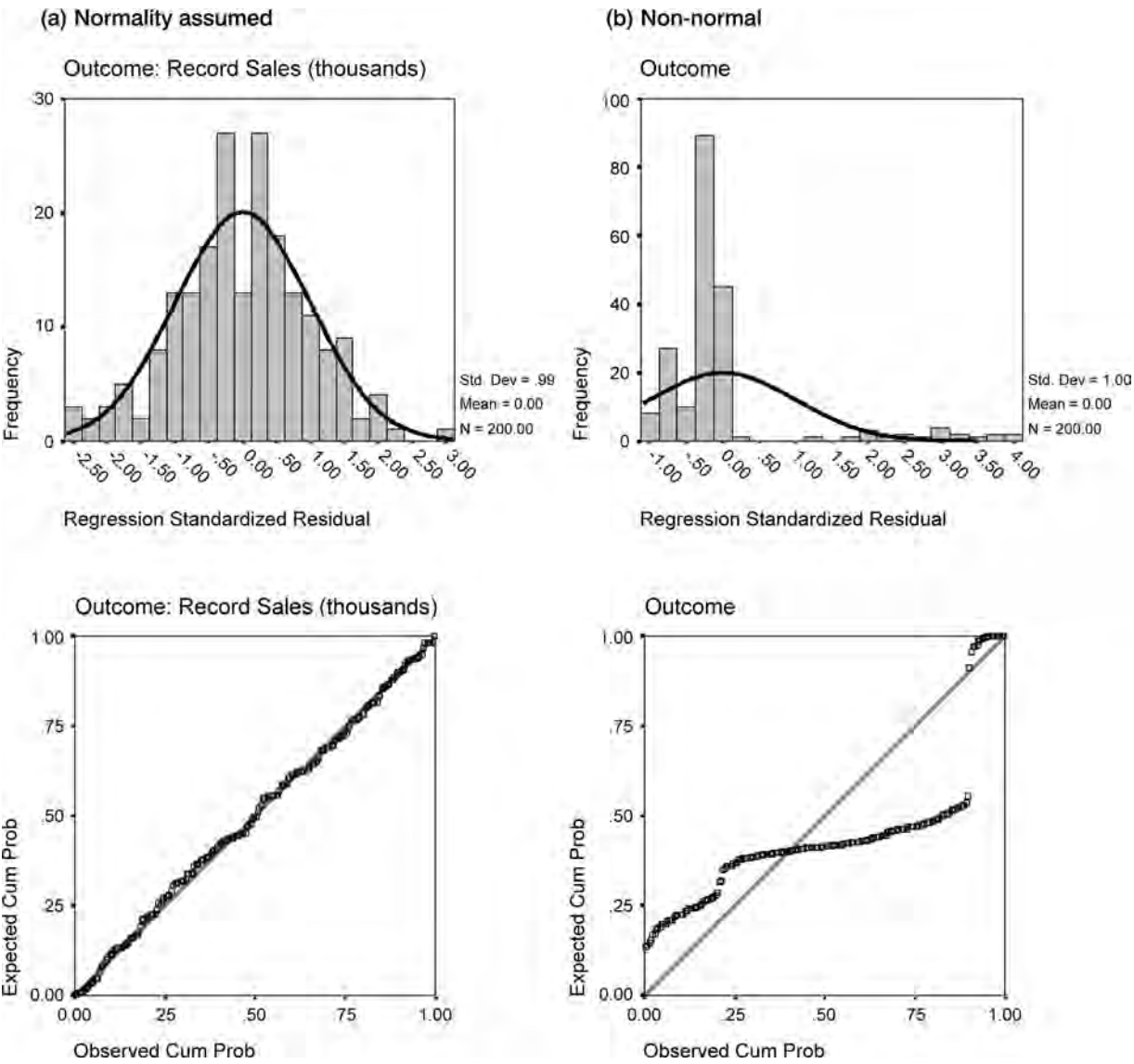
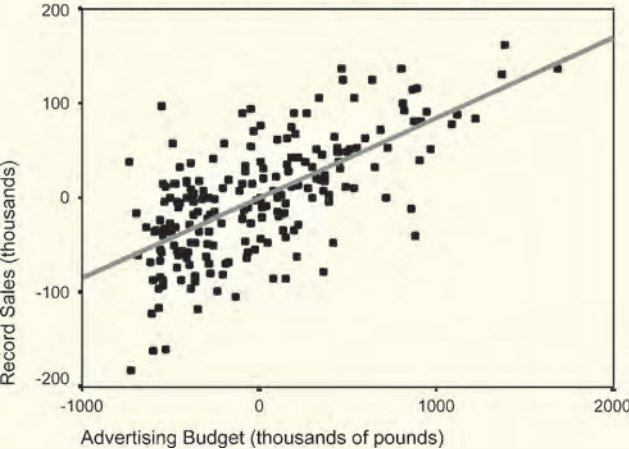
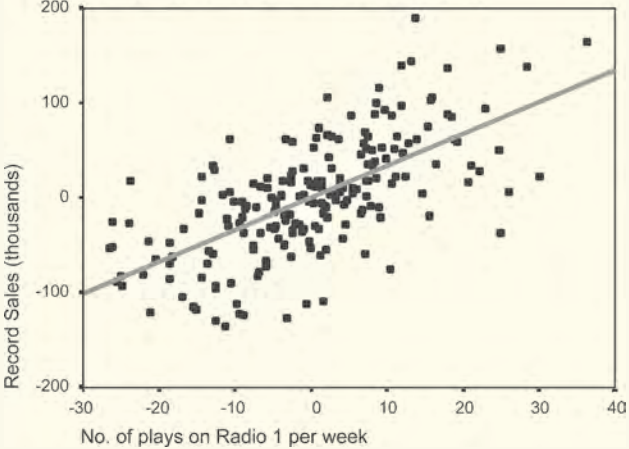
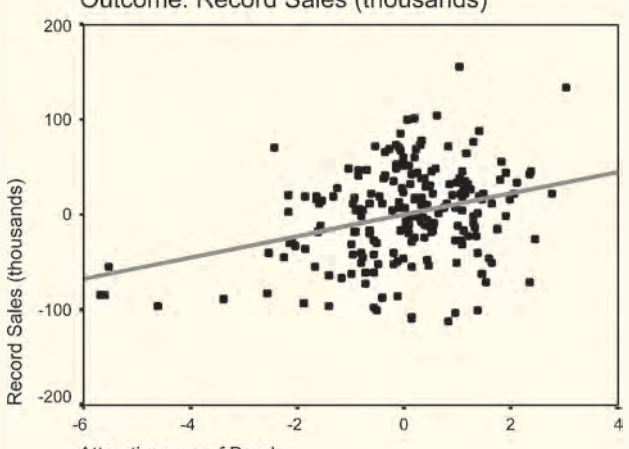


FIGURE 7.20 Histograms and normal P–P plots of normally distributed residuals (left-hand side) and non-normally distributed residuals (right-hand side)

are very distant from the line, which indicates a large deviation from normality. For both plots, the non-normal data are extreme cases and you should be aware that the deviations from normality are likely to be subtler. Of course, you can use what you learnt in Chapter 5 to do a K–S test on the standardized residuals to see whether they deviate significantly from normality.

A final set of plots specified in Figure 7.15 was the partial plots. These plots are scatterplots of the residuals of the outcome variable and each of the predictors when both variables are regressed separately on the remaining predictors. I mentioned earlier that obvious outliers on a partial plot represent cases that might have undue influence on a

predictor’s regression coefficient and that non-linear relationships and heteroscedasticity can be detected using these plots as well:

Outcome: Record Sales (thousands)  Advertising Budget (thousands of pounds)	For advertising budget the partial plot shows the strong positive relationship to record sales. The gradient of the line is b for advertising in the model (this line does not appear by default). There are no obvious outliers on this plot, and the cloud of dots is evenly spaced out around the line, indicating homoscedasticity.
Outcome: Record Sales (thousands)  No. of plays on Radio 1 per week	For airplay the partial plot shows a strong positive relationship to record sales. The gradient and pattern of the data are similar to advertising (which would be expected given the similarity of the standardized betas of these predictors). There are no obvious outliers on this plot, and the cloud of dots is evenly spaced around the line, indicating homoscedasticity.
Outcome: Record Sales (thousands)  Attractiveness of Band	For attractiveness the partial plot shows a positive relationship to record sales. The relationship looks less linear than the other predictors do and the dots seem to funnel out, indicating greater variance at high levels of attractiveness. There are no obvious outliers on this plot, but the funnel-shaped cloud of dots might indicate a violation of the assumption of homoscedasticity. We would be well advised to collect some more data for unattractive bands to verify the current model.



CRAMMING SAM'S TIPS

You need to check some of the assumptions of regression to make sure your model generalizes beyond your sample:

- Look at the graph of ZRESID plotted against ZPRED . If it looks like a random array of dots then this is good. If the dots seem to get more or less spread out over the graph (look like a funnel) then this is probably a violation of the assumption of homogeneity of variance. If the dots have a pattern to them (i.e. a curved shape) then this is probably a violation of the assumption of linearity. If the dots seem to have a pattern and are more spread out at some points on the plot than others then this probably reflects violations of both homogeneity of variance *and* linearity. Any of these scenarios puts the validity of your model into question. Repeat the above for all partial plots too.
- Look at histograms and P-P plots. If the histograms look like normal distributions (and the P-P plot looks like a diagonal line), then all is well. If the histogram looks non-normal and the P-P plot looks like a wiggly snake curving around a diagonal line then things are less good! Be warned, though: distributions can look very non-normal in small samples even when they are!

We could summarize by saying that the model appears, in most senses, to be both accurate for the sample and generalizable to the population. The only slight glitch is some concern over whether attractiveness ratings had violated the assumption of homoscedasticity. Therefore, we could conclude that in our sample advertising budget and airplay are fairly equally important in predicting record sales. Attractiveness of the band is a significant predictor of record sales but is less important than the other two predictors (and probably needs verification because of possible heteroscedasticity). The assumptions seem to have been met and so we can probably assume that this model would generalize to any record being released.

7.9. What if I violate an assumption? ②

It's worth remembering that you can have a perfectly good model for your data (no outliers, influential cases, etc.) and you can use that model to draw conclusions about your sample, even if your assumptions are violated. However, it's much more interesting to generalize your regression model and this is where assumptions become important. If they have been violated then you cannot generalize your findings beyond your sample. The options for correcting for violated assumptions are a bit limited. If residuals show problems with heteroscedasticity or non-normality you could try transforming the raw data – but this won't necessarily affect the residuals! If you have a violation of the linearity assumption then you could see whether you can do logistic regression instead (described in the next chapter). Finally, there are a series of robust regression techniques (see section 5.7.4), which are described extremely well by Rand Wilcox in Chapter 10 of his book. SPSS can't do these methods directly (well, technically it can do robust parameter estimates but it's not easy), but you can attempt a robust regression using some of Wilcox's files for the software R (Wilcox, 2005) using SPSS's R plugin.



OLIVER TWISTED

Please, Sir, can I have some more ... robust regression?

'R you going to show us how to do robust regression?,' mumbles Oliver as he dribbles down his shirt 'I want to do one.' I've prepared a flash movie on the companion website that shows you how to use the *R* plugin to do robust regression. You have no excuses now for not using it.

7.10. How to report multiple regression ②

If you follow the American Psychological Association guidelines for reporting multiple regression then the implication seems to be that tabulated results are the way forward. The APA also seem in favour of reporting, as a bare minimum, the standardized betas, their significance value and some general statistics about the model (such as the R^2). If you do decide to do a table then the beta values and their standard errors are also very useful. Personally I'd like to see the constant as well because then readers of your work can construct the full regression model if they need to. Also, if you've done a hierarchical regression you should report these values at each stage of the hierarchy. So, basically, you want to reproduce the table labelled **Coefficients** from the SPSS output and omit some of the non-essential information. For the example in this chapter we might produce a table like that in Table 7.2.

See if you can look back through the SPSS output in this chapter and work out from where the values came. Things to note are: (1) I've rounded off to 2 decimal places throughout; (2) for the standardized betas there is no zero before the decimal point (because these values cannot exceed 1) but for all other values less than 1 the zero is present; (3) the significance of the variable is denoted by an asterisk with a footnote to indicate the significance level being used (if there's more than one level of significance being used you can denote this with multiple asterisks, such as $*p < .05$, $**p < .01$, and $***p < .001$); and (4) the R^2 for the initial model and the change in R^2 (denoted as ΔR^2) for each subsequent step of the model are reported below the table.

TABLE 7.2 How to report multiple regression

	<i>B</i>	<i>SE B</i>	β
Step 1			
Constant	134.14	7.54	
Advertising Budget	0.10	0.01	.58*
Step 2			
Constant	-26.61	17.35	
Advertising Budget	0.09	0.01	.51*
Plays on BBC Radio 1	3.37	0.28	.51*
Attractiveness	11.09	2.44	.19*

Note: $R^2 = .34$ for Step 1, $\Delta R^2 = .33$ for Step 2 ($p < .001$). $*p < .001$.



LABCOAT LENI'S
REAL RESEARCH 7.1

*Why do you like your
lecturers?* ①

In the previous chapter we encountered a study by Chamorro-Premuzic et al. in which they measured students' personality characteristics and asked them to rate how much they wanted these same characteristics in their lecturers (see Labcoat Leni's Real Research 6.1 for a full description). In that chapter we correlated these scores; however, we could go a step further and see whether students' personality characteristics predict the characteristics that they would like to see in their lecturers.



The data from this study are in the file **Chamorro-Premuzic.sav**. Labcoat Leni wants you to carry out five multiple regression analyses: the outcome variable in each of the five analyses is the ratings of how much students want to see Neuroticism, Extroversion, Openness to experience, Agreeableness and Conscientiousness. For each of these outcomes, force Age and Gender into the analysis in the first step of the hierarchy, then in the second block force in the five student personality traits (Neuroticism, Extroversion, Openness to experience, Agreeableness and Conscientiousness). For each analysis create a table of the results.

Answers are in the additional material on the companion website (or look at Table 4 in the original article).

7.11. Categorical predictors and multiple regression ③

Often in regression analysis you'll collect data about groups of people (e.g. ethnic group, gender, socio-economic status, diagnostic category). You might want to include these groups as predictors in the regression model; however, we saw from our assumptions that variables need to be continuous or categorical with only two categories. We saw in section 6.5.5 that a point-biserial correlation is Pearson's r between two variables when one is continuous and the other has two categories coded as 0 and 1. We've also learnt that simple regression is based on Pearson's r , so it shouldn't take a great deal of imagination to see that, like the point-biserial correlation, we could construct a regression model with a predictor that has two categories (e.g. gender). Likewise, it shouldn't be too inconceivable that we could then extend this model to incorporate several predictors that had two categories. All that is important is that we code the two categories with the values of 0 and 1. Why is it important that there are only two categories and that they're coded 0 and 1? Actually, I don't want to get into this here because this chapter is already too long, the publishers are going to break my legs if it gets any longer, and I explain it anyway later in the book (sections 9.7 and 10.2.3) so, for the time being, just trust me!



SMART
ALEX
ONLY

7.11.1. Dummy coding ③

The obvious problem with wanting to use categorical variables as predictors is that often you'll have more than two categories. For example, if you'd measured religiosity you might have categories of Muslim, Jewish, Hindu, Catholic, Buddhist, Protestant, Jedi (for those of you not in the UK, we had a census here a few years back in which a significant portion of people put down Jedi as their religion). Clearly these groups cannot be distinguished using a single variable coded with zeros and ones. In these cases we have to use

what's called **dummy variables**. Dummy coding is a way of representing groups of people using only zeros and ones. To do it, we have to create several variables; in fact, the number of variables we need is one less than the number of groups we're recoding. There are eight basic steps:

- 1 Count the number of groups you want to recode and subtract 1.
- 2 Create as many new variables as the value you calculated in step 1. These are your dummy variables.
- 3 Choose one of your groups as a baseline (i.e. a group against which all other groups should be compared). This should usually be a control group, or, if you don't have a specific hypothesis, it should be the group that represents the majority of people (because it might be interesting to compare other groups against the majority).
- 4 Having chosen a baseline group, assign that group values of 0 for all of your dummy variables.
- 5 For your first dummy variable, assign the value 1 to the first group that you want to compare against the baseline group. Assign all other groups 0 for this variable.
- 6 For the second dummy variable assign the value 1 to the second group that you want to compare against the baseline group. Assign all other groups 0 for this variable.
- 7 Repeat this until you run out of dummy variables.
- 8 Place all of your dummy variables into the regression analysis!

Let's try this out using an example. In Chapter 4 we came across an example in which a biologist was worried about the potential health effects of music festivals. She collected some data at the Download Festival, which is a music festival specializing in heavy metal. The biologist was worried that the findings that she had were a function of the fact that she had tested only one type of person: metal fans. Perhaps it's not the festival that makes people smelly, maybe it's only metal fans at festivals who get smellier (as a metal fan, I would at this point sacrifice the biologist to Satan for being so prejudiced). Anyway, to answer this question she went to another festival that had a more eclectic clientele. So, she went to the Glastonbury Music Festival which attracts all sorts of people because it has all styles of music there. Again, she measured the hygiene of concert-goers over the three days of the festival using a technique that results in a score ranging between 0 (you smell like you've bathed in sewage) and 5 (you smell of freshly baked bread). Now, in Chapters 4 and 5, we just looked at the distribution of scores for the three days of the festival, but now the biologist wanted to look at whether the type of music you like (your cultural group) predicts whether hygiene decreases over the festival. The data are in the file called **GlastonburyFestivalRegression.sav**. This file contains the hygiene scores for each of three days of the festival, but it also contains a variable called **change** which is the change in hygiene over the three days of the festival (so it's the change from day 1 to day 3).¹³ Finally, the biologist categorized people according to their musical affiliation: if they mainly liked alternative music she called them 'indie kid', if they mainly liked heavy metal she called them a 'metaller' and if they mainly liked sort of hippy/folky/ambient type of stuff then she labelled them a 'crusty'. Anyone not falling into these categories was labelled 'no musical affiliation'. In the data file she coded these groups 1, 2, 3 and 4 respectively.



The first thing we should do is calculate the number of dummy variables. We have four groups, so there will be three dummy variables (one less than the number of groups). Next we need to choose a baseline group. We're interested in comparing those that have different musical affiliations against those that don't, so our baseline category will be 'no musical

¹³ Not everyone could be measured on day 3, so there is a change score only for a subset of the original sample.

TABLE 7.3 Dummy coding for the Glastonbury Festival data

	Dummy Variable 1	Dummy Variable 2	Dummy Variable 3
No Affiliation	0	0	0
Indie Kid	0	0	1
Metaller	0	1	0
Crusty	1	0	0

affiliation’. We give this group a code of 0 for all of our dummy variables. For our first dummy variable, we could look at the ‘crusty’ group, and to do this we give anyone that was a crusty a code of 1, and everyone else a code of 0. For our second dummy variable, we could look at the ‘metaller’ group, and to do this we give anyone that was a metaller a code of 1, and everyone else a code of 0. We have one dummy variable left and this will have to look at our final category: ‘indie kid’. To do this we give anyone that was an indie kid a code of 1, and everyone else a code of 0. The resulting coding scheme is shown in Table 7.3. The thing to note is that each group has a code of 1 on only one of the dummy variables (except the base category that is always coded as 0).

As I said, we’ll look at why dummy coding works in sections 9.7 and 10.2.3, but for the time being let’s look at how to recode our grouping variable into these dummy variables using SPSS. To recode variables you need to use the *Recode* function. Select **Transform** **→ Recode into Different Variables...** to access the dialog box in Figure 7.21.

The recode dialog box lists all of the variables in the data editor, and you need to select the one you want to recode (in this case **music**) and transfer it to the box labelled *Numeric Variable → Output Variable* by clicking on **→**. You then need to name the new variable (the output variable as SPSS calls it), so go to the part that says *Output Variable* and in the box below where it says *Name* write a name for your first dummy variable (you might call it **Crusty**). You can also give this variable a more descriptive name by typing something in the box labelled *Label* (for this first dummy variable I’ve called it **No Affiliation vs. Crusty**). When you’ve done this click on **Change** to transfer this new variable to the box labelled *Numeric Variable → Output Variable* (this box should now say *music → Crusty*).

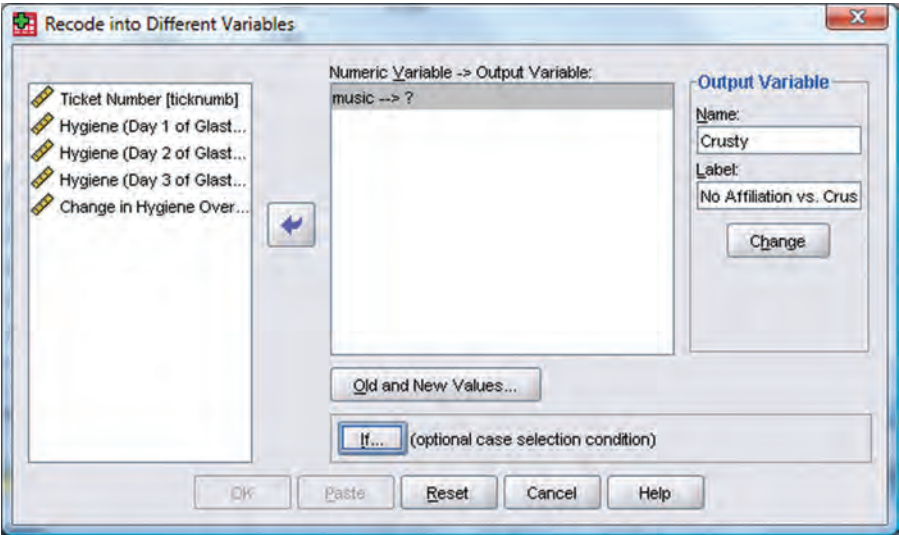


FIGURE 7.21
The recode
dialog box



OLIVER TWISTED

Please, Sir, can I have some more ... recode?

‘Our data set has missing values,’ worries Oliver. ‘What do we do if we only want to recode cases for which we have data?’ Well, we can set some other options at this point, that’s what we can do. This is getting a little bit more involved so if you want to know more, the additional material for this chapter on the companion website will tell you. Stop worrying Oliver, everything will be OK.

Having defined the first dummy variable, we need to tell SPSS how to recode the values of the variable **music** into the values that we want for the new variable, **music1**. To do this click on **Old and New Values...** to access the dialog box in Figure 7.22. This dialog box is used to change values of the original variable into different values for the new variable. For our first dummy variable, we want anyone who was a crusty to get a code of 1 and everyone else to get a code of 0. Now, crusty was coded with the value 3 in the original variable, so you need to type the value 3 in the section labelled *Old Value* in the box labelled *Value*. The new value we want is 1, so we need to type the value 1 in the section labelled *New Value* in the box labelled *Value*. When you’ve done this, click on **Add** to add this change to the list of changes (the list is displayed in the box labelled *Old → New*, which should now say 3 → 1 as in the diagram). The next thing we need to do is to change the remaining groups to have a value of 0 for the first dummy variable. To do this just select **All other values** and type the value 0 in the section labelled *New Value* in the box labelled *Value*.¹⁴ When you’ve done this, click on **Add** to add this change to the list of changes (this list will now also say *ELSE → 0*). When you’ve done this click on **Continue** to return to the main dialog box, and then click on **OK** to create the first dummy variable. This variable will appear as a new column in the data editor, and you should notice that it will have a value of 1 for anyone originally classified as a crusty and a value of 0 for everyone else.



SELF-TEST Try creating the remaining two dummy variables (call them **Metaller** and **Indie_Kid**) using the same principles.

7.11.2. SPSS output for dummy variables ③

Let’s assume you’ve created the three dummy coding variables (if you’re stuck there is a data file called **GlastonburyDummy.sav** (the ‘Dummy’ refers to the fact it has dummy variables in it – I’m not implying that if you need to use this file you’re a dummy!)). With dummy variables, you have to enter all related dummy variables in the same block (so use the *Enter* method).

¹⁴ Using this **All other values** option is fine when you don’t have missing values in the data, but just note that when you do (as is the case here) cases with both system-defined and user-defined missing values will be included in the recode. One way around this is to recode only cases for which there is a value (see Oliver Twisted Box). The alternative is specifically to recode missing values using the **Missing** option. It is also a good idea to use the *frequencies* or *crosstabs* commands after a recode and check that you have caught all of these missing values.

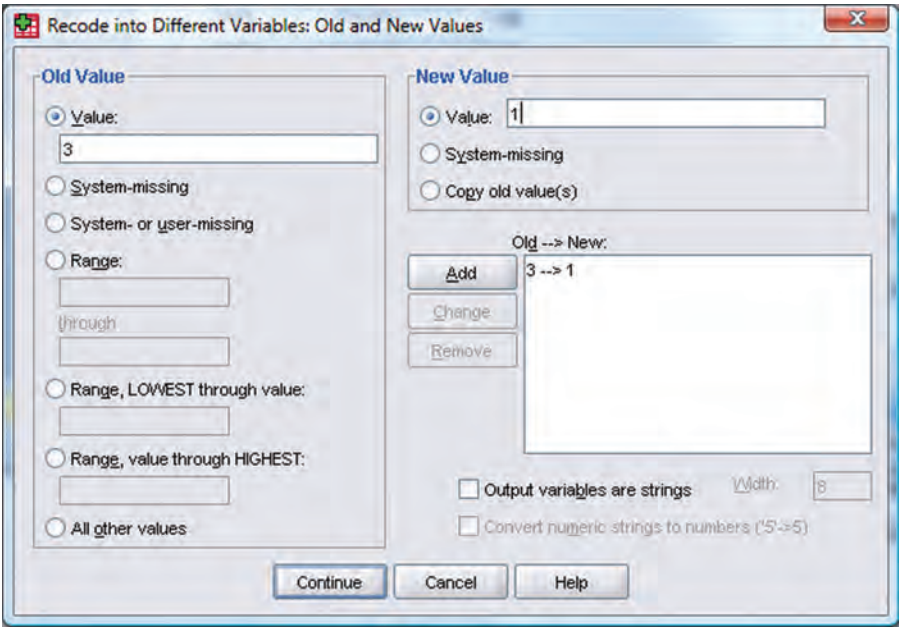


FIGURE 7.22
Dialog boxes
for the *Recode*
function (see
also SPSS
Tip 7.1)



SPSS TIP 7.1 Using syntax to recode ③



If you're doing a lot of recoding it soon becomes pretty tedious using the dialog boxes all of the time. I've written the syntax file, **RecodeGlastonburyData.sps**, to create all of the dummy variables we've discussed. Load this file and run the syntax, or open a syntax window (see Section 3.7) and type the following:

```
DO IF (1-MISSING(change)).  
RECODE music (3=1)(ELSE = 0) INTO Crusty.  
RECODE music (2=1)(ELSE = 0) INTO Metaller.  
RECODE music (1=1)(ELSE = 0) INTO Indie_Kid.  
END IF.  
VARIABLE LABELS Crusty 'No Affiliation vs. Crusty'.  
VARIABLE LABELS Metaller 'No Affiliation vs. Metaller'.  
VARIABLE LABELS Indie_Kid 'No Affiliation vs. Indie Kid'.  
VARIABLE LEVEL Crusty Metaller Indie_Kid (Nominal).  
FORMATS Crusty Metaller Indie_Kid (F1.0).  
EXECUTE.
```

Each RECODE command is doing the equivalent of what you'd do using the compute dialog box in Figure 7.22. So, the three lines beginning RECODE ask SPSS to create three new variables (**Crusty**, **Metaller** and **Indie_Kid**), which are based on the original variable **music**. For the first variable, if **music** is 3 then it becomes 1, and every other value becomes 0. For the second, if **music** is 2 then it becomes 1, and every other value becomes

(Continued)

(Continued)

0, and so on for the third dummy variable. Note that all of these RECODE commands are within an IF statement (beginning DO IF and ending with END IF). This tells SPSS to carry out the RECODE commands only if a certain condition is met. The condition we have set is (1- MISSING(change)). MISSING is a built-in command that returns 'true' (i.e. the value 1) for a case that has a system- or user-defined missing value for the specified variable; it returns 'false' (i.e. the value 0) if a case has a value. Hence, MISSING(change) returns a value of 1 for cases that have a missing value for the variable 'change' and 0 for cases that do have values. We want to recode the cases that *do* have a value for the variable change, therefore I have specified '1- MISSING(change)'. This command reverses MISSING(change) so that it returns 1 (true) for cases that have a value for the variable **change** and 0 (false) for system- or user-defined missing values. To sum up, the DO IF (1- MISSING(change)) tells SPSS 'Do the following RECODE commands if the case has a value for the variable **change**.'

The lines beginning VARIABLE LABELS just tells SPSS to assign the text in the quotations as labels for the variables **Crusty**, **Metaller** and **Indie_Kid** respectively. The line beginning VARIABLE LEVEL then sets these three variables to be 'nominal', and the line beginning FORMATS changes the three variables to have a width of 1 and 0 decimal places (hence the 1.0) – in other words, it changes the format to be a binary number.

The final line has the command EXECUTE without which none of the commands beforehand will be executed! Note also that every line ends with a full stop.

So, in this case we have to enter our dummy variables in the same block; however, if we'd had another variable (e.g. socio-economic status) that had been transformed into dummy variables, we could enter these dummy variables in a different block (so, it's only dummy variables that have recoded the same variable that need to be entered in the same block).



SELF-TEST Use what you've learnt in this chapter to run a multiple regression using the change scores as the outcome, and the three dummy variables (entered in the same block) as predictors.

Let's have a look at the output.

SPSS OUTPUT 7.12

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics					Durbin-Watson
					R Square Change	F Change	df1	df2	Sig. F Change	
1	.276 ^a	.076	.053	.68818	.076	3.270	3	119	.024	1.893

a. Predictors: (Constant), No Affiliation vs. Indie Kid, No Affiliation vs. Crusty, No Affiliation vs. Metaller
b. Dependent Variable: Change in Hygiene Over The Festival

ANOVA^b

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	4.646	3	1.549	3.270	.024 ^a
	Residual	56.358	119	.474		
	Total	61.004	122			

a. Predictors: (Constant), No Affiliation vs. Indie Kid, No Affiliation vs. Crusty, No Affiliation vs. Metaller
b. Dependent Variable: Change in Hygiene Over The Festival

SPSS Output 7.12 shows the model statistics. This shows that by entering the three dummy variables we can explain 7.6% of the variance in the change in hygiene scores (the R^2 -value $\times$ 100). In other words, 7.6% of the variance in the change in hygiene can be explained by the musical affiliation of the person. The ANOVA (which shows the same thing as the R^2 change statistic because there is only one step in this regression) tells us that the model is significantly better at predicting the change in hygiene scores than having no model (or, put another way, the 7.6% of variance that can be explained is a significant amount). Most of this should be clear from what you’ve read in this chapter already; what’s more interesting is how we interpret the individual dummy variables.

SPSS OUTPUT 7.13

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-.554	.090		-6.134	.000
	No Affiliation vs. Crusty	-.412	.167	-.232	-2.464	.015
	No Affiliation vs. Metaller	.028	.160	.017	.177	.860
	No Affiliation vs. Indie Kid	-.410	.205	-.185	-2.001	.048

a. Dependent Variable: Change in Hygiene Over The Festival

SPSS Output 7.13 shows a basic **Coefficients** table for the dummy variables (I’ve excluded the confidence intervals and collinearity diagnostics). The first thing to notice is that each dummy variable appears in the table with a useful label (such as No Affiliation vs. Crusty). This has happened because when we recoded our variables we gave each variable a label; if we hadn’t done this then the table would contain the rather less helpful variable names (crusty, metaller and Indie_Kid). The labels that I suggested giving to each variable give us a hint about what each dummy variable represents. The first dummy variable (No Affiliation vs. Crusty) shows the difference between the change in hygiene scores for the no affiliation group and the crusty group. Remember that the beta value tells us the change in the outcome due to a unit change in the predictor. In this case, a unit change in the predictor is the change from 0 to 1. As such it shows the shift in the change in hygiene scores that results from the dummy variable changing from 0 to 1 (Crusty). By including all three dummy variables at the same time, our baseline category is always zero, so this actually represents the difference in the change in hygiene scores if a person has no musical affiliation, compared to someone who is a crusty. This difference is the difference between the two group means.

To illustrate this fact, I’ve produced a table (SPSS Output 7.14) of the group means for each of the four groups and also the difference between the means for each group and the no affiliation group. These means represent the average change in hygiene scores for the three groups (i.e. the mean of each group on our outcome variable). If we calculate the difference in these means for the No Affiliation group and the crusty group we get, Crusty – no affiliation = $(-0.966) - (-0.554) = -0.412$. In other words, the change in hygiene scores is greater for the crusty group than it is for the no affiliation group (crusties’ hygiene decreases more over the festival than those with no musical affiliation). This value is the same as the *unstandardized* beta value in SPSS Output 7.13! So, the beta values tell us the relative difference between each group and the group that we chose as a baseline category. This beta value is converted to a *t*-statistic and the significance of this *t* reported. This *t*-statistic is testing, as we’ve seen before, whether the beta value is 0 and when we have two categories coded with 0 and 1, that means it’s testing whether the difference between group means is 0. If it is significant then it means that the group coded with 1 is significantly different from the baseline category – so, it’s testing the difference between

two means, which is the context in which students are most familiar with the *t*-statistic (see Chapter 9). For our first dummy variable, the *t*-test is significant, and the beta value has a negative value so we could say that the change in hygiene scores goes down as a person changes from having no affiliation to being a crusty. Bear in mind that a decrease in hygiene scores represents more change (you’re becoming smellier) so what this actually means is that hygiene decreased significantly more in crusties compared to those with no musical affiliation!

SPSS OUTPUT 7.14

OLAP Cubes			
Variables=Change in Hygiene Over The Festival			
Musical Affiliation	Mean	Std. Deviation	N
Indie Kid	-0.964	0.670	14
Metaller	-0.526	0.576	27
Crusty	-0.966	0.760	24
No Musical Affiliation	-0.554	0.708	58
Crusty - No Musical Affiliation	-0.412	0.052	-34
Metaller - No Musical Affiliation	0.028	-0.133	-31
Indie Kid - No Musical Affiliation	-0.410	-0.038	-44
Total	-0.675	0.707	123

Moving on to our next dummy variable, this compares metallers to those that have no musical affiliation. The beta value again represents the shift in the change in hygiene scores if a person has no musical affiliation, compared to someone who is a metaller. If we calculate the difference in the group means for the no affiliation group and the metaller group we get, metaller – no affiliation = $(-0.526) - (-0.554) = 0.028$. This value is again the same as the unstandardized beta value in SPSS Output 7.13! For this second dummy variable, the *t*-test is not significant, so we could say that the change in hygiene scores is the same if a person changes from having no affiliation to being a metaller. In other words, the change in hygiene scores is not predicted by whether someone is a metaller compared to if they have no musical affiliation.

For the final dummy variable, we’re comparing indie kids to those that have no musical affiliation. The beta value again represents the shift in the change in hygiene scores if a person has no musical affiliation, compared to someone who is an indie kid. If we calculate the difference in the group means for the no affiliation group and the indie kid group we get, indie kid – no affiliation = $(-0.964) - (-0.554) = -0.410$. It should be no surprise to you by now that this is the unstandardized beta value in SPSS Output 7.13! The *t*-test is significant, and the beta value has a negative value so, as with the first dummy variable, we could say that the change in hygiene scores goes down as a person changes from having no affiliation to being an indie kid. Bear in mind that a decrease in hygiene scores represents more change (you’re becoming smellier) so what this actually means is that hygiene decreased significantly more in indie kids compared to those with no musical affiliation!

So, overall this analysis has shown that compared to having no musical affiliation, crusties and indie kids get significantly smellier across the three days of the festival, but metallers don’t. This section has introduced some really complex ideas that I expand upon in Chapters 9 and 10. It might all be a bit much to take in, and so if you’re confused or want to know more about why dummy coding works in this way I suggest reading sections 9.7 and 10.2.3 and then coming back here. Alternatively, read Hardy’s (1993) excellent monograph!



What have I discovered about statistics? ①

This chapter is possibly the longest book chapter ever written, and if you feel like you aged several years while reading it then, well, you probably have (look around, there are cobwebs in the room, you have a long beard, and when you go outside you'll discover a second ice age has been and gone leaving only you and a few woolly mammoths to populate the planet). However, on the plus side, you now know more or less everything you ever need to know about statistics. Really, it's true; you'll discover in the coming chapters that everything else we discuss is basically a variation on the theme of regression. So, although you may be near death having spent your life reading this chapter (and I'm certainly near death having written it) you are a stats genius – it's official!

We started the chapter by discovering that at 8 years old I could have really done with regression analysis to tell me which variables are important in predicting talent competition success. Unfortunately I didn't have regression, but fortunately I had my dad instead (and he's better than regression). We then looked at how we could use statistical models to make similar predictions by looking at the case of when you have one predictor and one outcome. This allowed us to look at some basic principles such as the equation of a straight line, the method of least squares, and how to assess how well our model fits the data using some important quantities that you'll come across in future chapters: the model sum of squares, SS_M , the residual sum of squares, SS_R , and the total sum of squares, SS_T . We used these values to calculate several important statistics such as R^2 and the F -ratio. We also learnt how to do a regression on SPSS, and how we can plug the resulting beta values into the equation of a straight line to make predictions about our outcome.

Next, we saw that the question of a straight line can be extended to include several predictors and looked at different methods of placing these predictors in the model (hierarchical, forced entry, stepwise). Next, we looked at factors that can affect the accuracy of a model (outliers and influential cases) and ways to identify these factors. We then moved on to look at the assumptions necessary to generalize our model beyond the sample of data we've collected before discovering how to do the analysis on SPSS, and how to interpret the output, create our multiple regression model and test its reliability and generalizability. I finished the chapter by looking at how we can use categorical predictors in regression (and in passing we discovered the *recode* function). In general, multiple regression is a long process and should be done with care and attention to detail. There are a lot of important things to consider and you should approach the analysis in a systematic fashion. I hope this chapter helps you to do that!

So, I was starting to get a taste for the rock-idol lifestyle: I had friends, a fortune (well, two gold-plated winner's medals), fast cars (a bike) and dodgy-looking 8 year olds were giving me suitcases full of lemon sherbet to lick off mirrors. However, my parents and teachers were about to impress reality upon my young mind ...

Key terms that I've discovered

Adjusted predicted value
Adjusted R^2

Autocorrelation
 b_i

β_1	Multiple R
Cook's distance	Multiple regression
Covariance ratio (CVR)	Outcome variable
Cross-validation	Perfect collinearity
Deleted residual	Predictor variable
DFBeta	Residual
DFFit	Residual sum of squares
Dummy variables	Shrinkage
Durbin–Watson test	Simple regression
F-ratio	Standardized DFBeta
Generalization	Standardized DFFit
Goodness of fit	Standardized residuals
Hat values	Stepwise regression
Heteroscedasticity	Studentized deleted residuals
Hierarchical regression	Studentized residuals
Homoscedasticity	Suppressor effects
Independent errors	t-statistic
Leverage	Tolerance
Mahalanobis distances	Total sum of squares
Mean squares	Unstandardized residuals
Model sum of squares	Variance inflation factor (VIF)
Multicollinearity	

Smart Alex's tasks



- **Task 1:** A fashion student was interested in factors that predicted the salaries of catwalk models. She collected data from 231 models. For each model she asked them their salary per day on days when they were working (**salary**), their age (**age**), how many years they had worked as a model (**years**), and then got a panel of experts from modelling agencies to rate the attractiveness of each model as a percentage with 100% being perfectly attractive (**beauty**). The data are in the file **Supermodel.sav**. Unfortunately, this fashion student bought some substandard statistics text and so doesn't know how to analyse her data. 😊 Can you help her out by conducting a multiple regression to see which variables predict a model's salary? How valid is the regression model? ②
- **Task 2:** Using the Glastonbury data from this chapter (with the dummy coding in **GlastonburyDummy.sav**), which you should've already analysed, comment on whether you think the model is reliable and generalizable. ③
- **Task 3:** A study was carried out to explore the relationship between **Aggression** and several potential predicting factors in 666 children that had an older sibling. Variables measured were **Parenting_Style** (high score = bad parenting practices), **Computer_Games** (high score = more time spent playing computer games), **Television** (high score = more time spent watching television), **Diet** (high score = the child has a good diet low in E-numbers), and **Sibling_Aggression** (high score = more aggression seen in their older sibling). Past research indicated that parenting style and sibling aggression were good predictors of the level of aggression in the younger child. All other variables were treated in an exploratory fashion. The data are in the file **Child Aggression.sav**. Analyse them with multiple regression. ②



Answers can be found on the companion website.

Further reading

- Bowerman, B. L., & O'Connell, R. T. (1990). *Linear statistical models: An applied approach* (2nd ed.). Belmont, CA: Duxbury. (This text is only for the mathematically minded or postgraduate students but provides an extremely thorough exposition of regression analysis.)
- Hardy, M. A. (1993). *Regression with dummy variables*. Sage university paper series on quantitative applications in the social sciences, 07–093. Newbury Park, CA: Sage.
- Howell, D. C. (2006). *Statistical methods for psychology* (6th ed.). Belmont, CA: Duxbury. (Or you might prefer his *Fundamental Statistics for the Behavioral Sciences*, also in its 6th edition, 2007. Both are excellent introductions to the mathematics behind regression analysis.)
- Miles, J. N. V., & Shevlin, M. (2001). *Applying regression and correlation: a guide for students and researchers*. London: Sage. (This is an extremely readable text that covers regression in loads of detail but with minimum pain – highly recommended.)
- Stevens, J. (2002). *Applied multivariate statistics for the social sciences* (4th ed.). Hillsdale, NJ: Erlbaum. Chapter 3.

Online tutorial

The companion website contains the following Flash movie tutorials to accompany this chapter:

- Regression using SPSS
- Robust Regression

Interesting real research

- Chamorro-Premuzic, T., Furnham, A., Christopher, A. N., Garwood, J., & Martin, N. (2008). Birds of a feather: Students' preferences for lecturers' personalities as predicted by their own personality and learning approaches. *Personality and Individual Differences*, 44, 965–976.

8

Logistic regression

FIGURE 8.1

Practising for my career as a rock star by slaying the baying throng of Grove Primary School at the age of 10. (Note the girl with her hands covering her ears)



8.1. What will this chapter tell me? ①

We saw in the previous chapter that I had successfully conquered the holiday camps of Wales with my singing and guitar playing (and the Welsh know a thing or two about good singing). I had jumped on a snowboard called oblivion and thrown myself down the black run known as world domination. About 10 metres after starting this slippery descent I hit the lumpy patch of ice called ‘adults’. I was 9, life was fun, and yet every adult that I seemed to encounter was obsessed with my future. ‘What do you want to be when you grow up?’ they would ask. I was 9 and ‘grown-up’ was a lifetime away; all I knew was that I was going to marry Clair Sparks (more on her in the next chapter) and that I was a rock legend who didn’t need to worry about such adult matters as having a job. It was a difficult question, but adults require answers and I wasn’t going to let them know that I didn’t care about ‘grown-up’ matters. We saw in the previous chapter that we can use regression to predict future outcomes based on past data, when the outcome is a continuous variable,

but this question had a categorical outcome (e.g. would I be a fireman, a doctor, a pimp?). Luckily, though, we can use an extension of regression called logistic regression to deal with these situations. What a result; bring on the rabid wolves of categorical data. To make a prediction about a categorical outcome then, as with regression, I needed to draw on past data: I hadn't tried conducting brain surgery, neither had I experience of sentencing psychopaths to prison sentences for eating their husbands, nor had I taught anyone. I had, however, had a go at singing and playing guitar; 'I'm going to be a rock star' was my prediction. A prediction can be accurate (which would mean that I *am* a rock star) or it can be inaccurate (which would mean that I'm writing a statistics textbook). This chapter looks at the theory and application of **logistic regression**, an extension of regression that allows us to predict categorical outcomes based on predictor variables.

8.2. Background to logistic regression ①

In a nutshell, logistic regression is multiple regression but with an outcome variable that is a categorical variable and predictor variables that are continuous or categorical. In its simplest form, this means that we can predict which of two categories a person is likely to belong to given certain other information. A trivial example is to look at which variables predict whether a person is male or female. We might measure laziness, pig-headedness, alcohol consumption and number of burps that a person does in a day. Using logistic regression, we might find that all of these variables predict the gender of the person, but the technique will also allow us to predict whether a new person is likely to be male or female. So, if we picked a random person and discovered they scored highly on laziness, pig-headedness, alcohol consumption and the number of burps, then the regression model might tell us that, based on this information, this person is likely to be male. Admittedly, it is unlikely that a researcher would ever be interested in the relationship between flatulence and gender (it is probably too well established by experience to warrant research!), but logistic regression can have life-saving applications. In medical research logistic regression is used to generate models from which predictions can be made about the likelihood that a tumour is cancerous or benign (for example). A database of patients can be used to establish which variables are influential in predicting the malignancy of a tumour. These variables can then be measured for a new patient and their values placed in a logistic regression model, from which a probability of malignancy could be estimated. If the probability value of the tumour being malignant is suitably low then the doctor may decide not to carry out expensive and painful surgery that in all likelihood is unnecessary. We might not face such life-threatening decisions but logistic regression can nevertheless be a very useful tool. When we are trying to predict membership of only two categorical outcomes the analysis is known as **binary logistic regression**, but when we want to predict membership of more than two categories we use **multinomial (or polychotomous) logistic regression**.

8.3. What are the principles behind logistic regression? ③

I don't wish to dwell on the underlying principles of logistic regression because they aren't necessary to understand the test (I am living proof of this fact). However, I do wish to draw a few parallels to normal regression so that you can get the gist of what's going on using a framework that will be familiar to you already (what do you mean you haven't read the regression chapter yet?!). To keep things simple I'm going to explain binary logistic

regression, but most of the principles extend easily to when there are more than two outcome categories. Now would be a good time for the equation-phobes to look away. In simple linear regression, we saw that the outcome variable Y is predicted from the equation of a straight line:

$$Y_i = b_0 + b_1X_{1i} + \varepsilon_i \quad (8.1)$$

in which b_0 is the Y intercept, b_1 is the gradient of the straight line, X_1 is the value of the predictor variable and ε is a residual term. Given the values of Y and X_1 , the unknown parameters in the equation can be estimated by finding a solution for which the squared distance between the observed and predicted values of the dependent variable is minimized (the method of least squares).

This stuff should all be pretty familiar by now. In multiple regression, in which there are several predictors, a similar equation is derived in which each predictor has its own coefficient. As such, Y is predicted from a combination of each predictor variable multiplied by its respective regression coefficient.

$$Y_i = b_0 + b_1X_{1i} + b_2X_{2i} + \dots + b_nX_{ni} + \varepsilon_i \quad (8.2)$$

in which b_n is the regression coefficient of the corresponding variable X_n . In logistic regression, instead of predicting the value of a variable Y from a predictor variable X_1 or several predictor variables (X s), we predict the *probability* of Y occurring given known values of X_1 (or X s). The logistic regression equation bears many similarities to the regression equations just described. In its simplest form, when there is only one predictor variable X_1 , the logistic regression equation from which the probability of Y is predicted is given by:

$$P(Y) = \frac{1}{1 + e^{-(b_0 + b_1X_{1i})}} \quad (8.3)$$

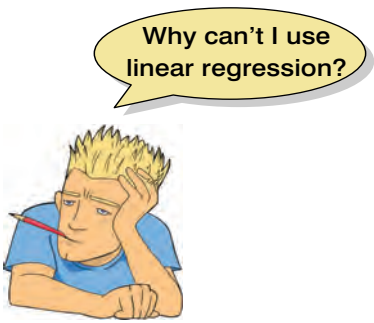
in which $P(Y)$ is the probability of Y occurring, e is the base of natural logarithms, and the other coefficients form a linear combination much the same as in simple regression. In fact, you might notice that the bracketed portion of the equation is identical to the linear regression equation in that there is a constant (b_0), a predictor variable (X_1) and a coefficient (or weight) attached to that predictor (b_1). Just like linear regression, it is possible to extend this equation so as to include several predictors. When there are several predictors the equation becomes:

$$P(Y) = \frac{1}{1 + e^{-(b_0 + b_1X_{1i} + b_2X_{2i} + \dots + b_nX_{ni})}} \quad (8.4)$$

Equation (8.4) is the same as the equation used when there is only one predictor except that the linear combination has been extended to include any number of predictors. So, whereas the one-predictor version of the logistic regression equation contained the simple linear regression equation within it, the multiple-predictor version contains the multiple regression equation.

Despite the similarities between linear regression and logistic regression, there is a good reason why we cannot apply linear regression directly to a situation in which the outcome variable is categorical. The reason is that one of the assumptions of linear regression is that the

relationship between variables is linear (see section 7.6.2.1). In that section we saw how important it is that the assumptions of a model are met for it to be accurate. Therefore, for linear regression to be a valid model, the observed data should contain a linear relationship. When the outcome variable is categorical, this assumption is violated (Berry, 1993). One way around this problem is to transform the data using the logarithmic transformation (see Berry & Feldman, 1985, and Chapter 5). This transformation is a way of expressing a non-linear relationship in a linear way. The logistic regression equation described above is based on this principle: it expresses the multiple linear regression equation in logarithmic terms (called the *logit*) and thus overcomes the problem of violating the assumption of linearity.



The exact form of the equation can be arranged in several ways but the version I have chosen expresses the equation in terms of the probability of Y occurring (i.e. the probability that a case belongs in a certain category). The resulting value from the equation, therefore, varies between 0 and 1. A value close to 0 means that Y is very unlikely to have occurred, and a value close to 1 means that Y is very likely to have occurred. Also, just like linear regression, each predictor variable in the logistic regression equation has its own coefficient. When we run the analysis we need to estimate the value of these coefficients so that we can solve the equation. These parameters are estimated by fitting models, based on the available predictors, to the observed data. The chosen model will be the one that, when values of the predictor variables are placed in it, results in values of Y closest to the observed values. Specifically, the values of the parameters are estimated using **maximum-likelihood estimation**, which selects coefficients that make the observed values most likely to have occurred. So, as with multiple regression, we try to fit a model to our data that allows us to estimate values of the outcome variable from known values of the predictor variable or variables.

8.3.1. Assessing the model: the log-likelihood statistic ③

We've seen that the logistic regression model predicts the probability of an event occurring for a given person (we would denote this as $P(Y_i)$ the probability that Y occurs for the i th person), based on observations of whether or not the event did occur for that person (we could denote this as Y_i , the actual outcome for the i th person). So, for a given person, Y will be either 0 (the outcome didn't occur) or 1 (the outcome did occur), and the predicted value, $P(Y)$, will be a value between 0 (there is no chance that the outcome will occur) and 1 (the outcome will certainly occur). We saw in multiple regression that if we want to assess whether a model fits the data we can compare the observed and predicted values of the outcome (if you remember, we use R^2 , which is the Pearson correlation between observed values of the outcome and the values predicted by the regression model). Likewise, in logistic regression, we can use the observed and predicted values to assess the fit of the model. The measure we use is the **log-likelihood**:

$$\text{log-likelihood} = \sum_{i=1}^N [Y_i \ln(P(Y_i)) + (1 - Y_i) \ln(1 - P(Y_i))] \tag{8.5}$$

The log-likelihood is based on summing the probabilities associated with the predicted and actual outcomes (Tabachnick & Fidell, 2007). The log-likelihood statistic is analogous to the residual sum of squares in multiple regression in the sense that it is an indicator of how much unexplained information there is after the model has been fitted. It, therefore, follows that large values of the log-likelihood statistic indicate poorly fitting statistical models, because the larger the value of the log-likelihood, the more unexplained observations there are.

Now, it's possible to calculate a log-likelihood for different models and to compare these models by looking at the difference between their log-likelihoods. One use of this is to compare the state of a logistic regression model against some kind of baseline state. The baseline state that's usually used is the model when only the constant is included. In multiple regression, the baseline model we use is the mean of all scores (this is our best guess of the outcome when we have no other information). In logistic regression, if we want to predict the outcome, what would our best guess be? Well, we can't use the mean score because our outcome is made of zeros and ones and so the mean is meaningless! However, if we know the frequency of zeros and ones, then the best guess will be the category with the largest number of cases. So, if the outcome occurs 107 times, and doesn't occur only 72 times, then our best guess of the outcome will be that it occurs (because it occurs 107 times compared to only 72 times when it doesn't occur). As such, like multiple regression, our baseline model is the model that gives us the best prediction when we know nothing other than the values of the outcome: in logistic regression this will be to predict the outcome that occurs most often. This is, the logistic regression model when only the constant is included. If we then add one or more predictors to the model, we can compute the improvement of the model as follows:

$$\chi^2 = 2[LL(\text{new}) - LL(\text{baseline})]$$

(8.6)

$(df = k_{\text{new}} - k_{\text{baseline}})$

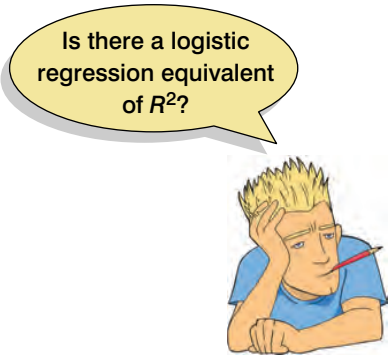
So, we merely take the new model and subtract from it the baseline model (the model when only the constant is included). You'll notice that we multiply this value by 2; this is because it gives the result a chi-square distribution (see Chapter 18 and the Appendix) and so makes it easy to calculate the significance of the value. The chi-square distribution we use has degrees of freedom equal to the number of parameters, k in the new model minus the number of parameters in the baseline model. The number of parameters in the baseline model will always be 1 (the constant is the only parameter to be estimated); any subsequent model will have degrees of freedom equal to the number of predictors plus 1 (i.e. the number of predictors plus one parameter representing the constant).

8.3.2. Assessing the model: R and R^2 ③

When we talked about linear regression, we saw that the multiple correlation coefficient R and the corresponding R^2 -value were useful measures of how well the model fits the data.

We've also just seen that the likelihood ratio is similar in the respect that it is based on the level of correspondence between predicted and actual values of the outcome. However, you can calculate a more literal version of the multiple correlation in logistic regression known as the R -statistic. This R -statistic is the partial correlation between the outcome variable and each of the predictor variables and it can vary between -1 and 1 . A positive value indicates, that as the predictor variable increases, so does the likelihood of the event occurring. A negative value implies that as the predictor variable increases, the likelihood of the outcome occurring decreases. If a variable has a small value of R then it contributes only a small amount to the model.

The equation for R is given in equation (8.7). The $-2LL$ is the -2 log-likelihood for the original model, the Wald statistic is calculated as described in the next section, and the degrees of freedom can be read from the summary table for the



variables in the equation. However, because this value of R is dependent upon the Wald statistic it is by no means an accurate measure (we'll see in the next section that the Wald statistic can be inaccurate under certain circumstances). For this reason the value of R should be treated with some caution, and it is invalid to square this value and interpret it as you would in linear regression:

$$R = \pm \sqrt{\left(\frac{\text{Wald} - (2 \times df)}{-2LL(\text{original})} \right)} \quad (8.7)$$

There is some controversy over what would make a good analogue to the R^2 -value in linear regression, but one measure described by Hosmer and Lemeshow (1989) can be easily calculated. In SPSS terminology, **Hosmer and Lemeshow's R_L^2** measure is calculated as:

$$R_L^2 = \frac{-2LL(\text{model})}{-2LL(\text{original})} \quad (8.8)$$

As such, R_L^2 is calculated by dividing the model chi-square (based on the log-likelihood) by the *original* $-2LL$ (the log-likelihood of the model before any predictors were entered). R_L^2 is the proportional reduction in the absolute value of the log-likelihood measure and as such it is a measure of how much the badness of fit improves as a result of the inclusion of the predictor variables. It can vary between 0 (indicating that the predictors are useless at predicting the outcome variable) and 1 (indicating that the model predicts the outcome variable perfectly).

However, this is not the measure used by SPSS. SPSS uses **Cox and Snell's R_{CS}^2** (1989), which is based on the log-likelihood of the model ($LL(\text{new})$) and the log-likelihood of the original model ($LL(\text{baseline})$), and the sample size, n :

$$R_{CS}^2 = 1 - e^{\left[-\frac{2}{n}(LL(\text{new})) - (LL(\text{baseline})) \right]} \quad (8.9)$$

However, this statistic never reaches its theoretical maximum of 1. Therefore, Nagelkerke (1991) suggested the following amendment (**Nagelkerke's R_N^2**):

$$R_N^2 = \frac{R_{CS}^2}{1 - e^{\left[\frac{2(LL(\text{baseline}))}{n} \right]}} \quad (8.10)$$

Although all of these measures differ in their computation (and the answers you get), conceptually they are somewhat the same. So, in terms of interpretation they can be seen as similar to the R^2 in linear regression in that they provide a gauge of the substantive significance of the model.

8.3.3. Assessing the contribution of predictors: the Wald statistic ②

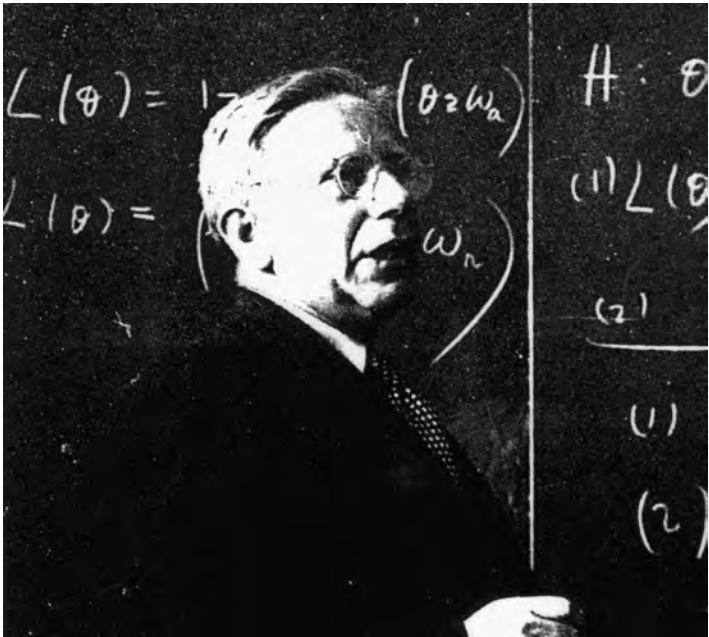
As in linear regression, we want to know not only how well the model overall fits the data, but also the individual contribution of predictors. In linear regression, we used the estimated regression coefficients (b) and their standard errors to compute a t -statistic. In logistic regression there is an analogous statistic known as the **Wald statistic**, which has a special distribution known as the **chi-square distribution**. Like the t -test in linear regression,

the Wald statistic tells us whether the b coefficient for that predictor is significantly different from zero. If the coefficient is significantly different from zero then we can assume that the predictor is making a significant contribution to the prediction of the outcome (Y):

$$\text{Wald} = \frac{b}{\text{SE}_b} \tag{8.11}$$

Equation (8.11) shows how the Wald statistic is calculated and you can see it's basically identical to the t -statistic in linear regression (see equation (7.6)): it is the value of the regression coefficient divided by its associated standard error. The Wald statistic (Figure 8.2) is usually used to ascertain whether a variable is a significant predictor of the outcome; however, it is probably more accurate to examine the likelihood ratio statistics. The reason why the Wald statistic should be used cautiously is because, when the regression coefficient (b) is large, the standard error tends to become inflated, resulting in the Wald statistic being underestimated (see Menard, 1995). The inflation of the standard error increases the probability of rejecting a predictor as being significant when in reality it is making a significant contribution to the model (i.e. you are more likely to make a Type II error).

FIGURE 8.2
Abraham Wald writing 'I must not devise test statistics prone to having inflated standard errors' on the blackboard 100 times



8.3.4. The odds ratio: $\text{Exp}(B)$ ③

More crucial to the *interpretation* of logistic regression is the value of the odds ratio ($\text{Exp}(B)$ in the SPSS output), which is an indicator of the change in odds resulting from a unit change in the predictor. As such, it is similar to the b coefficient in logistic regression but easier to understand (because it doesn't require a logarithmic transformation). When the predictor variable is categorical the odds ratio is easier to explain, so imagine we had a simple example in which we were trying to predict whether or not someone got pregnant from whether or not they used a condom last time they made love. The **odds** of an event occurring are defined as the probability of an event occurring divided by the probability of that event not occurring

(see equation (8.12)) and should not be confused with the more colloquial usage of the word to refer to probability. So, for example, the odds of becoming pregnant are the probability of becoming pregnant divided by the probability of not becoming pregnant:

$$\text{odds} = \frac{P(\text{event})}{P(\text{no event})}$$

$$P(\text{event } Y) = \frac{1}{1 + e^{-(b_0 + b_1 X_1)}} \quad (8.12)$$

$$P(\text{no event } Y) = 1 - P(\text{event } Y)$$

To calculate the change in odds that results from a unit change in the predictor, we must first calculate the odds of becoming pregnant given that a condom *wasn't* used. We then calculate the odds of becoming pregnant given that a condom *was* used. Finally, we calculate the proportionate change in these two odds.

To calculate the first set of odds, we need to use equation (8.3) to calculate the probability of becoming pregnant given that a condom wasn't used. If we had more than one predictor we would use equation (8.4). There are three unknown quantities in this equation: the coefficient of the constant (b_0), the coefficient for the predictor (b_1) and the value of the predictor itself (X). We'll know the value of X from how we coded the condom use variable (chances are we would've used 0 = condom wasn't used and 1 = condom was used). The values of b_1 and b_0 will be estimated for us. We can calculate the odds as in equation (8.12).

Next, we calculate the same thing after the predictor variable has changed by one unit. In this case, because the predictor variable is dichotomous, we need to calculate the odds of getting pregnant, given that a condom *was* used. So, the value of X is now 1 (rather than 0).

We now know the odds before and after a unit change in the predictor variable. It is a simple matter to calculate the proportionate change in odds by dividing the odds after a unit change in the predictor by the odds before that change:

$$\Delta \text{odds} = \frac{\text{odds after a unit change in the predictor}}{\text{original odds}} \quad (8.13)$$

This proportionate change in odds is the odds ratio, and we can interpret it in terms of the change in odds: if the value is greater than 1 then it indicates that as the predictor increases, the odds of the outcome occurring increase. Conversely, a value less than 1 indicates that as the predictor increases, the odds of the outcome occurring decrease. We'll see how this works with a real example shortly.

8.3.5. Methods of logistic regression ②

As with multiple regression (section 7.5.3), there are several different methods that can be used in logistic regression.

8.3.5.1. The forced entry method ②

The default method of conducting the regression is 'enter'. This is the same as forced entry in multiple regression in that all of the predictors are placed into the regression model in one block, and parameter estimates are calculated for each block.

8.3.5.2. Stepwise methods ②

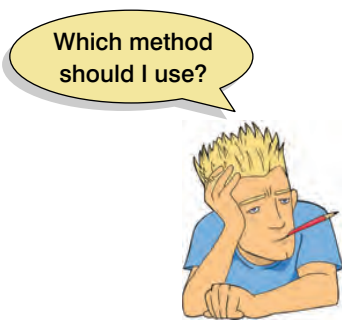
If you are undeterred by the criticisms of stepwise methods in the previous chapter, then you can select either a forward or a backward stepwise method. When the forward method is employed the computer begins with a model that includes only a constant and then adds single predictors to the model based on a specific criterion. This criterion is the value of the *score* statistic: the variable with the most significant score statistic is added to the model. The computer proceeds until none of the remaining predictors have a significant score statistic (the cut-off point for significance being .05). At each step, the computer also examines the variables in the model to see whether any should be removed. It does this in one of three ways. The first way is to use the likelihood ratio statistic described in section 18.3.3 (the *Forward:LR* method) in which case the current model is compared to the model when that predictor is removed. If the removal of that predictor makes a significant difference to how well the model fits the observed data, then the computer retains that predictor (because the model is better if the predictor is included). If, however, the removal of the predictor makes little difference to the model then the computer rejects that predictor. Rather than using the likelihood ratio statistic, which estimates how well the model fits the observed data, the computer could use the conditional statistic as a removal criterion (*Forward:Conditional*). This statistic is an arithmetically less intense version of the likelihood ratio statistic and so there is little to recommend it over the likelihood ratio method. The final criterion is the Wald statistic, in which case any predictors in the model that have significance values of the Wald statistic (above the default removal criterion of .1) will be removed. Of these methods the likelihood ratio method is the best removal criterion because the Wald statistic can, at times, be unreliable (see section 8.3.3).

The opposite of the forward method is the backward method. This method uses the same three removal criteria, but instead of starting the model with only a constant, it begins the model with all predictors included. The computer then tests whether any of these predictors can be removed from the model without having a substantial effect on how well the model fits the observed data. The first predictor to be removed will be the one that has the least impact on how the model fits the data.

8.3.5.3. How do I select a method? ②

As with ordinary regression (previous chapter), the method of regression chosen will depend on several things. The main consideration is whether you are testing a theory or merely carrying out exploratory work. As noted earlier, some people believe that stepwise methods have no value for theory testing. However, stepwise methods are defensible when used in situations in which no previous research exists on which to base hypotheses for testing, and in situations where causality is not of interest and you merely wish to find a model to fit your data (Agresti & Finlay, 1986; Menard, 1995). Also, as I mentioned for ordinary regression, if you do decide to use a stepwise method then the backward method is preferable to the forward method. This is because of **suppressor effects**, which occur when a predictor has a significant effect but only when another variable is held constant. Forward selection is more likely than backward elimination to exclude predictors

involved in suppressor effects. As such, the forward method runs a higher risk of making a Type II error. In terms of the test statistic used in stepwise methods, the Wald statistic, as we have seen, has a tendency to be inaccurate in certain circumstances and so the likelihood ratio method is best.



8.4. Assumptions and things that can go wrong ④

8.4.1. Assumptions ②

Logistic regression shares some of the assumptions of normal regression:

- 1 **Linearity:** In ordinary regression we assumed that the outcome had linear relationships with the predictors. In logistic regression the outcome is categorical and so this assumption is violated. As I explained before, this is why we use the log (or *logit*) of the data. The assumption of linearity in logistic regression, therefore, assumes that there is a linear relationship between any continuous predictors and the logit of the outcome variable. This assumption can be tested by looking at whether the interaction term between the predictor and its log transformation is significant (Hosmer & Lemeshow, 1989). We will go through an example in section 8.8.1.
- 2 **Independence of errors:** This assumption is the same as for ordinary regression (see section 7.6.2.1). Basically it means that cases of data should not be related; for example, you cannot measure the same people at different points in time. Violating this assumption produces overdispersion (see section 8.4.4).
- 3 **Multicollinearity:** Although not really an assumption as such, multicollinearity is a problem as it was for ordinary regression (see section 7.6.2.1). In essence, predictors should not be too highly correlated. As with ordinary regression, this assumption can be checked with tolerance and VIF statistics, the eigenvalues of the scaled, uncentred cross-products matrix, the condition indexes and the variance proportions. We go through an example in section 8.8.1.

Logistic regression also has some unique problems of its own (not assumptions, but things that can go wrong). SPSS solves logistic regression problems by an iterative procedure (SPSS Tip 8.1). Sometimes, instead of pouncing on the correct solution quickly, you'll notice nothing happening: SPSS begins to move infinitely slowly, or appears to have just got fed up with you asking it to do stuff and has gone on strike. If it can't find a correct solution, then sometimes it actually does give up, quietly offering you (without any apology) a result which is completely incorrect. Usually this is revealed by implausibly large standard errors. Two situations can provoke this situation, both of which are related to the ratio of cases to variables: incomplete information and complete separation.

8.4.2. Incomplete information from the predictors ④

Imagine you're trying to predict lung cancer from smoking (a foul habit believed to increase the risk of cancer) and whether or not you eat tomatoes (which are believed to reduce the risk of cancer). You collect data from people who do and don't smoke, and from people who do and don't eat tomatoes; however, this isn't sufficient unless you collect data from all combinations of smoking and tomato eating. Imagine you ended up with the following data:

Do you smoke?	Do you eat tomatoes?	Do you have cancer?
Yes	No	Yes
Yes	Yes	Yes
No	No	Yes
No	Yes	?????



SPSS TIP 8.1

Error messages about 'failure to coverage' ③

Many statistical procedures use an *iterative process*, which means that SPSS attempts to estimate the parameters of the model by finding successive approximations of those parameters. Essentially, it starts by estimating the parameters with a 'best guess'. It then attempts to approximate them more accurately (known as an *iteration*). It then tries again, and then again, and so on through many iterations. It stops either when the approximations of parameters converge (i.e. at each new attempt the 'approximations' of parameters are the same or very similar to the previous attempt), or when it reaches the maximum number of attempts (iterations).

Sometimes you will get an error message in the output that says something like *'Maximum number of iterations were exceeded, and the log-likelihood value and/or the parameter estimates cannot converge'*. What this means is that SPSS has attempted to estimate the parameters the maximum number of times (as specified in the options) but they are not converging (i.e. at each iteration SPSS is getting quite different estimates). This certainly means that you should ignore any output that SPSS has produced, and it might mean that your data are beyond help. You can try increasing the number of iterations that SPSS attempts, or make the criteria that SPSS uses to assess 'convergence' less strict.

Observing only the first three possibilities does not prepare you for the outcome of the fourth. You have no way of knowing whether this last person will have cancer or not based on the other data you've collected. Therefore, SPSS will have problems unless you've collected data from all combinations of your variables. This should be checked before you run the analysis using a crosstabulation table, and I describe how to do this in Chapter 18. While you're checking these tables, you should also look at the expected frequencies in each cell of the table to make sure that they are greater than 1 and no more than 20% are less than 5 (see section 18.4). This is because the goodness-of-fit tests in logistic regression make this assumption.

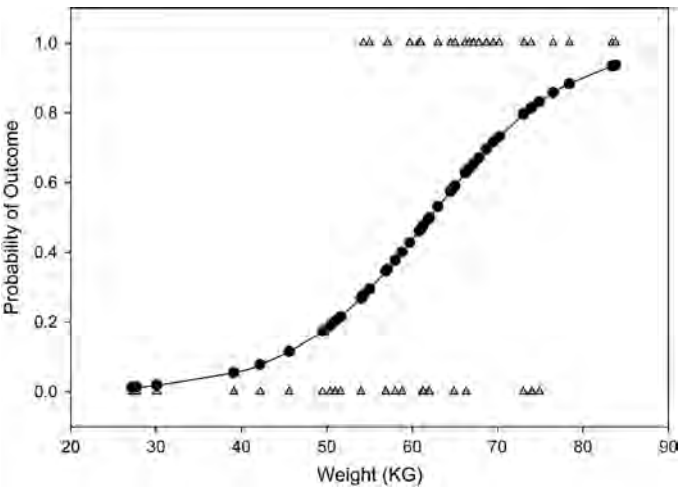
This point applies not only to categorical variables, but also to continuous ones. Suppose that you wanted to investigate factors related to human happiness. These might include age, gender, sexual orientation, religious beliefs, levels of anxiety and even whether a person is right-handed. You interview 1000 people, record their characteristics, and whether they are happy ('yes' or 'no'). Although a sample of 1000 seems quite large, is it likely to include an 80 year old, highly anxious, Buddhist left-handed lesbian? If you found one such person and she was happy, should you conclude that everyone else in the same category is happy? It would, obviously, be better to have several more people in this category to confirm that this combination of characteristics causes happiness. One solution is to collect more data.

As a general point, whenever samples are broken down into categories and one or more combinations are empty it creates problems. These will probably be signalled by coefficients that have unreasonably large standard errors. Conscientious researchers produce and check multiway crosstabulations of all categorical independent variables. Lazy but cautious ones don't bother with crosstabulations, but look carefully at the standard errors. Those who don't bother with either should expect trouble.

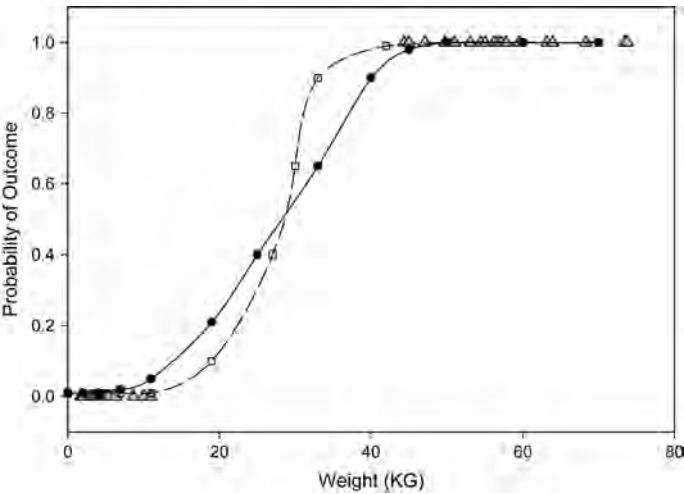
8.4.3. Complete separation ④

A second situation in which logistic regression collapses might surprise you: it's when the outcome variable can be perfectly predicted by one variable or a combination of variables! This is known as **complete separation**.

Let’s look at an example: imagine you placed a pressure pad under your door mat and connected it to your security system so that you could detect burglars when they creep in at night. However, because your teenage children (which you would have if you’re old enough and rich enough to have security systems and pressure pads) and their friends are often coming home in the middle of the night, when they tread on the pad you want it to work out the probability that the person is a burglar and not your teenager. Therefore, you could measure the weight of some burglars and some teenagers and use logistic regression to predict the outcome (teenager or burglar) from the weight. The graph would show a line of triangles at zero (the data points for all of the teenagers you weighed) and a line of triangles at 1 (the data points for burglars you weighed). Note that these lines of triangles overlap (some teenagers are as heavy as burglars). We’ve seen that in logistic regression, SPSS tries to predict the probability of the outcome given a value of the predictor. In this case, at low weights the fitted probability follows the bottom line of the plot, and at high weights it follows the top line. At intermediate values it tries to follow the probability as it changes.



Imagine that we had the same pressure pad, but our teenage children had left home to go to university. We’re now interested in distinguishing burglars from our pet cat based on weight. Again, we can weigh some cats and weigh some burglars. This time the graph still has a row of triangles at zero (the cats we weighed) and a row at 1 (the burglars) but this time the rows of triangles do not overlap: there is no burglar who weighs the same as a cat – obviously there were no cat burglars in the sample (groan now at that sorry excuse for a joke!). This is known as perfect separation: the outcome (cats and burglars) can be perfectly predicted from weight (anything less than 15 kg is a cat, anything more than 40 kg is a burglar). If we try to calculate the probabilities of the outcome given a certain weight then we run into trouble. When the weight is low, the probability is 0, and when the weight is high, the probability is 1, but what happens in between? We have no data in between 15 and 40 kg on which to base these probabilities. The figure shows two possible curves that we could fit to these data: one much steeper than the other. Either one of these curves is valid based on the data we have available. The lack of data means that SPSS will be uncertain about how steep it should make the intermediate slope and it will try to bring the centre as close to vertical as possible, but its estimates veer unsteadily towards infinity (hence large standard errors).



This problem often arises when too many variables are fitted to too few cases. Often the only satisfactory solution is to collect more data, but sometimes a neat answer is found by adopting a simpler model.

8.4.4. Overdispersion ④

I'm not a statistician, and most of what I've read on overdispersion doesn't make an awful lot of sense to me. From what I can gather, it is when the observed variance is bigger than expected from the logistic regression model. This can happen for two reasons. The first is correlated observations (i.e. when the assumption of independence is broken) and the second is due to variability in success probabilities. For example, imagine our outcome was whether a puppy in a litter survived or died. Genetic factors mean that within a given litter the chances of success (living) depend on the litter from which the puppy came. As such success probabilities vary across litters (Halekoh & Højsgaard, 2007), this example of dead puppies is particularly good – not because I'm a cat lover, but because it shows how variability in success probabilities can create correlation between observations (the survival rates of puppies from the same litter are not independent).

Overdispersion creates a problem because it tends to limit standard errors and result in narrower confidence intervals for test statistics of predictors in the logistic regression model. Given that the test statistics are computed by dividing by the standard error, if the standard error is too small then the test statistic will be bigger than it should be, and more likely to be deemed significant. Similarly, narrow confidence intervals will give us overconfidence in the effect of our predictors on the outcome. In short, there is more chance of Type I errors. The parameters themselves (i.e. the *b*-values) are unaffected.

SPSS produces a chi-square goodness-of-fit statistic, and overdispersion is present if the ratio of this statistic to its degrees of freedom is greater than 1 (this ratio is called the *dispersion parameter*, ϕ). Overdispersion is likely to be problematic if the dispersion parameter approaches or is greater than 2. (Incidentally, underdispersion is shown by values less than 1, but this problem is much less common in practice.) There is also the *deviance* goodness-of-fit statistic, and the dispersion parameter can be based on this statistic instead (again by dividing by the degrees of freedom). When the chi-square and deviance statistics are very discrepant, then overdispersion is likely.

The effects of overdispersion can be reduced by using the dispersion parameter to rescale the standard errors and confidence intervals. For example, the standard errors are multiplied by $\sqrt{\phi}$ to make them bigger (as a function of how big the overdispersion is). You can base these corrections on the deviance statistic too, and whether you rescale using this statistic or the Pearson chi-square statistic depends on which one is bigger. The bigger statistic will have the bigger dispersion parameter (because their degrees of freedom are the same), and will make the bigger correction; therefore, correct by the bigger of the two.



CRAMMING SAM'S TIPS

Issues in logistic regression

- In logistic regression, like ordinary regression, we assume linearity, no multicollinearity and independence of errors.
- The linearity assumption is that each predictor has a linear relationship with the log of the outcome variable.
- If we created a table that combined all possible values of all variables then we should ideally have some data in every cell of this table. If you don't then watch out for big standard errors.
- If the outcome variable can be predicted perfectly from one predictor variable (or a combination of predictor variables) then we have *complete separation*. This problem creates large standard errors too.
- *Overdispersion* is where the variance is larger than expected from the model. This can be caused by violating the assumption of independence. This problem makes the standard errors too small!

8.5. Binary logistic regression: an example that will make you feel eel ②

It's amazing what you find in academic journals sometimes. It's a bit of a hobby of mine trying to unearth bizarre academic papers (really, if you find any email them to me). I believe that science should be fun, and so I like finding research that makes me laugh. A research paper by Lo and colleagues is the one that (so far) has made me laugh the most (Lo, Wong, Leung, Law, & Yip, 2004). Lo and colleagues report the case of a 50 year old man who reported to the Accident and Emergency Department (ED for the Americans) with abdominal pain. A physical examination revealed peritonitis so they took an X-ray of the man's abdomen. Although it somehow slipped the patient's mind to mention this to the receptionist upon arrival at the hospital, the X-ray revealed the shadow of an eel. The authors don't directly quote the man's response to this news, but I like to imagine it was something to the effect of 'Oh, that! Erm, yes, well I didn't think it was terribly relevant to my abdominal pain so I didn't mention it, but I did insert an eel into my anus. Do you think that's the problem?' Whatever he *did* say, the authors report that he admitted inserting an eel up his anus to 'relieve constipation'.

I can have a lively imagination at times, and when I read this article I couldn't help thinking about the poor eel. There it was, minding it's own business swimming about in a river (or fish tank possibly), thinking to itself 'Well, today seems like a nice day, there are no eel-eating sharks about, the sun is out, the water is nice, what could possibly go wrong?' The next thing it knows, it's being shoved up the anus of a man from Hong Kong. 'Well, I didn't see that coming,' thinks the eel. Putting myself in the mindset of an eel for a moment, he has found himself in a tight dark tunnel, there's no light, there's a distinct lack of water compared to his usual habitat, and he's probably fearing for his life. His day has gone *very* wrong. How can he escape this horrible fate? Well, doing what any self-respecting eel would do, he notices that his prison cell is fairly soft and decides 'bugger this,¹ I'll *eat* my way out of here'. Unfortunately he didn't make it, but he went out with a fight (there's a fairly unpleasant photograph in the article of the eel biting the splenic flexure). The authors conclude that 'Insertion of a live animal into the rectum causing rectal perforation has never been reported. This may be related to a bizarre healthcare belief, inadvertent sexual behaviour, or criminal assault. However, the true reason may never be known.' Quite.

OK, so this is a really grim tale.² It's not really very funny for the man or the eel, but it was so unbelievably bizarre that I did laugh. Of course my instant reaction was that sticking an eel up your anus to 'relieve constipation' is the poorest excuse for bizarre sexual behaviour I have ever heard. But upon reflection I wondered if I was being harsh on the man – maybe an eel up the anus really can cure constipation. If we wanted to test this, we could collect some data. Our outcome might be 'constipated' vs. 'not constipated', which is a dichotomous variable that we're trying to predict. One predictor variable would be

Can an eel cure constipation?



¹ Literally.

² As it happens, it isn't an isolated grim tale. Through this article I found myself hurtling down a road of morbid curiosity that was best left untravelled. Although the eel was my favourite example, I could have chosen from a very large stone (Sachdev, 1967), a test tube (Hughes, Marice, & Gathright, 1976), a baseball (McDonald & Rosenthal, 1977), an aerosol deodorant can, hose pipe, iron bar, broomstick, penknife, marijuana, bank notes, blue plastic tumbler, vibrator and primus stove (Clarke, Buccimazza, Anderson, & Thomson, 2005), or (a close second place to the eel) a toy pirate ship, with or without pirates I'm not sure (Bemelman & Hammacher, 2005). So, although I encourage you to send me bizarre research, if it involves objects in the rectum then probably don't, unless someone has managed to put Buckingham Palace up there.



intervention (eel up the anus) vs. waiting list (no treatment). We might also want to factor how many days the patient had been constipated before treatment. This scenario is perfect for logistic regression (but not for eels). The data are in **Eel.sav**.

I'm quite aware that many statistics lecturers would rather not be discussing eel-created rectal perforations with their students, so I have named the variables in the file more generally:

- **Outcome (dependent variable): Cured** (cured or not cured).
- **Predictor (independent variable): Intervention** (intervention or no treatment).
- **Predictor (independent variable): Duration** (the number of days before treatment that the patient had the problem).

In doing so, your lecturer can adapt the example to something more palatable if they wish to, but you will secretly know that it's all about having eels up your bum!

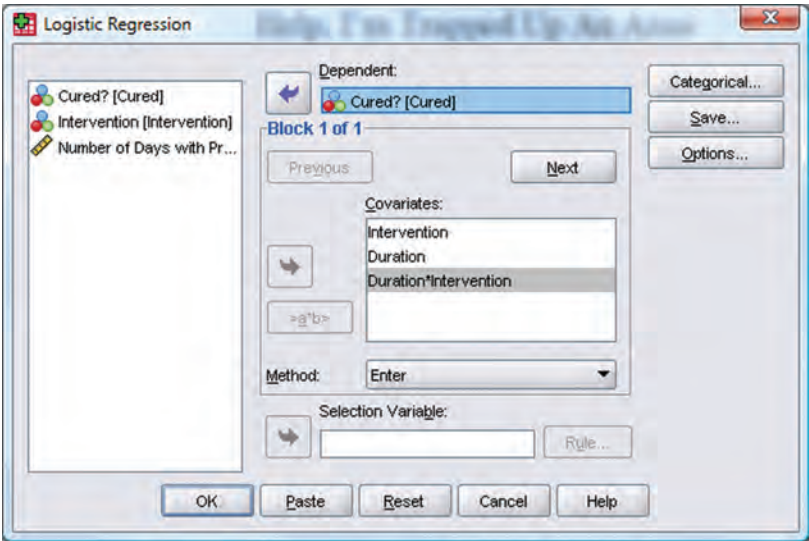
8.5.1. The main analysis ②

To carry out logistic regression, the data must be entered as for normal regression: they are arranged in the data editor in three columns (one representing each variable). Looking at the data editor you should notice that both of the categorical variables have been entered as coding variables (see section 3.4.2.3); that is, numbers have been specified to represent categories. For ease of interpretation, the outcome variable should be coded 1 (event occurred) and 0 (event did not occur); in this case, 1 represents being cured and 0 represents not being cured. For the intervention a similar coding has been used (1 = intervention, 0 = no treatment). To open the main *Logistic Regression* dialog box (Figure 8.3) select

Analyze Regression Binary Logistic...

The main dialog box is very similar to the standard *regression* dialog box. There is a space to place a dependent variable (or outcome variable). In this example, the outcome was whether or not the patient was cured, so we can simply click on **Cured** and drag it to the Dependent box or click on . There is also a box for specifying the covariates (the predictor variables). It is possible to specify the **main effect** of a predictor variable, which

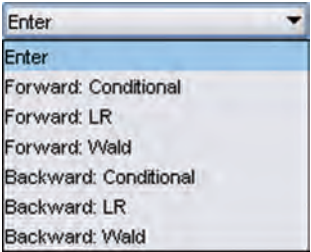
FIGURE 8.3
Logistic Regression main dialog box



is simply the effect (on an outcome variable) of a variable *on its own*. You can also specify an **interaction effect**, which is the effect (on an outcome variable) of two or more variables *in combination*. We will discover more about main effects and interactions in Chapter 12. To specify a main effect, simply select one predictor (e.g. **Duration**) and then drag it to the *Covariates* box or click on . To input an interaction, click on more than one variable on the left-hand side of the dialog box (i.e. click on several variables while holding down the *Ctrl* key) and then click on  to move them to the *Covariates* box. In this example there are only two predictors and therefore there is only one possible interaction (the **Duration** × **Intervention** interaction), but if you have three predictors then you can select several interactions using two predictors, and an interaction involving all three. In the figure I have selected the two main effects of **Duration**, **Intervention** and the **Duration** × **Intervention** interaction. Select these variables too.

8.5.2. Method of regression ②

As with multiple regression, there are different ways of doing logistic regression (see section 8.3.5). You can select a particular method of regression by clicking on *Enter*  and then clicking on a method in the resulting drop-down menu. For this analysis select a *Forward:LR* method of regression. Having spent a vast amount of time telling you never to do stepwise analyses, it’s probably a bit strange to hear me suggest doing forward regression here. Well, for one thing this study is the first in the field and so we have no past research to tell us which variables to expect to be reliable predictors. Second, I didn’t show you a stepwise example in the regression chapter and so this will be a useful way to demonstrate how a stepwise procedure operates!



8.5.3. Categorical predictors ②

It is necessary to ‘tell’ SPSS which predictor variables, if any, are categorical by clicking on *Categorical...* in the main *Logistic Regression* dialog box to activate the dialog box in Figure 8.4. In this dialog box, the covariates are listed on the left-hand side, and there is a space on the right-hand side in which categorical covariates can be placed. Highlight any categorical variables you have (in this example we have only one, so click on **Intervention**) and drag them to the *Categorical Covariates* box or click on .

There are many ways in which you can treat categorical predictors. In section 7.11 we saw that categorical predictors could be incorporated into regression by recoding them using zeros and ones (known as dummy coding). Actually, there are different ways that you can code categorical variables depending on what you want to compare, and SPSS has several ‘standard’ ways built into it that you can select. By default SPSS uses *Indicator* coding, which is the standard dummy variable coding that I explained in section 7.11 (and you can choose to have either the first or last category as your baseline). To change to a different kind of contrast click on *Indicator*  to access a drop-down list of possible contrasts: it is possible to select simple contrasts; difference contrasts, Helmert contrasts, repeated contrasts, polynomial contrasts and deviation contrasts. These techniques will be discussed in detail in Chapter 10 and I’ll explain what these contrasts actually do

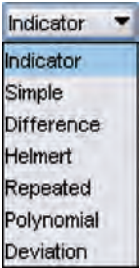
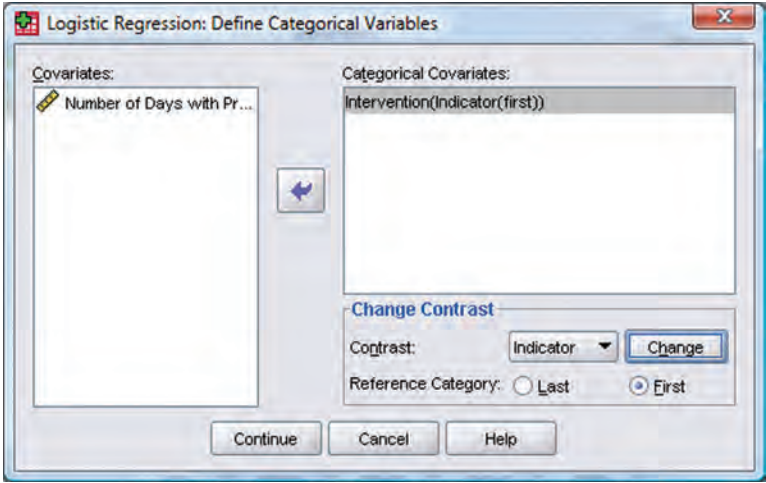


FIGURE 8.4
Defining
categorical
variables
in logistic
regression



then (see Table 10.6). However, you won’t come across indicator contrasts in that chapter and so we’ll use them here. To reiterate, when an indicator contrast is used, levels of the categorical variable are recoded using standard dummy variable coding (see sections 7.11 and 9.7). We do need to decide whether to use the first category as our baseline (☒ **First**) or the last category (☐ **Last**). In this case it doesn’t make much difference because we have only two categories, but if you had a categorical predictor with more than two categories, then you should either use the highest number to code your control category and then select ☐ **Last** for your indicator contrast, or use the lowest number to code your control category and then change the indicator contrast to compare against ☒ **First**. In our data, I coded ‘cured’ as 1 and ‘not cured’ (our control category) as 0; therefore, select the contrast, then click on ☒ **First** and then **Change** so that the completed dialog box looks like Figure 8.4 .

8.5.4. Obtaining residuals ②

As with linear regression, it is possible to save a set of residuals (see section 7.6.1.1) as new variables in the data editor. These residual variables can then be examined to see how well the model fits the observed data. To save residuals click on **Save** in the main *Logistic Regression* dialog box (Figure 8.3). SPSS saves each of the selected variables into the data editor but they can be listed in the output viewer by using the *Case Summaries* command (see section 7.8.6) and selecting the residual variables of interest. The residuals dialog box in Figure 8.5 gives us several options and most of these are the same as those in multiple regression (refer to section 7.7.5). Two residuals that are unique to logistic regression are the *predicted probabilities* and the *predicted group memberships*. The predicted probabilities are the probabilities of Y occurring given the values of each predictor for a given participant. As such, they are derived from equation (8.4) for a given case. The predicted group membership is self-explanatory in that it predicts to which of the two outcome categories a participant is most likely to belong based on the model. The group memberships are based on the predicted probabilities and I will explain these values in more detail when we consider how to interpret the residuals. It is worth selecting all of the available options, or as a bare minimum select the same options as in Figure 8.5.

To reiterate a point from the previous chapter, running a regression without checking how well the model fits the data is like buying a new pair of trousers without trying them on – they might look fine on the hanger but get them home and you find you’re Johnny-tight-pants. The trousers do their job (they cover your legs and keep you warm) but they

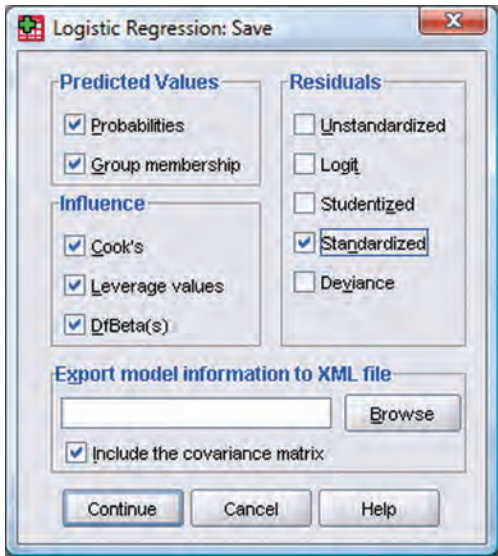


FIGURE 8.5
Dialog box
for obtaining
residuals
for logistic
regression

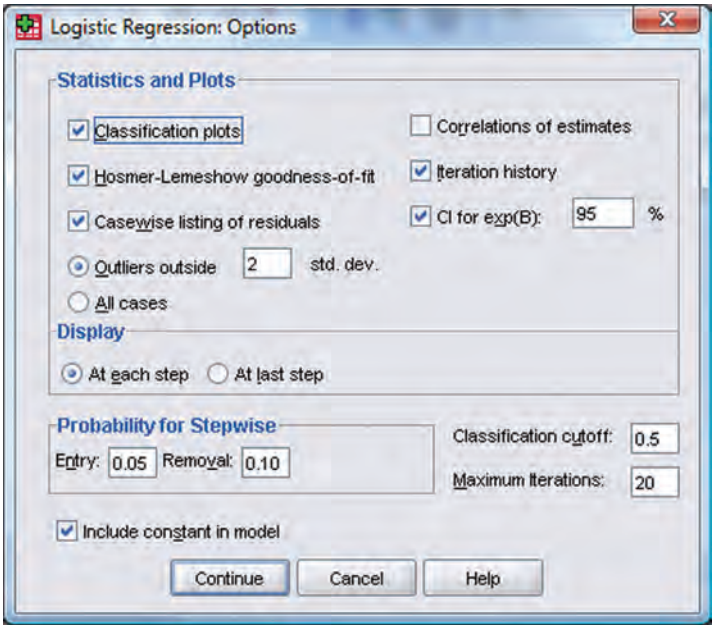
have no real-life value (because they cut off the blood circulation to your legs, which then have to be amputated). Likewise, regression does its job regardless of the data – it will create a model – but the real-life value of the model may be limited (see section 7.6).

8.5.5. Further options ②

There is a final dialog box that offers further options. This box is shown in Figure 8.6 and is accessed by clicking on `Options...` in the main *Logistic Regression* dialog box. For the most part, the default settings in this dialog box are fine. I mentioned in section 8.5.2 that when a stepwise method is used there are default criteria for selecting and removing predictors from the model. These default settings are displayed in the options dialog box under *Probability for Stepwise*. The probability thresholds can be changed, but there is really no need unless you have a good reason for wanting harsher criteria for variable selection. Another default is to arrive at a model after a maximum of 20 iterations (SPSS Tip 8.1). Unless you have a very complex model, 20 iterations will be more than adequate. We saw in Chapter 7 that regression equations contain a constant that represents the Y intercept (i.e. the value of Y when the value of the predictors is 0). By default SPSS includes this constant in the model, but it is possible to run the analysis without this constant and this has the effect of making the model pass through the origin (i.e. Y is 0 when X is 0). Given that we are usually interested in producing a model that best fits the data we have collected, there is little point in running the analysis without the constant included.

A classification plot is a histogram of the actual and predicted values of the outcome variable. This plot is useful for assessing the fit of the model to the observed data. It is also possible to do a *Casewise listing of residuals* either for any cases for which the standardized residual is greater than 2 standard deviations (this value can be changed but the default is sensible), or for all cases. I recommend a more thorough examination of residuals but this option can be useful for a quick inspection. You can ask SPSS to display a confidence interval (see section 2.5.2) for the odds ratio (see section 8.3.4) $Exp(B)$, and by default a 95% confidence interval is used, which is appropriate (if it says anything else then change it to 95%) and a useful statistic to have. More important, you can request the *Hosmer-Lemeshow goodness-of-fit* statistic, which can be used to assess how well the chosen model fits the data.

FIGURE 8.6
Dialog box
for logistic
regression
options



The remaining options are fairly unimportant: you can choose to display all statistics and graphs at each stage of an analysis (the default), or only after the final model has been fitted. Finally, you can display a correlation matrix of parameter estimates for the terms in the model, and you can display coefficients and log-likelihood values at each iteration of the parameter estimation process – the practical function of doing this is lost on most of us mere mortals!

8.6. Interpreting logistic regression ②

When you have selected all of the options that I’ve just described, click on **OK** and watch the output spew out in the viewer window.

8.6.1. The initial model ②

SPSS Output 8.1 tells us two things. First it tells us how we coded our outcome variable, and this table merely reminds us that 0 = not cured, and 1 = cured.³ The second is that it tell us how SPSS has coded the categorical predictors. The parameter codings are also given for the categorical predictor variable (**Intervention**). Indicator coding was chosen with two categories, and so the coding is the same as the values in the data editor (0 = no treatment, 1 = treatment). If *deviation* coding had been chosen then the coding would have been –1 (Treatment) and 1 (No Treatment). With a *simple* contrast the codings would have been –0.5 (**Intervention** = No Treatment) and 0.5 (**Intervention** = Treatment) if **First** was selected as the reference category or vice versa if **Last** was selected as the reference category. The parameter codings are important for calculating the probability of the outcome variable ($P(Y)$), but we will come to that later.

³ These values are the same as the data editor so this table might seem pointless; however, had we used codes other than 0 and 1 (e.g. 1 = not cured, 2 = cured) then SPSS changes these to zeros and ones and this table informs you of which category is represented by 0 and which by 1. This is important when it comes to interpretation.

SPSS OUTPUT 8.1

Dependent Variable Encoding	
Original Value	Internal Value
Not Cured	0
Cured	1

Categorical Variables Codings			
		Frequency	Parameter coding
			(1)
Intervention	No Treatment	56	.000
	Intervention	57	1.000

For this first analysis we requested a forward stepwise method⁴ and so the initial model is derived using only the constant in the regression equation. SPSS Output 8.2 tells us about the model when only the constant is included (i.e. all predictor variables are omitted). The table labelled **Iteration History** tells us that the log-likelihood of this baseline model (see section 8.3.1) is 154.08. This represents the fit of the most basic model to the data. When including only the constant, the computer bases the model on assigning every participant to a single category of the outcome variable. In this example, SPSS can decide either to predict that the patient was cured, or that every patient was not cured. It could make this decision arbitrarily, but because it is crucial to try to maximize how well the model predicts the observed data, SPSS will predict that every patient belongs to the category in which most observed cases fell. In this example there were 65 patients who were cured, and only 48 who were not cured. Therefore, if SPSS predicts that every patient was cured then this prediction will be correct 65 times out of 113 (i.e. 58% approx.). However, if SPSS predicted that every patient was not cured, then this prediction would be correct only 48 times out of 113 (42% approx.). As such, of the two available options it is better to predict that all patients were cured because this results in a greater number of correct predictions. The output shows a contingency table for the model in this basic state. You can see that SPSS has predicted that all patients are cured, which results in 0% accuracy for the patients who were not cured, and 100% accuracy for those observed to be cured. Overall, the model correctly classifies 57.5% of patients.

SPSS OUTPUT 8.2

Iteration History^{a,b,c}

Iteration		-2 Log likelihood	Coefficients
			Constant
Step 0	1	154.084	.301
	2	154.084	.303
	3	154.084	.303

a. Constant is included in the model.

b. Initial -2 Log Likelihood: 154.084

c. Estimation terminated at iteration number 3 because parameter estimates changed by less than .001.

Classification Table^{a,b}

Observed			Predicted		Percentage Correct	
			Cured?			
			Not Cured	Cured		
Step 0	Cured?	Not Cured	0	48	.0	
		Cured	0	65	100.0	
Overall Percentage					57.5	

a. Constant is included in the model.

b. The cut value is .500

⁴ Actually, this is a *really* bad idea when you have an interaction term because to look at an interaction you need to include the main effects of the variables in the interaction term. I chose this method *only* to illustrate how stepwise methods work.

SPSS Output 8.3 summarizes the model (**Variables in the Equation**), and at this stage this entails quoting the value of the constant (b_0), which is equal to 0.30. The table labelled **Variables not in the Equation** tells us that the residual chi-square statistic is 9.83 which is significant at $p < .05$ (it labels this statistic *Overall Statistics*). This statistic tells us that the coefficients for the variables not in the model are significantly different from zero – in other words, that the addition of one or more of these variables to the model will significantly affect its predictive power. If the probability for the residual chi-square had been greater than .05 it would have meant that forcing all of the variables excluded from the model into the model would not have made a significant contribution to its predictive power.

The remainder of this table lists each of the predictors in turn with a value of **Roa's efficient score statistic** for each one (column labelled *Score*). In large samples when the null hypothesis is true, the score statistic is identical to the Wald statistic and the likelihood ratio statistic. It is used at this stage of the analysis because it is computationally less intensive than the Wald statistic and so can still be calculated in situations when the Wald statistic would prove prohibitive. Like any test statistic, Roa's score statistic has a specific distribution from which statistical significance can be obtained. In this example, **Intervention** and the **Intervention × Duration** interaction both have significant score statistics at $p < .01$ and could potentially make a contribution to the model, but **Duration** alone does not look likely to be a good predictor because its score statistic is non-significant $p > .05$. As mentioned in section 8.5.2, the stepwise calculations are relative and so the variable that will be selected for inclusion is the one with the highest value for the score statistic that has a significance below .05. In this example, that variable will be **Intervention** because its score statistic (9.77) is the biggest.

SPSS OUTPUT 8.3

Variables in the Equation						
	B	S.E.	Wald	df	Sig.	Exp(B)
Step 0 Constant	.303	.190	2.538	1	.111	1.354

Variables not in the Equation				
Step 0	Variables	Score	df	Sig.
	Intervention(1)	9.771	1	.002
	Duration	.609	1	.435
	Duration by Intervention (1)	9.052	1	.003
	Overall Statistics	9.827	3	.020

8.6.2. Step 1: intervention ③

As I predicted in the previous section, whether or not an intervention was given to the patient (**Intervention**) is added to the model in the first step. As such, a patient is now classified as being cured or not based on whether they had an intervention or not (waiting list). This can be explained easily if we look at the crosstabulation for the variables **Intervention** and **Cured**.⁵ The model will use whether a patient had an intervention or not to predict whether they were cured or not by applying the crosstabulation table shown in Table 8.1.

The model predicts that all of the patients who had an intervention were cured. There were 57 patients who had an intervention, so the model predicts that these 57 patients were cured; it is correct for 41 of these patients, but misclassifies 16 people as ‘cured’ who were not cured – see Table 8.1. In addition, this new model predicts that all of the 56 patients who received no treatment were not cured; for these patients the model is correct 32 times but misclassifies as ‘not cured’ 24 people who were.

⁵ The dialog box to produce this table can be obtained by selecting **Analyze Descriptive Statistics Crosstabs...**

TABLE 8.1 Crosstabulation of intervention with outcome status (cured or not)

		Intervention or Not (Intervention)	
		No Treatment	Intervention
Cured? (Cured)	Not Cured	32	16
	Cured	24	41
	Total	56	57

Omnibus Tests of Model Coefficients

		Chi-Square	df	Sig.
Step 1	Step	9.926	1	.002
	Block	9.926	1	.002
	Model	9.926	1	.002

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	144.158 ^a	.084	.113

a. Estimation terminated at iteration number 3 because parameter estimates changed by less than .001.

Observed		Predicted		Percentage Correct
		Cured?		
		Not Cured	Cured	
Step 1	Cured?	32	16	66.7
	Not Cured	24	41	63.1
Overall Percentage				64.6

a. The cut value is .500

SPSS OUTPUT 8.4

SPSS Output 8.4 shows summary statistics about the new model (which we’ve already seen contains **Intervention**). The overall fit of the new model is assessed using the log-likelihood statistic (see section 8.3.1). In SPSS, rather than reporting the log-likelihood itself, the value is multiplied by -2 (and sometimes referred to as $-2LL$): this multiplication is done because $-2LL$ has an approximately chi-square distribution and so it makes it possible to compare values against those that we might expect to get by chance alone. Remember that large values of the log-likelihood statistic indicate poorly fitting statistical models.

At this stage of the analysis the value of $-2LL$ should be less than the value when only the constant was included in the model (because lower values of $-2LL$ indicate that the model is predicting the outcome variable more accurately). When only the constant was included, $-2LL = 154.08$, but now **Intervention** has been included this value has been reduced to 144.16. This reduction tells us that the model is better at predicting whether someone was cured than it was before **Intervention** was added. The question of how much better the model predicts the outcome variable can be assessed using the *model chi-square statistic*, which measures the difference between the model as it currently stands and the model when only the constant was included. We saw in section 8.3.1 that we could assess the significance of the change in a model by taking the log-likelihood of the new model and subtracting the log-likelihood of the baseline model from it. The value of the model chi-square statistic works on this principle and is, therefore, equal to $-2LL$ with **Intervention** included minus the value of $-2LL$ when only the constant was in the model (154.08

– 144.16 = 9.92). This value has a chi-square distribution and so its statistical significance can be calculated easily.⁶ In this example, the value is significant at a .05 level and so we can say that overall the model is predicting whether a patient is cured or not significantly better than it was with only the constant included. The model chi-square is an analogue of the *F*-test for the linear regression (see Chapter 7). In an ideal world we would like to see a non-significant overall –2LL (indicating that the amount of unexplained data is minimal) and a highly significant model chi-square statistic (indicating that the model including the predictors is significantly better than without those predictors). However, in reality it is possible for both statistics to be highly significant.

There is a second statistic called the *step* statistic that indicates the improvement in the predictive power of the model since the last stage. At this stage there has been only one step in the analysis and so the value of the improvement statistic is the same as the model chi-square. However, in more complex models in which there are three or four stages, this statistic gives a measure of the improvement of the predictive power of the model since the last step. Its value is equal to –2LL at the current step minus –2LL at the previous step. If the improvement statistic is significant then it indicates that the model now predicts the outcome significantly better than it did at the last step, and in a forward regression this can be taken as an indication of the contribution of a predictor to the predictive power of the model. Similarly, the *block* statistic provides the change in –2LL since the last block (for use in hierarchical or blockwise analyses).

SPSS Output 8.4 also tells us the values of Cox and Snell’s and Nagelkerke’s *R*², but we will discuss these a little later. There is also a classification table that indicates how well the model predicts group membership; because the model is using **Intervention** to predict the outcome variable, this classification table is the same as Table 8.1. The current model correctly classifies 32 patients who were not cured but misclassifies 16 others (it correctly classifies 66.7% of cases). The model also correctly classifies 41 patients who were cured but misclassifies 24 others (it correctly classifies 63.1% of cases). The overall accuracy of classification is, therefore, the weighted average of these two values (64.6%). So, when only the constant was included, the model correctly classified 57.5% of patients, but now, with the inclusion of **Intervention** as a predictor, this has risen to 64.6%.

SPSS OUTPUT 8.5

Variables in the Equation								
		B	S.E.	Wald	df	Sig.	Exp(B)	95.0% C.I. for EXP(B)
								Lower Upper
Step 1 ^a	Intervention(1)	1.229	.400	9.447	1	.002	3.417	1.561 7.480
	Constant	-.288	.270	1.135	1	.287	.750	

a. Variable(s) entered on step 1: Intervention.

The next part of the output (SPSS Output 8.5) is crucial because it tells us the estimates for the coefficients for the predictors included in the model. This section of the output gives us the coefficients and statistics for the variables that have been included in the model at this point (namely **Intervention** and the constant). The *b*-value is the same as the *b*-value in linear regression: they are the values that we need to replace in equation (8.4) to establish the probability that a case falls into a certain category. We saw in linear regression that the value of *b* represents the change in the outcome resulting from a unit change in the predictor variable. The interpretation of this coefficient in logistic regression is very similar in that it represents the change in the logit of the outcome variable associated with a one-unit change in the predictor variable. The logit of the outcome is simply the natural logarithm of the odds of *Y* occurring.

⁶ The degrees of freedom will be the number of parameters in the new model (the number of predictors plus 1, which in this case with one predictor means 2) minus the number of parameters in the baseline model (which is 1, the constant). So, in this case, *df* = 2 – 1 = 1.

The crucial statistic is the Wald statistic⁷ which has a chi-square distribution and tells us whether the *b* coefficient for that predictor is significantly different from zero. If the coefficient is significantly different from zero then we can assume that the predictor is making a significant contribution to the prediction of the outcome (*Y*). We came across the Wald statistic in section 8.3.3 and saw that it should be used cautiously because when the regression coefficient (*b*) is large, the standard error tends to become inflated, resulting in the Wald statistic being underestimated (see Menard, 1995). However, for these data it seems to indicate that having the intervention (or not) is a significant predictor of whether the patient is cured (note that the significance of the Wald statistic is less than .05).

In section 8.3.2 we saw that we could calculate an analogue of *R* using equation (8.7). For these data, the Wald statistic and its *df* can be read from SPSS Output 8.5 (9.45 and 1 respectively), and the original $-2LL$ was 154.08. Therefore, *R* can be calculated as:

$$R = \pm \sqrt{\frac{9.45 - (2 \times 1)}{154.08}} \tag{8.14}$$
$$= .22$$

In the same section we saw that Hosmer and Lemeshow’s measure (R^2_L) is calculated by dividing the model chi-square by the *original* $-2LL$. In this example the model chi-square after **Intervention** has been entered into the model is 9.93, and the original $-2LL$ (before any variables were entered) was 154.08. So, $R^2_L = 9.93/154.08 = .06$, which is different to the value we would get by squaring the value of *R* given above ($R^2 = .22^2 = 0.05$). Earlier on in SPSS Output 8.4, SPSS gave us two other measures of R^2 that were described in section 8.3.2. The first is Cox and Snell’s measure, which SPSS reports as .084, and the second is Nagelkerke’s adjusted value, which SPSS reports as .113. As you can see, all of these values differ, but they can be used as effect size measures for the model.



SELF-TEST Using equations (8.9) and (8.10) calculate the values of Cox and Snell’s and Nagelkerke’s R^2 reported by SPSS. [Hint: These equations use the log-likelihood, whereas SPSS reports $-2 \times \text{log-likelihood}$. $LL(\text{new})$ is, therefore, $144.16/-2 = -72.08$, and $LL(\text{baseline}) = 154.08/-2 = -77.04$. The sample size, *n*, is 113.]



The final thing we need to look at is the odds ratio ($Exp(B)$ in the SPSS output), which was described in section 8.3.4. To calculate the change in odds that results from a unit change in the predictor for this example, we must first calculate the odds of a patient being cured given that they *didn’t* have the intervention. We then calculate the odds of a patient being cured given that they *did* have the intervention. Finally, we calculate the proportionate change in these two odds.

To calculate the first set of odds, we need to use equation (8.12) to calculate the probability of a patient being cured given that they *didn’t* have the intervention. The parameter coding at the beginning of the output told us that patients who did not have the intervention were coded with a 0, so we can use this value in place of *X*. The value of b_1 has been estimated for us as 1.229 (see **Variables in the Equation** in SPSS Output 8.5), and the

⁷ As we have seen, this is simply *b* divided by its standard error ($1.229/.40 = 3.0725$); however, SPSS actually quotes the Wald statistic squared. For these data $3.0725^2 = 9.44$ as reported (within rounding error) in the table.

coefficient for the constant can be taken from the same table and is -0.288 . We can calculate the odds as:

$$\begin{aligned}
 P(\text{Cured}) &= \frac{1}{1 + e^{-(b_0 + b_1 X_1)}} \\
 &= \frac{1}{1 + e^{-[-0.288 + (1.299 \times 0)]}} \\
 &= 0.428 \\
 P(\text{Not Cured}) &= 1 - P(\text{Cured}) \\
 &= 1 - 0.428 \\
 &= 0.572 \\
 \text{odds} &= \frac{0.428}{0.572} \\
 &= 0.748
 \end{aligned}
 \tag{8.15}$$

Now, we calculate the same thing after the predictor variable has changed by one unit. In this case, because the predictor variable is dichotomous, we need to calculate the odds of a patient being cured, given that they have had the intervention. So, the value of the intervention variable, X , is now 1 (rather than 0). The resulting calculations are as follows:

$$\begin{aligned}
 P(\text{Cured}) &= \frac{1}{1 + e^{-(b_0 + b_1 X_1)}} \\
 &= \frac{1}{1 + e^{-[-0.288 + (1.299 \times 1)]}} \\
 &= 0.719 \\
 P(\text{Not Cured}) &= 1 - P(\text{Cured}) \\
 &= 1 - 0.719 \\
 &= 0.281 \\
 \text{odds} &= \frac{0.719}{0.281} \\
 &= 2.559
 \end{aligned}
 \tag{8.16}$$

We now know the odds before and after a unit change in the predictor variable. It is now a simple matter to calculate the proportionate change in odds by dividing the odds after a unit change in the predictor by the odds before that change:

$$\begin{aligned}
 \Delta \text{odds} &= \frac{\text{odds after a unit change in the predictor}}{\text{original odds}} \\
 &= \frac{2.56}{0.75} \\
 &= 3.41
 \end{aligned}
 \tag{8.17}$$

You should notice that the value of the proportionate change in odds is the same as the value that SPSS reports for $\text{Exp}(B)$ (allowing for differences in rounding). We can interpret the odds ratio in terms of the change in odds. If the value is greater than 1 then it indicates that as the predictor increases, the odds of the outcome occurring increase. Conversely, a value less than 1 indicates that as the predictor increases, the odds of the outcome occurring decrease. In this example, we can say that the odds of a patient who is treated being cured are 3.41 times higher than those of a patient who is not treated.

In the options (see section 8.5.5), we requested a confidence interval for the odds ratio and it can also be found in the output. The way to interpret this confidence interval is the same as any other confidence interval (section 2.5.2): if we calculated confidence intervals for the value of the odds ratio in 100 different samples, then these intervals would encompass the actual value of the odds ratio in the population (rather than the sample) in 95 of those samples. In this case, we can be fairly confident that the population value of the odds ratio lies between 1.56 and 7.48. However, our sample could be one of the 5% that produces a confidence interval that ‘misses’ the population value.

The important thing about this confidence interval is that it doesn’t cross 1 (both values are greater than 1). This is important because values greater than 1 mean that as the predictor variable increases, so do the odds of (in this case) being cured. Values less than 1 mean the opposite: as the predictor variable increases, the odds of being cured decrease. The fact that both limits of our confidence interval are above 1 gives us confidence that the direction of the relationship that we have observed is true in the population (i.e. it’s likely that having an intervention compared to not increases the odds of being cured). If the lower limit had been below 1 then it would tell us that there is a chance that in the population the direction of the relationship is the opposite to what we have observed. This would mean that we could not trust that our intervention increases the odds of being cured.

SPSS OUTPUT 8.6

Model if Term Removed					
Variable		Model Log Likelihood	Change in -2 Log Likelihood	df	Sig. of the Change
Step 1	Intervention	-77.042	9.926	1	.002

Variables not in the Equation					
			Score	df	Sig.
Step 1	Variables	Duration	.002	1	.964
		Duration by Intervention (1)	.043	1	.835
	Overall Statistics		.063	2	.969

The test statistics for **Intervention** if it were removed from the model are in SPSS Output 8.6. Now, remember that earlier on I said how the regression would place variables into the equation and then test whether they then met a removal criterion. Well, the **Model if Term Removed** part of the output tells us the effects of removal. The important thing to note is the significance value of the log-likelihood ratio (log LR). The log LR for this model is significant ($p < .01$) which tells us that removing **Intervention** from the model would have a significant effect on the predictive ability of the model – in other words, it would be a very bad idea to remove it!

Finally, we are told about the variables currently not in the model. First of all, the residual chi-square (labelled *Overall Statistics* in the output), which is non-significant, tells us that none of the remaining variables have coefficients significantly different from zero. Furthermore, each variable is listed with its score statistic and significance value, and for both variables their coefficients are not significantly different from zero (as can be seen from the significance values of .964 for **Duration** and .835 for the **Duration**×**Intervention** interaction). Therefore, no further variables will be added to the model.

SPSS Output 8.7 displays the classification plot that we requested in the options dialog box. This plot is a histogram of the predicted probabilities of a patient being cured. If the model perfectly fits the data, then this histogram should show all of the cases for which the event has occurred on the right-hand side, and all the cases for which the event hasn’t occurred on the left-hand side. In other words, all of the patients who were cured should appear on the right and all those who were not cured should appear on the left. In this example, the only significant predictor is dichotomous and so there are only two columns of cases on the plot. If the

8.6.3. Listing predicted probabilities ②

It is possible to list the expected probability of the outcome variable occurring based on the final model. In section 8.5.4 we saw that SPSS could save residuals and also predicted probabilities. SPSS saves these predicted probabilities and predicted group memberships as variables in the data editor and names them PRE_1 and PGR_1 respectively. These probabilities can be listed using the **Analyze Reports**  **Case Summaries...** dialog box (see section 7.8.6).



SELF-TEST Use the *case summaries* function in SPSS to create a table for the first 15 cases in the file **eel.sav** showing the values of **Cured**, **Intervention**, **Duration**, the predicted probability (**PRE_1**) and the predicted group membership (**PGR_1**) for each case.



SPSS Output 8.8 shows a selection of the predicted probabilities (because the only significant predictor was a dichotomous variable, there will be only two different probability values). It is also worth listing the predictor variables as well to clarify from where the predicted probabilities come.

Case Summaries^a

SPSS OUTPUT 8.8

	Case Number	Cured?	Intervention	Number of Days with Problem before Treatment	Predicted probability	Predicted group
1	1	Not Cured	No Treatment	7	.42857	Not Cured
2	2	Not Cured	No Treatment	7	.42857	Not Cured
3	3	Not Cured	No Treatment	6	.42857	Not Cured
4	4	Cured	No Treatment	8	.42857	Not Cured
5	5	Cured	Intervention	7	.71930	Cured
6	6	Cured	No Treatment	6	.42857	Not Cured
7	7	Not Cured	Intervention	7	.71930	Cured
8	8	Cured	Intervention	7	.71930	Cured
9	9	Cured	No Treatment	8	.42857	Not Cured
10	10	Not Cured	No Treatment	7	.42857	Not Cured
11	11	Cured	Intervention	7	.71930	Cured
12	12	Cured	No Treatment	7	.42857	Not Cured
13	13	Cured	No Treatment	5	.42857	Not Cured
14	14	Not Cured	Intervention	9	.71930	Cured
15	15	Not Cured	No Treatment	6	.42857	Not Cured
Total	N	15	15	15	15	15

a. Limited to first 15 cases.

We found from the model that the only significant predictor of being cured was having the intervention. This could have a value of either 1 (have the intervention) or 0 (no intervention). If these two values are placed into equation (8.4) with the respective regression coefficients, then the two probability values are derived. In fact, we calculated these values as part of equation (8.15) and equation (8.16) and you should note that the calculated probabilities – $P(\text{Cured})$ in these equations – correspond to the values in PRE_1. These values tells us that when a patient is not treated (**Intervention** = 0, No Treatment), there is a

probability of .429 that they will be cured – basically, about 43% of people get better without any treatment. However, if the patient does have the intervention (**Intervention** = 1, yes), there is a probability of .719 that they will get better – about 72% of people treated get better. When you consider that a probability of 0 indicates no chance of getting better, and a probability of 1 indicates that the patient will definitely get better, the values obtained provide strong evidence that having the intervention increases your chances of getting better (although the probability of recovery without the intervention is still not bad).

Assuming we are content that the model is accurate and that the intervention has some substantive significance, then we could conclude that our intervention (which, to remind you, was putting an eel up the anus) is the single best predictor of getting better (not being constipated). Furthermore, the duration of the constipation pre-intervention and its interaction with the intervention did not significantly predict whether a person got better.



SELF-TEST Rerun this analysis using the forced entry method of analysis – how do your conclusions differ?

8.6.4. Interpreting residuals ②

Our conclusions so far are fine in themselves, but to be sure that the model is a good one, it is important to examine the residuals. In section 8.5.4 we saw how to get SPSS to save various residuals in the data editor. We can now interpret them.

We saw in the previous chapter that the main purpose of examining residuals in any regression is to (1) isolate points for which the model fits poorly, and (2) isolate points that exert an undue influence on the model. To assess the former we examine the residuals, especially the Studentized residual, standardized residual and deviance statistics. To assess the latter we use influence statistics such as Cook’s distance, DFBeta and leverage statistics. These statistics were explained in detail in section 7.6 and their interpretation in logistic regression is the same; therefore, Table 8.2 summarizes the main statistics that you should look at and what to look for, but for more detail consult the previous chapter.

If you request these residual statistics, SPSS saves them as new columns in the data editor. You can look at the values in the data editor, or produce a table using the **Analyze Reports** ▶ **Case Summaries...** dialog box.

The basic residual statistics for this example (Cook’s distance, leverage, standardized residuals and DFBeta values) are pretty good: note that all cases have DFBetas less than 1, and leverage statistics (LEV_1) are very close to the calculated expected value of 0.018. There are



OLIVER TWISTED
Please, Sir, can I have some more ... diagnostics?

‘What about the trees?’ protests eco-warrior Oliver. ‘These SPSS outputs take up so much room, why don’t you put them on the website instead?’ It’s a valid point so I have produced a table of the diagnostic statistics for this example, but it’s in the additional material for this chapter on the companion website

TABLE 8.2 Summary of residual statistics saved by SPSS

Label	Name	Comment
PRE_1	Predicted Value	
PGR_1	Predicted Group	
COO_1	Cook's Distance	Should be less than 1
LEV_1	Leverage	Lies between 0 (no influence) and 1 (complete influence). The expected leverage is $(k + 1)/N$, where k is the number of predictors and N is the sample size. In this case it would be $2/113 = .018$
SRE_1	Studentized Residual	Only 5% should lie outside ± 1.96 , and about 1% should lie outside ± 2.58 . Cases above 3 are cause for concern and cases close to 3 warrant inspection
ZRE_1	Standardized Residual	
DEV_1	Deviance	
DFB0_1	DFBeta for the Constant	Should be less than 1
DFB1_1	DFBeta for the First Predictor (Intervention)	

also no unusually high values of Cook's distance (COO_1) which, all in all, means that there are no influential cases having an effect on the model. The standardized residuals all have values of less than ± 2 and so there seems to be very little here to concern us.

You should note that these residuals are slightly unusual because they are based on a single predictor that is categorical. This is why there isn't a lot of variability in the values of the residuals. Also, if substantial outliers or influential cases had been isolated, you are not justified in eliminating these cases to make the model fit better. Instead these cases should be inspected closely to try to isolate a good reason why they were unusual. It might simply be an error in inputting data, or it could be that the case was one which had a special reason for being unusual: for example, there were other medical complications that might contribute to the constipation that were noted during the patient's assessment. In such a case, you may have good reason to exclude the case and duly note the reasons why.



CRAMMING SAM'S TIPS

Diagnostic statistics

You need to look for cases that might be influencing the logistic regression model:

- Look at standardized residuals and check that no more than 5% of cases have absolute values above 2, and that no more than about 1% have absolute values above 2.5. Any case with a value above about 3 could be an outlier.
- Look in the data editor for the values of Cook's distance: any value above 1 indicates a case that might be influencing the model.
- Calculate the average leverage (the number of predictors plus 1, divided by the sample size) and then look for values greater than twice or three times this average value.
- Look for absolute values of DFBeta greater than 1.

8.6.5. Calculating the effect size ②

We’ve already seen (section 8.3.2) that SPSS produces a value of *r* for each predictor, based on the Wald statistic. This can be used as your effect size measure for a predictor; however, it’s worth bearing in mind what I’ve already said about it: it won’t be accurate when the Wald statistic is inaccurate. The better effect size to use is the odds ratio (see section 18.5.5).

8.7. How to report logistic regression ②

Logistic regression is fairly rarely used in my discipline of psychology, so it’s difficult to find any concrete guidelines about how to report one. My personal view is that you should report it much the same as linear regression (see section 7.9). I’d be inclined to tabulate the results, unless it’s a very simple model. As a bare minimum, report the beta values and their standard errors and significance value and some general statistics about the model (such as the *R*² and goodness-of-fit statistics). I’d also highly recommend reporting the odds ratio and its confidence interval. I’d also include the constant so that readers of your work can construct the full regression model if they need to. You might also consider reporting the variables that were not significant predictors because this can be as valuable as knowing about which predictors were significant.

For the example in this chapter we might produce a table like that in Table 8.3. Hopefully you can work out from where the values came by looking back through the chapter so far. As with multiple regression, I’ve rounded off to 2 decimal places throughout; for the *R*² there is no zero before the decimal point (because these values cannot exceed 1) but for all other values less than 1 the zero is present; the significance of the variable is denoted by an asterisk with a footnote to indicate the significance level being used.

TABLE 8.3 How to report logistic regression

	95% CI for Odds Ratio			
	B (SE)	Lower	Odds Ratio	Upper
Included				
Constant	−0.29 (0.27)			
Intervention	1.23* (0.40)	1.56	3.42	7.48

Note: *R*² = .06 (Hosmer & Lemeshow), .08 (Cox & Snell), .11 (Nagelkerke). Model $\chi^2(1) = 9.93, p < .01$. **p* < .01.

8.8. Testing assumptions: another example ②

This example was originally inspired by events in the soccer World Cup of 1998 (a long time ago now, but such crushing disappointments are not easily forgotten). Unfortunately for me (being an Englishman), I was subjected to watching England get knocked out of the competition by losing a penalty shootout. Reassuringly, six years later I watched England get

knocked out of the European Championship in another penalty shootout. Even more reassuring, a few months ago I saw them fail to even qualify for the European Championship (not a penalty shootout this time, just playing like cretins).

Now, if I were the England coach, I'd probably shoot the spoilt over-paid prima donnas, or I might be interested in finding out what factors predict whether or not a player will score a penalty. Those of you who hate football can read this example as being factors that predict success in a free-throw in basketball or netball, a penalty in hockey or a penalty kick in rugby⁸ or field goal in American football. Now, this research question is perfect for logistic regression because our outcome variable is a dichotomy: a penalty can be either scored or missed. Imagine that past research (Eriksson, Beckham, & Vassell, 2004; Hoddle, Batty, & Ince, 1998) had shown that there are two factors that reliably predict whether a penalty kick will be missed or scored. The first factor is whether the player taking the kick is a worrier (this factor can be measured using a measure such as the Penn State Worry Questionnaire, PSWQ). The second factor is the player's past success rate at scoring (so, whether the player has a good track record of scoring penalty kicks). It is fairly well accepted that anxiety has detrimental effects on the performance of a variety of tasks and so it was also predicted that state anxiety might be able to account for some of the unexplained variance in penalty success.

This example is a classic case of building on a well-established model, because two predictors are already known and we want to test the effect of a new one. So, 75 football players were selected at random and before taking a penalty kick in a competition they were given a state anxiety questionnaire to complete (to assess anxiety before the kick was taken). These players were also asked to complete the PSWQ to give a measure of how much they worried about things generally, and their past success rate was obtained from a database. Finally, a note was made of whether the penalty was scored or missed. The data can be found in the file **penalty.sav**, which contains four variables – each in a separate column:

Why do the England football team always miss penalties?



- **Scored:** This variable is our outcome and it is coded such that 0 = penalty missed and 1 = penalty scored.
- **PSWQ:** This variable is the first predictor variable and it gives us a measure of the degree to which a player worries.
- **Previous:** This variable is the percentage of penalties scored by a particular player in their career. As such, it represents previous success at scoring penalties.
- **Anxious:** This variable is our third predictor and it is a variable that has not previously been used to predict penalty success. **Anxious** is a measure of state anxiety before taking the penalty.



SELF-TEST We learnt how to do hierarchical regression in the previous chapter. Try to conduct a hierarchical logistic regression analysis on these data. Enter **Previous** and **PSWQ** in the first block and **Anxious** in the second (forced entry). There is a full guide on how to do the analysis and its interpretation in the additional material on the companion website.



⁸ Although this would be an unrealistic example because our rugby team, unlike their football counterparts, have Jonny Wilkinson who is the lord of penalty kicks and we bow at his great left foot in wonderment (well, I do).

8.8.1. Testing for linearity of the logit ③

In this example we have three continuous variables, therefore we have to check that each one is linearly related to the log of the outcome variable (**Scored**). I mentioned earlier in this chapter that to test this assumption we need to run the logistic regression but include predictors that are the interaction between each predictor and the log of itself (Hosmer & Lemeshow, 1989). To create these interaction terms, we need to use **Transform > Compute Variable...** (see section 5.7.3). For each variable create a new variable that is the log of the original variable. For example, for **PSWQ**, create a new variable called **LnPSWQ** by entering this name into the box labelled **Target Variable** and then click on **Type & Label...** and give the variable a more descriptive label such as **Ln(PSWQ)**. In the list box labelled **Function group** click on **Arithmetic** and then in the box labelled **Functions and Special Variables** click on **Ln** (this is the natural log transformation) and transfer it to the command area by clicking on **➤**. When the command is transferred, it appears in the command area as **LN(?)** and the question mark should be replaced with a variable name (which can be typed manually or transferred from the variables list). So replace the question mark with the variable **PSWQ** by either selecting the variable in the list and clicking on **➤** or just typing **PSWQ** where the question mark is. Click on **OK** to create the variable.



SELF-TEST Try creating two new variables that are the natural logs of **Anxious** and **Previous**.

To test the assumption we need to redo the analysis exactly the same as before except that we should force all variables in a single block (i.e. we don't need to do it hierarchically), and we also need to put in three new interaction terms of each predictor and their logs. Select **Analyze > Regression > Binary Logistic...**, then in the main dialog box click on **Scored** and drag it to the **Dependent** box or click on **➤**. Specify the main effects by clicking on **PSWQ**, **Anxious** and **Previous** while holding down the **Ctrl** key and then drag them to the **Covariates** box or click on **➤**. To input the interactions, click on the two variables in the interaction while holding down the **Ctrl** key (e.g. click on **PSWQ** then, while holding down **Ctrl**, click on **Ln(PSWQ)**) and then click on **➤** to move them to the **Covariates** box. This specifies the **PSWQ × Ln(PSWQ)** interaction; specify the **Anxious × Ln(Anxious)** and **Previous × Ln(Previous)** interactions in the same way. The completed dialog box is in Figure 8.7 (note that the final **Previous × Ln(Previous)** interaction isn't visible but it is there!).

SPSS Output 8.9 shows the part of the output that tests the assumption. We're interested only in whether the interaction terms are significant. Any interaction that is significant indicates that the main effect has violated the assumption of linearity of the logit. All three interactions have significance values greater than .05 indicating that the assumption of linearity of the logit has been met for **PSWQ**, **Anxious** and **Previous**.

SPSS OUTPUT 8.9

Variables in the Equation							
Step		B	S.E.	Wald	df	Sig.	Exp(B)
1 ^a	PSWQ	-.422	1.103	.147	1	.702	.656
	Anxious	-2.645	2.797	.894	1	.344	.071
	Previous	1.666	1.482	1.264	1	.261	5.291
	LnPSWQ by PSWQ	.044	.297	.022	1	.882	1.045
	Anxious by LnAnxious	.681	.653	1.088	1	.297	1.975
	LnPrevious by Previous	-.319	.317	1.008	1	.315	.727
	Constant	-3.879	14.924	.068	1	.795	.021

a. Variable(s) entered on step 1: PSWQ, Anxious, Previous, LnPSWQ * PSWQ, Anxious * LnAnxious, LnPrevious * Previous.

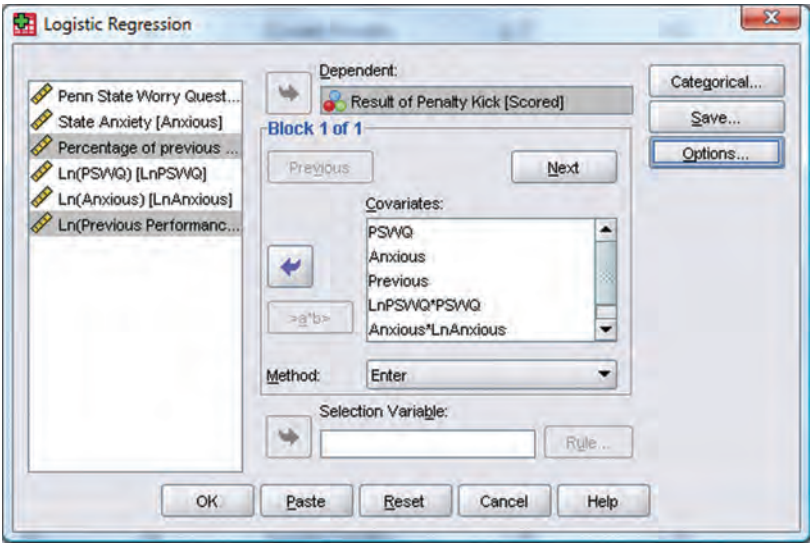


FIGURE 8.7
Dialog box
for testing the
assumption
of linearity
in logistic
regression

8.8.2. Testing for multicollinearity ③

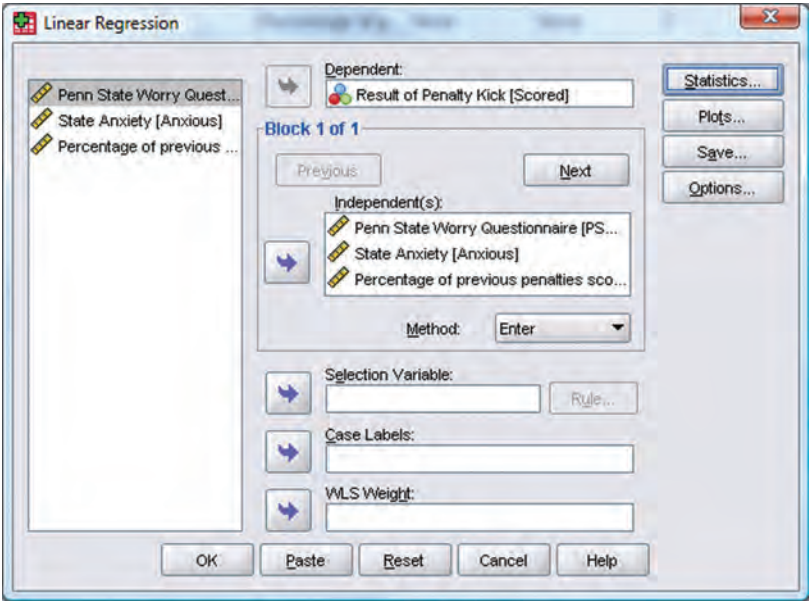
In section 7.6.2.4 we saw how multicollinearity can affect the parameters of a regression model. Logistic regression is just as prone to the biasing effect of collinearity and it is essential to test for collinearity following a logistic regression analysis. Unfortunately, SPSS does not have an option for producing collinearity diagnostics in logistic regression (which can create the illusion that it is unnecessary to test for it!). However, you can obtain statistics such as the tolerance and VIF by simply running a linear regression analysis using the same outcome and predictors. For the penalty example in the previous section, access the *Linear Regression* dialog box by selecting **Analyze** **Regression** **Linear...**. The completed dialog box is shown in Figure 8.8. It is unnecessary to specify lots of options (we are using this technique only to obtain tests of collinearity) but it is essential that you click on **Statistics...** and then select *Collinearity diagnostics* in the dialog box. Once you have selected ☒ *Collinearity diagnostics*, switch off all of the default options, click on **Continue** to return to the *Linear Regression* dialog box, and then click on **OK** to run the analysis.

The results of the linear regression analysis are shown in SPSS Output 8.10. From the first table we can see that the tolerance values are 0.014 for **Previous** and **Anxious** and 0.575 for **PSWQ**. In Chapter 7 we saw various criteria for assessing collinearity. To recap, Menard (1995) suggests that a tolerance value less than 0.1 almost certainly indicates a serious collinearity problem. Myers (1990) also suggests that a VIF value greater than 10 is cause for concern and in these data the values are over 70 for both **Anxious** and **Previous**. It seems from these values that there is an issue of collinearity between the predictor variables. We can investigate this issue further by examining the collinearity diagnostics.

SPSS Output 8.10 also shows a table labelled **Collinearity Diagnostics**. In this table, we are given the eigenvalues of the scaled, uncentred cross-products matrix, the condition index and the variance proportions for each predictor. If any of the eigenvalues in this table are much larger than others then the uncentred cross-products matrix is said to be ill-conditioned, which means that the solutions of the regression parameters can be greatly affected by small changes in the predictors or outcome. In plain English, these values give us some idea as to how accurate our regression model is: if the eigenvalues are fairly similar then the derived model is likely to be unchanged by small changes in the measured variables. The *condition indexes* are another way of expressing these eigenvalues and represent the square root of the ratio of the largest eigenvalue to the eigenvalue of interest (so, for the dimension with the

largest eigenvalue, the condition index will always be 1). For these data the final dimension has a condition index of 81.3, which is massive compared to the other dimensions. Although there are no hard and fast rules about how much larger a condition index needs to be to indicate collinearity problems, this case clearly shows that a problem exists.

FIGURE 8.8
Linear Regression
dialog box for the
penalty data



SPSS OUTPUT 8.10
Collinearity
diagnostics for the
penalty data

Coefficients^a

Model		Collinearity Statistics	
		Tolerance	VIF
1	Penn State Worry Questionnaire	.575	1.741
	State Anxiety	.014	71.764
	Percentage of previous penalties scored	.014	70.479

a. Dependent Variable: Result of Penalty Kick

Collinearity Diagnostics^a

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions			
				(Constant)	Penn State Worry Questionnaire	State Anxiety	Percentage of previous penalties scored
1	1	3.434	1.000	.00	.01	.00	.00
	2	.492	2.641	.00	.04	.00	.00
	3	.073	6.871	.00	.95	.01	.00
	4	.001	81.303	1.00	.00	.99	.99

a. Dependent Variable: Result of Penalty Kick

The final step in analysing this table is to look at the variance proportions. The variance of each regression coefficient can be broken down across the eigenvalues and the variance proportions tell us the proportion of the variance of each predictor's regression coefficient that is attributed to each eigenvalue. These proportions can be converted to percentages by multiplying them by 100 (to make them more easily understood). So, for example, for **PSWQ** 95% of the variance of the regression coefficient is associated with eigenvalue number 3, 4% is associated with eigenvalue number 2 and 1% is associated with eigenvalue number 1. In terms of collinearity, we are looking for predictors that have high proportions

on the same *small* eigenvalue, because this would indicate that the variances of their regression coefficients are dependent. So we are interested mainly in the bottom few rows of the table (which represent small eigenvalues). In this example, 99% of the variance in the regression coefficients of both **Anxiety** and **Previous** is associated with eigenvalue number 4 (the smallest eigenvalue), which clearly indicates dependency between these variables.

The result of this analysis is pretty clear cut: there is collinearity between state anxiety and previous experience of taking penalties and this dependency results in the model becoming biased.



SELF-TEST Using what you learned in Chapter 6, carry out a Pearson correlation between all of the variables in this analysis. Can you work out why we have a problem with collinearity?



If you have identified collinearity then, unfortunately, there's not much that you can do about it. One obvious solution is to omit one of the variables (so, for example, we might stick with the model from block 1 that ignored state anxiety). The problem with this should be obvious: there is no way of knowing which variable to omit. The resulting theoretical



**LABCOAT LENI'S
REAL RESEARCH 8.1**

Mandatory suicide? ②

Although I have fairly eclectic tastes in music, my favourite kind of music is heavy metal. One thing that is mildly irritating about liking heavy music is that everyone assumes that you're a miserable or aggressive bastard. When not listening (and often while listening) to heavy metal, I spend most of my time researching clinical psychology: I research how anxiety develops in children. Therefore, I was literally beside myself with excitement when a few years back I stumbled on a paper that combined these two interests: Lacourse, Claes, and Villeneuve (2001) carried out a study to see whether a love of heavy metal could predict suicide risk. Fabulous stuff!

Eric Lacourse and his colleagues used questionnaires to measure several variables: suicide risk (yes or no), marital status of parents (together or divorced/separated), the extent to which the person's mother and father were neglectful, self-estrangement/powerlessness (adolescents who have negative self-perceptions, bored with life, etc.), social isolation (feelings of a lack of support), normlessness (beliefs that socially disapproved behaviours can be used

to achieve certain goals), meaninglessness (doubting that school is relevant to gain employment) and drug use. In addition, the author measured liking of heavy metal; they included the sub-genres of classic (Black Sabbath, Iron Maiden), thrash metal (Slayer, Metallica), death/black metal (Obituary, Burzum) and gothic (Marilyn Manson). As well as liking they measured behavioural manifestations of worshipping these bands (e.g. hanging posters, hanging out with other metal fans) and vicarious music listening (whether music was used when angry or to bring out aggressive moods). They used logistic regression to predict suicide risk from these predictors for males and females separately.

The data for the female sample are in the file **Lacourse et al. (2001) Females.sav**. Labcoat Leni wants you to carry out a logistic regression predicting **Suicide_Risk** from all of the predictors (forced entry). (To make your results easier to compare to the published results, enter the predictors in the same order as Table 3 in the paper: **Age, Marital_Status, Mother_Negligence, Father_Negligence, Self_Estrangement, Isolation, Normlessness, Meaninglessness, Drug_Use, Metal, Worshipping, Vicarious**). Create a table of the results. Does listening to heavy metal make girls suicidal? If not, what does?

Answers are in the additional material on the companion website (or look at Table 3 in the original article).



conclusions are meaningless because, statistically speaking, any of the collinear variables could be omitted. There are no statistical grounds for omitting one variable over another. Even if a predictor is removed, Bowerman and O’Connell (1990) recommend that another equally important predictor that does not have such strong multicollinearity replaces it. They also suggest collecting more data to see whether the multicollinearity can be lessened. Another possibility when there are several predictors involved in the multicollinearity is to run a factor analysis on these predictors and to use the resulting factor scores as a predictor (see Chapter 17). The safest (although unsatisfactory) remedy is to acknowledge the unreliability of the model. So, if we were to report the analysis of which factors predict penalty success, we might acknowledge that previous experience significantly predicted penalty success in the first model, but propose that this experience might affect penalty taking by increasing state anxiety. This statement would be highly speculative because the correlation between **Anxious** and **Previous** tells us nothing of the direction of causality, but it would acknowledge the inexplicable link between the two predictors. I’m sure that many of you may find the lack of remedy for collinearity grossly unsatisfying – unfortunately statistics is frustrating sometimes!

8.9. Predicting several categories: multinomial logistic regression ③

I mentioned earlier that it is possible to use logistic regression to predict membership of more than two categories and that this is called multinomial logistic regression.

Essentially, this form of logistic regression works in the same way as binary logistic regression so there’s no need for any additional equations to explain what is going on (hooray!). The analysis breaks the outcome variable down into a series of comparisons between two categories (which helps explain why no extra equations are really necessary). For example, if you have three outcome categories (A, B and C), then the analysis will consist of two comparisons. The form that these comparisons take depends on how you specify the analysis: you can compare everything against your first category (e.g. A vs. B and A vs. C), or your last category (e.g. A vs. C and B vs. C), or a custom category, for example category B (e.g. B vs. A and B vs. C). In practice, this means that you have to select a baseline category.

The important parts of the analysis and output are much the same as we have just seen for binary logistic regression.

Let’s look at an example. There has been some recent work looking at how men and women evaluate chat-up lines (Bale, Morrison, & Caryl, 2006; Cooper, O’Donnell, Caryl, Morrison, & Bale, 2007). This research has looked at how the content (e.g. whether the chat-up line is funny, has sexual content or reveals desirable personality characteristics) affects how favourably the chat-up line is viewed. To sum up this research, it has found that men and women like different things in chat-up lines: men prefer chat-up lines with a high sexual content and women prefer chat-up lines that are funny and show good moral fibre!

Imagine that we wanted to assess how *successful* these chat-up lines were. We did a study in which we recorded the chat-up lines used by 348 men and 672 women in a night-club. Our outcome was whether the chat-up line resulted in one of the following three events: the person got no response or the recipient walked away, the person obtained the recipient’s phone number, or the person left the night-club with the recipient. Afterwards, the chat-up lines used in each case were rated by a panel of judges for how funny they were

What do I do when I have more than two outcome categories?



(0 = not funny at all, 10 = the funniest thing that I have ever heard), sexuality (0 = no sexual content at all, 10 = very sexually direct) and whether the chat-up line reflected good moral values (0 = the chat-up line does not reflect good characteristics, 10 = the chat-up line is very indicative of good characteristics). For example, ‘I may not be Fred Flintstone, but I bet I could make your bed rock’ would score high on sexual content, low on good characteristics and medium on humour; ‘I’ve been looking all over for YOU, the woman of my dreams’ would score high on good characteristics, low on sexual content and low on humour (but high on cheese had it been measured). We predict based on past research that the success of different types of chat-up line will interact with gender.

This situation is perfect for multinomial regression. The data are in the file **Chat-Up Lines.sav**. There is one outcome variable (**Success**) with three categories (no response, phone number, go home with recipient) and four predictors: funniness of the chat-up line (**Funny**), sexual content of the chat-up line (**Sex**), degree to which the chat-up line reflects good characteristics (**Good_Mate**) and the gender of the person being chatted up (**Gender**).



8.9.1. Running multinomial logistic regression in SPSS ③

To run multinomial logistic regression in SPSS, first select the main dialog box by selecting **Analyze > Regression > Multinomial Logistic...**. In this dialog box there are spaces to place the outcome variable (*Dependent*), any categorical predictors (*Factor(s)*) and any continuous predictors (*Covariate(s)*). In this example, the outcome variable is **Success** so select this variable from the list and transfer it to the box labelled *Dependent* by dragging it there or clicking on . We also have to tell SPSS whether we want to compare categories against the first category or the last and we do this by clicking on **Reference Category...**.

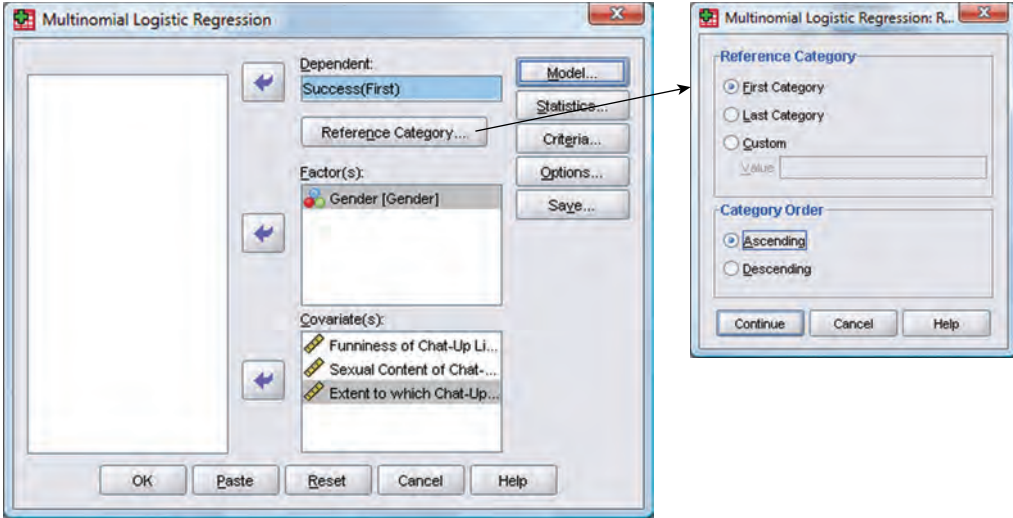


SELF-TEST Think about the three categories that we have as an outcome variable. Which of these categories do you think makes most sense to use as a baseline category?

By default SPSS uses the last category, but in our case it makes most sense to use the first category (No response/walk off) because this category represents failure (the chat-up line did not have the desired effect) whereas the other two categories represent some form of success (getting a phone number or leaving the club together). To change the reference category to be the first category, click on **First Category** and then click on **Continue** to return to the main dialog box (Figure 8.9).

Next we have to specify the predictor variables. We have one categorical predictor variable, which is **Gender**, so select this variable next and transfer it to the box labelled *Factor(s)* by dragging it there or clicking on . Finally, we have three continuous predictors or covariates (**Funny**, **Sex** and **Good_Mate**). You can select all of these variables simultaneously by holding down the *Ctrl* key as you click on each one. Drag all three to the box labelled *Covariate(s)* or click on . For a basic analysis in which all of these predictors are forced into the model, this is all we really need to do. However, as we saw in the regression chapter you will often want to do a hierarchical regression and so for this analysis we’ll look at how to do this in SPSS.

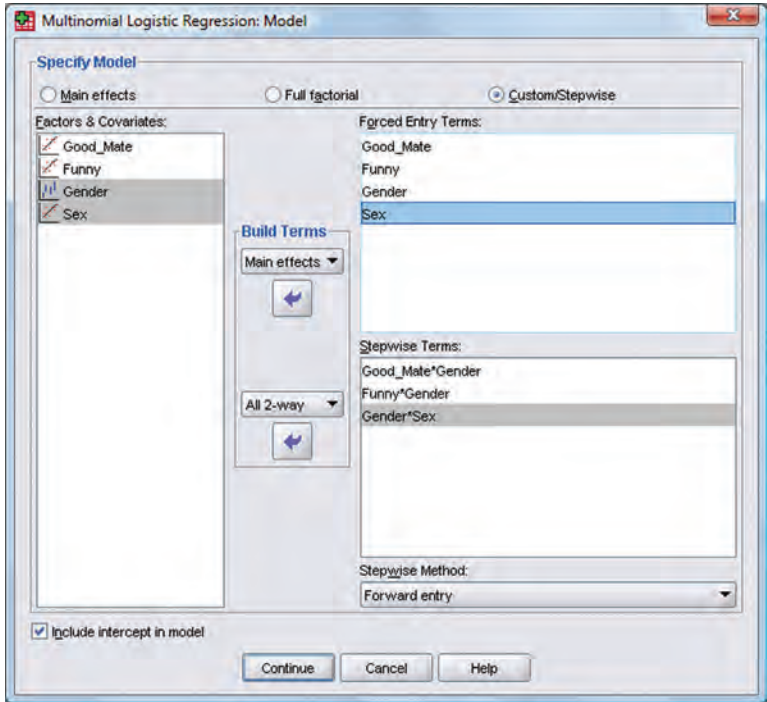
FIGURE 8.9
Main dialog box
for multinomial
logistic
regression



8.9.1.1. Customizing the model ③

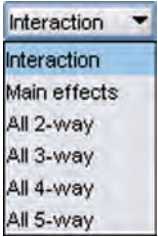
In binary logistic regression, SPSS allowed us to specify interactions between predictor variables in the main dialog box, but with multinomial logistic regression we cannot do this. Instead we have to specify what SPSS calls a ‘custom model’, and this is done by clicking on **Model...** to open the dialog box in Figure 8.10. You’ll see that, by default, SPSS just looks at the main effects of the predictor variables. In this example, however, the main effects are not particularly interesting: based on past research we don’t necessarily expect funny chat-up lines to be successful, but we do expect them to be more successful when

FIGURE 8.10
Specifying a
custom model



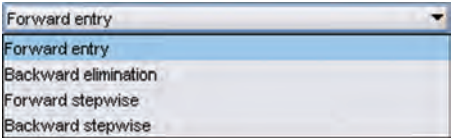
used on women than on men. What this prediction implies is that the *interaction* of **Gender** and **Funny** will be significant. Similarly, chat-up lines with a high sexual content might not be successful overall, but expect them to be relatively successful when used on men. Again, this means that we might not expect the **Sex** main effect to be significant, but we do expect the **Sex**×**Gender** interaction to be significant. As such, we need to enter some interaction terms into the model.

To customize the model we first have to select **Custom/Stepwise** to activate the rest of the dialog box. There are two main ways that we can specify terms: we can force them in (by moving them to the box labelled *Forced Entry Terms*) or we can put them into the model using a stepwise procedure (by moving them into the box labelled *Stepwise terms*). If we want to look at interaction terms, we must force the main effects into the model. If we look at interactions without the corresponding main effects being in the model then we allow the interaction term to explain variance that might otherwise be attributed to the main effect (in other words, we’re not really looking at the interaction any more). So, select all of the variables in the box labelled *Factors and covariates* by clicking on them while holding down **Ctrl** (or selecting the first variable and then clicking on the last variable while holding down **Shift**). There is a drop-down list that determines whether you transfer these effects as main effects or interactions. We want to transfer them as main effects so set this box to **Main effects** and click on **➤**.



To specify interactions we can do much the same: we can select two or more variables and then set the drop-down box to be **Interaction** and then click on **➤**. If, for example, we selected **Funny** and **Sex**, then doing this would specify the **Funny**×**Sex** interaction. We can also specify multiple interactions at once. For example, if we selected **Funny**, **Sex** and **Gender** and then set the drop-down box to **All 2-way**, it would transfer *all* of the interactions involving two variables (i.e. **Funny**×**Sex**, **Funny**×**Gender** and **Sex**×**Gender**). You get the general idea. We could also select **Full factorial** which would automatically enter all main effects (**Funny**, **Sex**, **Good_Mate**, **Gender**), all interactions with two variables (**Funny**×**Sex**, **Funny**×**Gender**, **Funny**×**Good_Mate**, **Sex**×**Gender**, **Sex**×**Good_Mate**, **Gender**×**Good_Mate**), all interactions with three variables (**Funny**×**Sex**×**Gender**, **Funny**×**Sex**×**Good_Mate**, **Good_Mate**×**Sex**×**Gender**, **Funny**×**Good_Mate**×**Gender**) and the interaction of all four variables (**Funny**×**Sex**×**Gender**×**Good_Mate**).

In this scenario, we want to specify interactions between the ratings of the chat-up lines and gender only (we’re not interested in any interactions involving three variables, or all four variables). We can either force these interaction terms into the model by putting them in the box labelled *Forced Entry Terms* or we can put them into the model using a stepwise procedure (by moving them into the box labelled *Stepwise terms*). We’re going to do the latter, so interactions will be entered into the model only if they are significant predictors of the success of the chat-up line. Let’s first enter the **Funny**×**Gender** interaction first. Click on **Funny** and then **Gender** in the *Factors and covariates* box while holding down the **Ctrl** key. Then next to the box labelled *Stepwise terms* change the drop-down menu to be **Interaction** and then click on **➤**. You should now see **Gender**×**Funny** (SPSS orders the variables in reverse alphabetical order for some reason) listed in the *Stepwise terms* box. Specify the **Sex**×**Gender** and **Good_Mate**×**Gender** in the same way. Once the three interaction terms have been specified we can decide how we want to carry out the stepwise analysis. There is a drop-down list of methods under the heading *Stepwise Method* and this list enables you to choose between forward and stepwise entry, or backward and stepwise elimination (i.e. terms are removed from the model if they do not make a significant contribution). I’ve described these methods elsewhere, so select forward entry for this analysis.



8.9.2. Statistics ③

If you click on **Statistics...** you will see the dialog box in Figure 8.11, in which you can specify certain statistics:

- *Pseudo R-square*: This produces the Cox and Snell and Nagelkerke R^2 statistics. These can be used as effect sizes so this is a useful option to select.
- *Step summary*: This option should be selected for the current analysis because we have a stepwise component to the model; this option produces a table that summarizes the predictors entered or removed at each step.
- *Model fitting information*: This option produces a table that compares the model (or models in a stepwise analysis) to the baseline (the model with only the intercept term in it and no predictor variables). This table can be useful to compare whether the model had improved (from the baseline) as a result of entering the predictors that you have.
- *Information criteria*: This option produces Akaike's information criterion (AIC) and Schwarz's Bayesian information criterion (BIC). Both of these statistics are a useful way to compare models. Therefore, if you're using stepwise methods, or if you want to compare different models containing different combinations of predictors, then select this option. Low values of the AIC and BIC indicate good fit; therefore, models with lower values of the AIC or BIC fit the data relatively better than models with higher values.
- *Cell probabilities*: This option produces a table of the observed and expected frequencies. This is basically the same as the classification table produced in binary logistic regression and is probably worth inspecting.
- *Classification table*: This option produces a contingency table of observed versus predicted responses for all combinations of predictor variables. I wouldn't select this option, unless you're running a relatively small analysis (i.e. a small number of predictors made up of a small number of possible values). In this example, we have three covariates with 11 possible values and one predictor (gender) with 2 possible values. Tabulating all combinations of these variables will create a very big table indeed.
- *Goodness-of-fit*: This option is important because it produces Pearson and likelihood-ratio chi-square statistics for the model.
- *Monotonicity measures*: This option is worth selecting only if your outcome variable has two outcomes (which in our case it doesn't). It will produce measures of monotonic association such as the concordance index, which measures the probability that, using a previous example, a person who scored a penalty kick is classified by the model as having scored and can range from .5 (guessing) to 1 (perfect prediction).
- *Estimates*: This option produces the beta values, test statistics and confidence intervals for predictors in the model. This option is very important.
- *Likelihood ratio tests*: The model overall is tested using likelihood ratio statistics, but this option will compute the same test for individual effects in the model. (Basically it tells us the same as the significance values for individual predictors.)
- *Asymptotic correlations and covariances*: These produce a table of correlations (or covariances) between the betas in the model.

Set the options as in Figure 8.11 and click on **Continue** to return to the main dialog box.

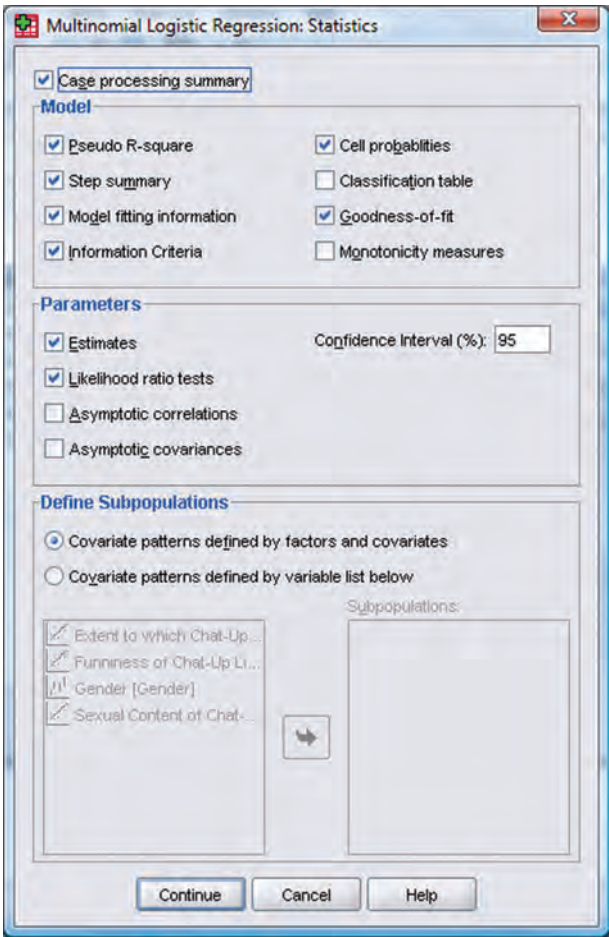


FIGURE 8.11
Statistics options
for multinomial
logistic
regression

8.9.3. Other options ③

If you click on **Criteria...** you'll access the dialog box in Figure 8.12 (left). Logistic regression works through an iterative process (SPSS Tip 8.1). The options available here relate to this process. For example, by default, SPSS will make 100 attempts (iterations) and the threshold for how similar parameter estimates have to be to 'converge' can be made more or less strict (the default is .0000001). You should leave these options alone unless when you run the analysis you get an error message saying something about 'failing to converge', in which case you could try increasing the *Maximum iterations* (to 150 or 200), the *Parameter convergence* (to .00001) or *Log-likelihood convergence* (to greater than 0). However, bear in mind that a failure to converge can reflect messy data and forcing the model to converge does not necessarily mean that parameters are accurate or stable across samples.

You can also click on **Options...** in the main dialog box to access the dialog box in Figure 8.12 (right). The *Scale* option here can be quite useful; I mentioned in section 8.4.4 that overdispersion can be a problem in logistic regression because it reduces the standard errors that are used to test the significance and construct the confidence intervals of the parameter estimates for individual predictors in the model. I also mentioned that this problem could be counteracted by rescaling the standard errors. Should you be in a situation where you need to do this (i.e. you have run the analysis and found evidence of overdispersion)

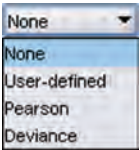


FIGURE 8.12
Criteria and options for multinomial logistic regression

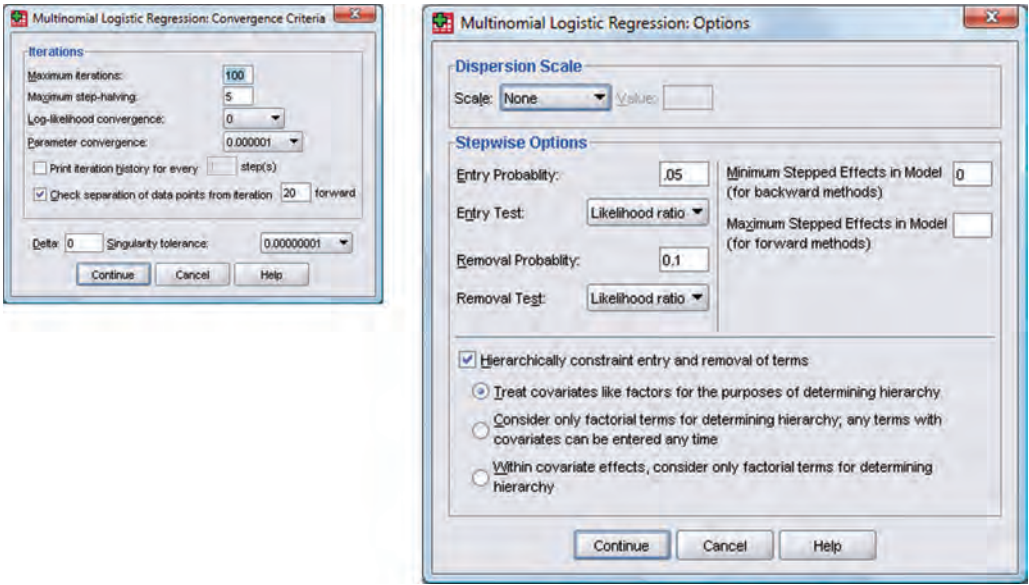
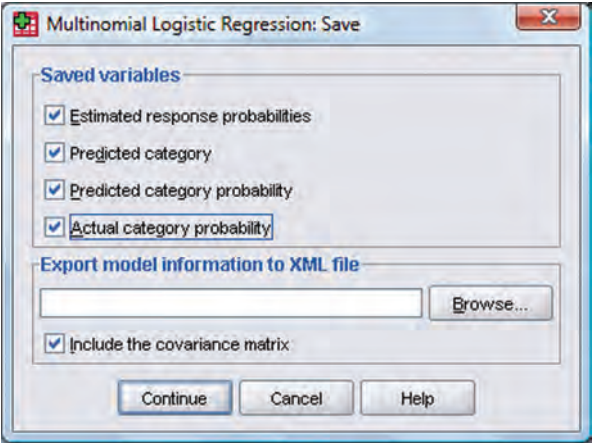


FIGURE 8.13
Save options for multinomial logistic regression



then you need to come to this dialog box and use the drop-down list to select to correct the standard errors by the dispersion parameter based on either the *Deviance* or *Pearson* statistic. You should select whichever of these two statistics was bigger in the original analysis (because this will produce the bigger correction).

Finally, if you click on *Save...* you can opt to save predicted probabilities and predicted group membership (the same as for binary logistic regression except that they are called *Estimated response probabilities* and *Predicted category* (Figure 8.13)).

8.9.4. Interpreting the multinomial logistic regression output ③

Our SPSS output from this analysis begins with a warning (SPSS Tip 8.2). It's always nice after months of preparation, weeks entering data, years reading chapters of stupid statistics



SPSS TIP 8.2

Warning! Zero frequencies ③

Warnings

There are 504 (53.5%) cells (i.e., dependent variable levels by subpopulations) with zero frequencies.

Sometimes in logistic regression you get a warning about zero frequencies. This relates to the problem that I discussed in section 8.4.2 of 80 year old, highly anxious, Buddhist left-handed lesbians (well, incomplete information). Imagine we had just looked at gender as a predictor of chat-up line success. We have three outcome categories and two gender categories. There are six possible combinations of these two variables and ideally we would like a large number of observations in each of these combinations. However, in this case, we have three variables (**Funny**, **Sex** and **Good_Mate**) with 11 possible outcomes, and **Gender** with 2 possible outcomes and an outcome variable with 3 outcomes. It should be clear that by including the three covariates, the number of combinations of these variables has escalated considerably. This error message tells us that there are some combinations of these variables for which there are no observations. So, we really have a situation where we didn't find an 80 year old, highly anxious, Buddhist left-handed lesbian; well, we didn't find (for example) a chat-up line that was the most funny, showed the most good characteristics, had the most sexual content and was used on both a man and woman! In fact 53.5% of our possible combinations of variables had no data!

Whenever you have covariates it is inevitable that you will have empty cells, so you will get this kind of error message. To some extent, given its inevitability, we can just ignore it (in this study, for example, we have 1020 cases of data and half of our cells are empty, so we would need to at least double the sample size to stand any chance of filling those cells). However, it is worth reiterating what I said earlier that empty cells create problems and that when you get a warning like this you should look for coefficients that have unreasonably large standard errors and if you find them be wary of them.

textbooks, and sleepless nights with equations slicing at your brain with little pick-axes, to see at the start of your analysis: 'Warning! Warning! abandon ship! flee for your life! bad data alert! bad data alert!' Still, such is life.

Once, like all good researchers, we have ignored the warnings then the first part of the output tells us about our model overall (SPSS Output 8.11). First, because we requested a stepwise analysis for our interaction terms, we get a table summarizing the steps in the analysis. You can see here that after the main effects were entered (Model 0), the **Gender**×**Funny** interaction term was entered (Model 1) followed by the **Sex**×**Gender** interaction (Model 2). The chi-square statistics for each of these steps are highly significant, indicating that these interactions have a significant effect on predicting whether a chat-up line was significant (this fact is also self-evident because these terms wouldn't have been entered into the model had they not been significant). Also note that the AIC gets smaller as these terms are added to the model, indicating that the fit of the model is getting better as these terms are added (the BIC changes less, but still shows that having the interaction terms in the model results in a better fit than when just the main effects are present). Underneath the step summary, we see the statistics for the final model, which replicates the model-fitting criteria from the last line of the step summary table. This table also produces a likelihood ratio test of the overall model.



SELF-TEST What does the log-likelihood measure?



Remember that the log-likelihood is a measure of how much unexplained variability there is in the data; therefore, the difference or change in log-likelihood indicates how much new variance has been explained by the model. The chi-square test tests the decrease in unexplained variance from the baseline model (1149.53) to the final model (871.00), which is a difference of $1149.53 - 871 = 278.53$. This change is significant, which means that our final model explains a significant amount of the original variability (in other words, it's a better fit than the original model).

SPSS OUTPUT 8.11

Step Summary								
Model	Action	Effect(s)	Model Fitting Criteria			Effect Selection Tests		
			AIC	BIC	-2 Log Likelihood	Chi-Square ^a	df	Sig.
0	Entered	Intercept, Good_Male, Funny, Gender, Sex	937.572	986.848	917.572			
1	Entered	Gender * Funny	908.451	967.582	884.451	33.121	2	.000
2	Entered	Gender * Sex	899.002	967.987	871.002	13.450	2	.001

Stepwise Method: Forward Entry

a. The chi-square for entry is based on the likelihood ratio test.

SPSS OUTPUT 8.12

Model Fitting Information						
Model	Model Fitting Criteria			Likelihood Ratio Tests		
	AIC	BIC	-2 Log Likelihood	Chi-Square	df	Sig.
Intercept Only	1.154E3	1.163E3	1149.526			
Final	899.002	967.987	871.002	278.525	12	.000

Goodness-of-Fit			
	Chi-Square	df	Sig.
Pearson	886.616	614	.000
Deviance	617.481	614	.453

Pseudo R-Square	
Cox and Snell	.239
Nagelkerke	.277
McFadden	.138

The next part of the output (SPSS Output 8.12) relates to the fit of the model to the data. We know that the model is significantly better than no model, but is it a good fit of the data? The Pearson and deviance statistics test the same thing, which is whether the predicted values from the model differ significantly from the observed values. If these statistics are not significant then the predicted values are not significantly different from the observed values; in other words, the model is a good fit. Here we have contrasting results: the deviance statistic says that the model is a good fit of the data ($p = .45$, which is much higher than $.05$), but the Pearson test indicates the opposite, namely that predicted values are significantly different from the observed values ($p < .001$). Oh dear.



SELF-TEST Why might the Pearson and deviance statistics be different? What could this be telling us?

One answer is that differences between these statistics can be caused by overdispersion. This is a possibility that we need to look into. However, there are other reasons for this conflict: for example, the Pearson statistic can be very inflated by low expected frequencies (which could

happen because we have so many empty cells as indicated by our warning). One thing that is certain is that conflicting deviance and Pearson chi-square statistics are not good news!

Let’s look into the possibility of overdispersion. We can compute the dispersion parameters from both statistics:

$$\phi_{\text{Pearson}} = \frac{\chi^2_{\text{Pearson}}}{df} = \frac{886.62}{614} = 1.44$$
$$\phi_{\text{Deviance}} = \frac{\chi^2_{\text{Deviance}}}{df} = \frac{617.48}{614} = 1.01$$

Neither of these is particularly high, and the one based on the deviance statistic is close to the ideal value of 1. The value based on Pearson is greater than 1, but not close to 2, so again does not give us an enormous cause for concern that the data are overdispersed.⁹

The output also shows us the two other measures of R^2 that were described in section 8.3.2. The first is Cox and Snell’s measure, which SPSS reports as .24, and the second is Nagelkerke’s adjusted value, which SPSS reports as .28. As you can see, they are reasonably similar values and represent relatively decent-sized effects.

SPSS Output 8.13 shows the results of the likelihood ratio tests and these can be used to ascertain the significance of predictors to the model. The first thing to note is that no significance values are produced for covariates that are involved in higher-order interactions (this is why there are blank spaces in the Sig. column for the effects of **Funny** and **Sex**). This table tells us, though, that gender had a significant main effect on success rates of chat-up lines, $\chi^2(2) = 18.54, p < .001$, as did whether the chat-up lined showed evidence of being a good partner, $\chi^2(2) = 6.32, p < .042$. Most interesting are the interactions which showed that the humour in the chat-up line interacted with gender to predict success at getting a date, $\chi^2(2) = 35.81, p < .001$; also the sexual content of the chat-up line interacted with the gender of the person being chatted up in predicting their reaction, $\chi^2(2) = 13.45, p < .001$. These likelihood statistics can be seen as sorts of overall statistics that tell us which predictors significantly enable us to predict the outcome category, but they don’t really tell us specifically what the effect is. To see this we have to look at the individual parameter estimates.

Likelihood Ratio Tests

SPSS OUTPUT 8.13

Effect	Model Fitting Criteria			Likelihood Ratio Tests		
	AIC of Reduced Model	BIC of Reduced Model	-2 Log Likelihood of Reduced Model	Chi-Square	df	Sig.
Intercept	899.002	967.987	8.710E2 ^a	.000	0	
Good_Mate	901.324	960.454	877.324	6.322	2	.042
Funny	899.002	967.987	8.710E2 ^a	.000	0	
Gender	913.540	972.671	889.540	18.538	2	.000
Sex	899.002	967.987	8.710E2 ^a	.000	0	
Gender * Funny	930.810	989.941	906.810	35.808	2	.000
Gender * Sex	908.451	967.582	884.451	13.450	2	.001

The chi-square statistic is the difference in -2 log-likelihoods between the final model and a reduced model. The reduced model is formed by omitting an effect from the final model. The null hypothesis is that all parameters of that effect are 0.

a. This reduced model is equivalent to the final model because omitting the effect does not increase the degrees of freedom.

⁹ Incidentally, large dispersion parameters can occur for reasons other than overdispersion, for example omitted variables or interactions (in this example there were several interaction terms that we could have entered but chose not to), and predictors that violate the linearity of the logit assumption.

SPSS Output 8.14 shows the individual parameter estimates. Note that the table is split into two halves. This is because these parameters compare pairs of outcome categories. We specified the first category as our reference category; therefore, the part of the table labelled *Get Phone Number* is comparing this category against the ‘No response/walked away’ category. Let’s look at the effects one by one; because we are just comparing two categories the interpretation is the same as for binary logistic regression (so if you don’t understand my conclusions reread the start of this chapter):

- **Good_Mate:** Whether the chat-up line showed signs of good moral fibre significantly predicted whether you got a phone number or no response/walked away, $b = 0.13$, Wald $\chi^2(1) = 6.02$, $p < .05$. The odds ratio tells us that as this variable increases, so as chat-up lines show one more unit of moral fibre, the change in the odds of getting a phone number (rather than no response/walked away) is 1.14. In short, you’re more likely to get a phone number than not if you use a chat-up line that demonstrates good moral fibre. (Note that this effect is superseded by the interaction with gender below.)
- **Funny:** Whether the chat-up line was funny did not significantly predict whether you got a phone number or no response, $b = 0.14$, Wald $\chi^2(1) = 1.60$, $p > .05$. Note that although this predictor is not significant, the odds ratio is approximately the same as for the previous predictor (which was significant). So, the effect size is comparable, but the non-significance stems from a relatively higher standard error.
- **Gender:** The gender of the person being chatted up significantly predicted whether they gave out their phone number or gave no response, $b = -1.65$, Wald $\chi^2(1) = 4.27$, $p < .05$. Remember that 0 = female and 1 = male, so this is the effect of females compared to males. The odds ratio tells us that as gender changes from female (0) to male (1) the change in the odds of giving out a phone number compared to not responding is 0.19. In other words, the odds of a man giving out his phone number compared to not responding are $1/0.19 = 5.26$ times more than for a woman. Men are cheap.
- **Sex:** The sexual content of the chat-up line significantly predicted whether you got a phone number or no response/walked away, $b = 0.28$, Wald $\chi^2(1) = 9.59$, $p < .01$. The odds ratio tells us that as the sexual content increased by a unit, the change in the odds of getting a phone number (rather than no response) is 1.32. In short, you’re more likely to get a phone number than not if you use a chat-up line with high sexual content. (But this effect is superseded by the interaction with gender.)
- **Funny×Gender:** The success of funny chat-up lines depended on whether they were delivered to a man or a woman because in interaction these variables predicted whether or not you got a phone number, $b = 0.49$, Wald $\chi^2(1) = 12.37$, $p < .001$. Bearing in mind how we interpreted the effect of gender above, the odds ratio tells us that as gender changes from female (0) to male (1) in combination with funniness increasing, the change in the odds of giving out a phone number compared to not responding was 1.64. In other words, as funniness increases, women become more likely to hand out their phone number than men. Funny chat-up lines are more successful when used on women than men.
- **Sex×Gender:** The success of chat-up lines with sexual content depended on whether they were delivered to a man or a woman because in interaction these variables predicted whether or not you got a phone number, $b = -0.35$, Wald $\chi^2(1) = 10.82$, $p < .01$. Bearing in mind how we interpreted the interaction above (note that b is negative here but positive above), the odds ratio tells us that as gender changes from female (0) to male (1) in combination with the sexual content increasing, the change in the odds of giving out a phone number compared to not responding is 0.71. In other words, as sexual content increases, women become *less* likely than men to hand out their phone number. Chat-up lines with a high sexual content are more successful when used on men than women.

SPSS OUTPUT 8.14

Parameter Estimates								
		B	Std. Error	Wald	df	Sig.	Exp(B)	95% Confidence Interval for Exp(B) Lower Bound Upper Bound
Success of Chat-Up Line ^a								
Get Phone Number	Intercept	-1.783	.870	7.087	1	.008		
	Good_Male	.132	.054	6.022	1	.014	1.141	1.027 1.268
	Funny	.139	.110	1.602	1	.206	1.150	.926 1.427
	[Gender=0]	-1.646	.796	4.274	1	.039	.193	.040 .918
	[Gender=1]	0 ^b			0			
	Sex	.276	.089	9.589	1	.002	1.318	1.107 1.570
	[Gender=0] * Funny	.492	.140	12.374	1	.000	1.636	1.244 2.153
	[Gender=1] * Funny	0 ^b			0			
	[Gender=0] * Sex	-.348	.106	10.824	1	.001	.706	.574 .869
	[Gender=1] * Sex	0 ^b			0			
Go Home with Person								
	Intercept	-4.286	.941	20.731	1	.000		
	Good_Male	.130	.084	2.423	1	.120	1.139	.967 1.341
	Funny	.318	.125	6.459	1	.011	1.375	1.076 1.758
	[Gender=0]	-5.626	1.329	17.934	1	.000	.004	.000 .049
	[Gender=1]	0 ^b			0			
	Sex	.417	.122	11.683	1	.001	1.518	1.195 1.928
	[Gender=0] * Funny	1.172	.199	34.627	1	.000	3.230	2.186 4.773
	[Gender=1] * Funny	0 ^b			0			
	[Gender=0] * Sex	-.477	.163	8.505	1	.004	.621	.451 .855
	[Gender=1] * Sex	0 ^b			0			

a. The reference category is: No response/Walk Off.

b. This parameter is set to zero because it is redundant.

The bottom half of SPSS Output 8.14 shows the individual parameter estimates for the *Go Home with Person* category compared to the ‘No response/walked away’ category. We can interpret these effects as follows:

- **Good_Mate:** Whether the chat-up line showed signs of good moral fibre did not significantly predict whether you went home with the date or got a slap in the face, $b = 0.13$, Wald $\chi^2(1) = 2.42$, $p > .05$. In short, you’re not significantly more likely to go home with the person if you use a chat-up line that demonstrates good moral fibre.
- **Funny:** Whether the chat-up line was funny significantly predicted whether you went home with the date or no response, $b = 0.32$, Wald $\chi^2(1) = 6.46$, $p < .05$. The odds ratio tells us that as chat-up lines are one more unit funnier, the change in the odds of going home with the person (rather than no response) is 1.38. In short, you’re more likely to go home with the person than get no response if you use a chat-up line that is funny. (This effect, though, is superseded by the interaction with gender below.)
- **Gender:** The gender of the person being chatted up significantly predicted whether they went home with the person or gave no response, $b = -5.63$, Wald $\chi^2(1) = 17.93$, $p < .001$. The odds ratio tells us that as gender changes from female (0) to male (1), the change in the odds of going home with the person compared to not responding is 0.004. In other words, the odds of a man going home with someone compared to not responding are $1/0.004 = 250$ times more likely than for a woman. Men are *really* cheap.
- **Sex:** The sexual content of the chat-up line significantly predicted whether you went home with the date or got a slap in the face, $b = 0.42$, Wald $\chi^2(1) = 11.68$, $p < .01$. The odds ratio tells us that as the sexual content increased by a unit, the change in the odds of going home with the person (rather than no response) is 1.52: you’re more likely to go home with the person than not if you use a chat-up line with high sexual content.
- **Funny×Gender:** The success of funny chat-up lines depended on whether they were delivered to a man or a woman because in interaction these variables predicted whether or not you went home with the date, $b = 1.17$, Wald $\chi^2(1) = 34.63$, $p < .001$. The odds ratio tells us that as gender changes from female (0) to male (1) in combination with funniness increasing, the change in the odds of going home with the person compared to not responding is 3.23. As funniness increases, women become more likely to go home with the person than men. Funny chat-up lines are more successful when used on women compared to men.

- **Sex × Gender:** The success of chat-up lines with sexual content depended on whether they were delivered to a man or a woman because in interaction these variables predicted whether or not you went home with the date, $b = -0.48$, Wald $\chi^2(1) = 8.51$, $p < .01$. The odds ratio tells us that as gender changes from female (0) to male (1) in combination with the sexual content increasing, the change in the odds of going home with the date compared to not responding is 0.62. As sexual content increases, women become less likely than men to go home with the person. Chat-up lines with sexual content are more successful when used on men than women.



SELF-TEST Use what you learnt earlier in this chapter to check the assumptions of multicollinearity and linearity of the logit

8.9.5. Reporting the results

We can report the results as with binary logistic regression using a table (see Table 8.4). Note that I have split the table by the outcome categories being compared, but otherwise it is the same as before. These effects are interpreted as in the previous section.

TABLE 8.4 How to report multinomial logistic regression

		95% CI for Odds Ratio		
	B (SE)	Lower	Odds Ratio	Upper
Phone Number vs. No Response				
Intercept	−1.78 (0.67)**			
Good Mate	.13 (0.05)*	1.03	1.14	1.27
Funny	.14 (0.11)	.93	1.15	1.43
Gender	−1.65 (0.80)*	.04	.19	.92
Sexual Content	.28 (0.09)**	1.11	1.32	1.57
Gender × Funny	.49 (0.14)***	1.24	1.64	2.15
Gender × Sex	−.35 (0.11)**	.57	.71	.87
Going Home vs. No Response				
Intercept	−4.29 (0.94)***			
Good Mate	.13 (0.08)	.97	1.14	1.34
Funny	.32 (0.13)*	1.08	1.38	1.76
Gender	−5.63 (1.33)***	.00	.00	.05
Sexual Content	.42 (0.12)**	1.20	1.52	1.93
Gender × Funny	1.17 (0.20)***	2.19	3.23	4.77
Gender × Sex	−.48 (0.16)**	.45	.62	.86

Note: $R^2 = .24$ (Cox & Snell), $.28$ (Nagelkerke). Model $\chi^2(12) = 278.53$, $p < .001$. * $p < .05$, ** $p < .01$, *** $p < .001$.

What have I discovered about statistics? ①

At the age of 10 I thought I was going to be a rock star. Such was my conviction about this that even today (many years on) I’m still not entirely sure how I ended up *not* being a rock star (lack of talent, not being a very cool person, inability to write songs that don’t make people want to throw rotting vegetables at you, are all possible explanations). Instead of the glitzy and fun life that I anticipated I am instead reduced to writing chapters about things that I don’t even remotely understand.

We began the chapter by looking at why we can’t use linear regression when we have a categorical outcome, but instead have to use binary logistic regression (two outcome categories) or multinomial logistic regression (several outcome categories). We then looked into some of the theory of logistic regression by looking at the regression equation and what it means. Then we moved onto assessing the model and talked about the log-likelihood statistic and the associated chi-square test. I talked about different methods of obtaining equivalents to R^2 in regression (Hosmer & Lemeshow, Cox & Snell and Nagelkerke). We also discovered the Wald statistic and $Exp(B)$. The rest of the chapter looked at three examples using SPSS to carry out various logistic regression. So, hopefully, you should have a pretty good idea of how to conduct and interpret a logistic regression by now.

Having decided that I was going to be a rock star I put on my little denim jacket with Iron Maiden patches sewn onto it and headed off down the rocky road of stardom. The first stop was ... my school.

Key terms that I’ve discovered

-2LL	Log-likelihood
Binary logistic regression	Main effect
Chi-square distribution	Maximum-likelihood estimation
Complete separation	Multinomial logistic regression
Cox and Snell’s R^2_{CS}	Nagelkerke’s R^2_N
$Exp(B)$	Odds
Hosmer and Lemeshow’s R^2_L	Polychotomous logistic regression
Interaction effect	Roa’s efficient score statistic
Likelihood	Suppressor effects
Logistic regression	Wald statistic

Smart Alex’s tasks

- **Task 1:** A psychologist was interested in whether children’s understanding of display rules can be predicted from their age, and whether the child possesses a theory of mind. A display rule is a convention of displaying an appropriate emotion in a given situation. For example, if you receive a Christmas present that you don’t like, the appropriate emotional display is to smile politely and say ‘Thank you Auntie Kate,



I've always wanted a rotting cabbage.' The inappropriate emotional display is to start crying and scream 'Why did you buy me a rotting cabbage you selfish old bag?' Using appropriate display rules has been linked to having a theory of mind (the ability to understand what another person might be thinking). To test this theory, children were given a false belief task (a task used to measure whether someone has a theory of mind), a display rule task (which they could either pass or fail) and their age in months was measured. The data are in **Display.sav**. Run a logistic regression to see whether possession of display rule understanding (did the child pass the test: Yes/No?) can be predicted from possession of a theory of mind (did the child pass the false belief task: Yes/No?), age in months and their interaction. ③

- **Task 2:** Recent research has shown that lecturers are among the most stressed workers. A researcher wanted to know exactly what it was about being a lecturer that created this stress and subsequent burnout. She took 467 lecturers and administered several questionnaires to them that measured: **Burnout** (burnt out or not), **Perceived Control** (high score = low perceived control), **Coping Style** (high score = high ability to cope with stress), **Stress from Teaching** (high score = teaching creates a lot of stress for the person), **Stress from Research** (high score = research creates a lot of stress for the person) and **Stress from Providing Pastoral Care** (high score = providing pastoral care creates a lot of stress for the person). The outcome of interest was burnout, and Cooper, Sloan, and Williams' (1988) model of stress indicates that perceived control and coping style are important predictors of this variable. The remaining predictors were measured to see the unique contribution of different aspects of a lecturer's work to their burnout. Can you help her out by conducting a logistic regression to see which factors predict burnout? The data are in **Burnout.sav**. ③
- **Task 3:** A health psychologist interested in research into HIV wanted to know the factors that influenced condom use with a new partner (relationship less than 1 month old). The outcome measure was whether a condom was used (Use: condom used = 1, not used = 0). The predictor variables were mainly scales from the Condom Attitude Scale (CAS) by Sacco, Levine, Reed, and Thompson (*Psychological Assessment: A Journal of Consulting and Clinical Psychology*, 1991): **Gender** (gender of the person); **Safety** (relationship safety, measured out of 5, indicates the degree to which the person views this relationship as 'safe' from sexually transmitted disease); **Sexexp** (sexual experience, measured out of 10, indicates the degree to which previous experience influences attitudes towards condom use); **Previous** (a measure not from the CAS, this variable measures whether or not the couple used a condom in their previous encounter, 1 = condom used, 0 = not used, 2 = no previous encounter with this partner); **selfcon** (self-control, measured out of 9, indicates the degree of self-control that a person has when it comes to condom use, i.e. do they get carried away with the heat of the moment, or do they exert control?); **Perceive** (perceived risk, measured out of 6, indicates the degree to which the person feels at risk from unprotected sex). Previous research (Sacco, Rickman, Thompson, Levine, and Reed, 1993) has shown that **Gender**, **relationship safety** and **perceived risk** predict condom use. Carry out an appropriate analysis to verify these previous findings, and to test whether self-control, previous usage and sexual experience can predict any of the remaining variance in condom use. (1) Interpret all important parts of the SPSS output. (2) How reliable is the final model? (3) What are the probabilities that participants 12, 53 and 75 will use a condom? (4) A female who used a condom in her previous encounter with her new partner scores 2 on all variables except perceived risk (for which she scores 6). Use the model to estimate the probability that she will use a condom in her next encounter. Data are in the file **condom.sav**. ③



Answers can be found on the companion website.

Further reading

- Hutcheson, G., & Sofroniou, N. (1999). *The multivariate social scientist*. London: Sage. Chapter 4.
- Menard, S. (1995). *Applied logistic regression analysis*. Sage university paper series on quantitative applications in the social sciences, 07–106. Thousand Oaks, CA: Sage. (This is a fairly advanced text, but great nevertheless. Unfortunately, few basic-level texts include logistic regression so you'll have to rely on what I've written!)
- Miles, J. N. V., & Shevlin, M. (2001). *Applying regression and correlation: a guide for students and researchers*. London: Sage. (Chapter 6 is a nice introduction to logistic regression.)

Online tutorial

The companion website contains the following Flash movie tutorial to accompany this chapter:

- Logistic regression using SPSS

Interesting real research

- Bale, C., Morrison, R., & Caryl, P. G. (2006). Chat-up lines as male sexual displays. *Personality and Individual Differences*, 40(4), 655–664.
- Bemelman, M., & Hammacher, E. R. (2005). Rectal impalement by pirate ship: A case report. *Injury Extra*, 36, 508–510.
- Cooper, M., O'Donnell, D., Caryl, P. G., Morrison, R., & Bale, C. (2007). Chat-up lines as male displays: Effects of content, sex, and personality. *Personality and Individual Differences*, 43(5), 1075–1085.
- Lacourse, E., Claes, M., & Villeneuve, M. (2001). Heavy metal music and adolescent suicidal risk. *Journal of Youth and Adolescence*, 30(3), 321–332.
- Lo, S. F., Wong, S. H., Leung, L. S., Law, I. C., & Yip, A. W. C. (2004). Traumatic rectal perforation by an eel. *Surgery*, 135(1), 110–111.